M.Sc. Engg. (CSE) Thesis

Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

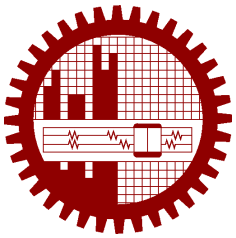
Submitted by

Md. Sakif Khan

0421052099

Supervised by

Dr. Sadia Sharmin



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Candidate's Declaration

I, do, hereby, certify that the work presented in this thesis, titled, "Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models", is the outcome of the investigation and research carried out by me under the supervision of Dr. Sadia Sharmin, Associate Professor, Department of CSE, BUET.

I also declare that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

Md. Sakif Khan

0421052099

The thesis titled "**Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models**", submitted by Md. Sakif Khan, Student ID 0421052099, Session April 2021, to the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology, has been accepted as satisfactory in partial fulfilment of the requirements for the degree of Master of Science in Computer Science and Engineering and approved as to its style and contents on June 6, 2026.

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Md. Sakif Khan
0421052099

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Abstract

Large language models answer factual questions fluently, and just as fluently when they do not hold the fact. Retrieval-augmented generation grounds them in retrieved text, but it retrieves once, before reasoning begins, so a multi-hop question must be answered from whatever that query returned. Agentic systems that navigate a knowledge graph interleave the two, but they still emit whatever their final generation call produces, with nothing checking what the answer asserts before it is delivered.

To address this, we propose Agentic Graph Reasoning (AGR), a knowledge-graph question-answering framework in which an agent plans sub-objectives and navigates the graph through a constrained, deterministic tool API. Its distinguishing component is a Structural Verification Layer, which splits the draft answer into atomic claims and checks each against the triples the agent actually traversed, before the answer is emitted. Claims it cannot ground are re-explored or withdrawn, so AGR hedges rather than asserts and returns every answer with its supporting triples.

We evaluate AGR on a Freebase-derived environment of $2{,}592{,}892$ entities and $8{,}309{,}194$ triples, over 400 certified questions from each of WebQSP and ComplexWebQuestions, against four baselines: parametric answering, vector retrieval, GraphRAG, and an agentic navigator. All five run on one frozen backbone under the same budget, so architecture rather than model capacity accounts for the differences.

AGR reaches Hits@1 of 0.755 and F1 of 0.642 on WebQSP and 0.522 and 0.469 on ComplexWebQuestions, ahead of every baseline — including the agentic one at 0.660 and 0.435 — at half the language-model calls. Across both datasets it asserts no ungrounded entity in $1{,}709$ assertions, against 22.1% of the parametric control's $1{,}001$ — a property we attribute to graph navigation, not to the verification layer. On ComplexWebQuestions its accuracy rises with hop count while every other system decays.

A four-condition ablation with paired significance testing detects an accuracy effect in only one component, and its sign is not the expected one: removing the planner *improves* WebQSP by 0.083 F1 at $p = 0.006$ and cuts tokens by 30%, while trending the other way on ComplexWebQuestions. Decomposition should therefore be gated on question structure, not applied unconditionally. We read all 259 remaining failures and name the characteristic error of a graph-navigating system: not invention, but the *echo attractor* — a real, grounded entity one hop from the answer, which any grounding check passes because it is true, and shared across systems. It also confirms 58 questions where the benchmark, not the system, was wrong.

Chapter 1

Introduction

Large language models answer factual questions fluently, and often enough they answer them incorrectly. The failure is not random noise but a systematic and confident one, and it is hard to catch from the output alone, because a fabricated answer reads in exactly the same register as a correct one. Its rate rises sharply once an answer requires composing several facts rather than recalling a single one. This thesis closes that gap in two steps: we give the model a symbolic knowledge graph to walk through, and we then check what the model is about to assert before it asserts it.

This chapter motivates the problem and examines why the obvious remedies remain incomplete. It then states the problem and the research questions, summarises our contributions, and sets out how the rest of the document is organised.

1.1 Hallucination in Large Language Models

A *hallucination* is a generated statement that is fluent and syntactically well formed but that no source the model can point to supports; frequently it is simply false. Huang et al. [1] survey the phenomenon at length and treat it as an intrinsic consequence of how these models are built rather than as an incidental defect to be patched away. That framing matters here, because it decides what kind of remedy is available at all. If hallucination were a bug, the response would be to fix it. Because it is structural, the response has to be architectural.

1.1.1 Where Hallucination Originates: Training Data, Model, and Prompt

Hallucinations arise at three distinct loci, and separating them is worthwhile because each admits a very different intervention.

The first is the *training data*. A model trained on a corpus that carries errors, staleness, or thin coverage of a domain reproduces those defects faithfully. Correcting them means curating the

corpus and retraining, which only the model's producer can do.

The second is the *model* itself — its parameterisation and its decoding procedure. A model stores what it knows diffusely across its weights with no explicit index, so it holds no reliable internal signal separating a fact it has memorised from a plausible continuation it is constructing. Intervening here means changing the architecture or the training objective, which is again largely out of reach.

The third is the *prompt*, and it is the locus this thesis addresses. When the context supplied alongside a question already carries the facts needed to answer it, the model need not fall back on parametric recall, and structured context can override defects inherited from the training data without touching the model at all. Retrieval augmentation [2] rests on that premise, and so does the present work. Our contribution lies in *how* we assemble that context, and in *what we check* before the answer is emitted.

1.1.2 Why Multi-Hop Questions Are the Hard Case

Single-fact questions are comparatively benign: "Where was Ada Lovelace born?" takes one lookup, and the model either knows it or it does not.

Multi-hop questions — "Who directed the film that won Best Picture in 1998?" — instead demand an ordered chain of lookups, each one conditioned on the result of the last, and two properties make them the hard case. First, errors compound: a wrong resolution at the first hop rarely surfaces downstream, so the rest of the chain proceeds confidently from a false premise. Second, a question-level correctness check cannot see the intermediate results at all, which leaves a system that answers correctly by accident indistinguishable from one that answers correctly by sound reasoning.

Such questions also carry temporal, categorical, or ordinal *constraints* that a system has to satisfy jointly rather than sequentially. Dropping one silently is among the commonest ways a system answers such a question wrongly instead of declining to answer it.

1.2 From Retrieval Augmentation to Graph-Structured Grounding

1.2.1 Vector Retrieval and Its Structural Blind Spot

Retrieval-Augmented Generation [2] attaches the model to an external corpus: the system embeds the question, retrieves semantically similar passages by vector similarity, and lets the model answer from them. Where the answer sits in a single retrievable passage, this works well.

Its weakness is structural. Vector retrieval treats a corpus as flat chunks of text and ranks them by similarity to the question, and for a multi-hop question that signal misleads. The passage holding the *intermediate* entity may bear little lexical or semantic resemblance to the original question, and the system does not yet know which entity to look for. Worse, flat retrieval does not represent relationships between entities at all — they survive at best as adjacency inside a retrieved chunk — and relationships are precisely what a compositional question traverses. Surveys of the intersection between graphs and language models identify this as the central limitation of flat retrieval [3].

1.2.2 Knowledge Graphs and Static GraphRAG

A *knowledge graph* makes those relationships explicit. It stores facts as typed edges between identified entities, so composing two facts becomes a traversal rather than an inference over prose. GraphRAG [4] exploits this by building a graph over a corpus and retrieving structured context in place of passages, and related systems extend the idea to long-term memory [5] and to temporally-qualified facts [6].

These systems nonetheless retrieve *statically*: they fetch a subgraph around the linked entities in one shot and hand it to the model. Because retrieval runs first and runs only once, it cannot know what the reasoning step will need. Consider a question whose second hop depends on the answer to its first. A fixed-radius neighbourhood is then either too small to contain that answer, or so large that the relevant edges lie buried among thousands of irrelevant ones.

1.2.3 Agentic Retrieval: Reasoning as Navigation

The natural response is to interleave retrieval with reasoning, letting the model take one step, observe what it found, and decide where to look next. Retrieval then becomes *navigation*, and the system becomes an agent operating on the graph rather than a pipeline reading from it.

Think-on-Graph [7] established the pattern with LLM-guided beam search over the graph. Plan-on-Graph [8] added sub-objective decomposition and reflection-driven backtracking, and Reasoning-on-Graphs [9] generates grounded relation paths as explicit plans. Further systems add tool abstractions [10], handle incomplete graphs [11], or generalise across structured sources [12]. Recent surveys treat this convergence of language models and knowledge graphs as a distinct research programme [13, 14].

Agentic navigation is the right shape for the problem, and this thesis adopts it. What remains open is what it leaves unsolved.

1.3 Limitations of Existing Agentic KGQA Systems

1.3.1 No Check on What the Final Answer Asserts

Agentic systems ground their answers in a traversed subgraph, and our own measurements confirm that they do so effectively: the entities they assert exist in the graph, and they are reachable along the paths the systems walked (Section 8.6). Fabrication is not the deficit.

The deficit is *precision*. Every system we survey in Chapter 3 emits whatever its final generation call produces from the traversed context, and nothing checks that each asserted entity actually answers the question that was asked. An entity can sit in the retrieved subgraph, have been retrieved correctly, and still be the wrong answer: an intermediate node from the reasoning chain, a container where the question asked for a member, or one of several plausible neighbours chosen arbitrarily. A system may equally assert many entities where the question admits few, which inflates recall at the cost of precision. A system that verifies retrieval but never verifies assertion cannot see either failure, and both surface as a measurable precision gap against a system that does check (Section 8.2).

Chain-of-Verification [15] and multi-agent debate [16] establish that post-hoc self-checking reduces factual error in free-form generation, and Hu et al. [17] demonstrate a self-correcting agentic graph pipeline in a clinical setting. What none of them supplies is a claim-level check against the retrieved triples, performed *before* the answer is emitted, inside a graph-navigating agent.

1.3.2 Unquantified Contribution of Individual Agentic Mechanisms

Agentic systems are assemblies of mechanisms — decomposition, scoring, backtracking, self-correction — and the literature reports them as wholes. When such a system outperforms a static baseline, published work rarely establishes which mechanism produced the gain, or whether some contribute nothing at all. Without that attribution, subsequent work inherits the entire assembly on faith, and its cost along with it. Component-level ablation is the standard remedy, and this literature conspicuously underuses it.

1.3.3 Accuracy Reported Without Cost

Navigation is expensive. Each exploration step costs at least one model call, and agentic systems routinely spend an order of magnitude more tokens per question than a single-shot baseline. An accuracy table that omits this makes agentic methods look uniformly preferable, when the honest picture is a trade-off — and that trade-off is exactly what a practitioner deciding whether to deploy such a system needs to see.

1.4 Problem Statement

We address the following problem. Given a natural-language question q and a knowledge graph $G = (E, R, T)$ with entities E, relation types R, and triples $T \subseteq E \times R \times E$, produce an answer set $A \subseteq E$ together with a supporting triple set $S \subseteq T$, such that S justifies every entity in A and contains only triples the system actually traversed in reaching A.

Three properties distinguish this from standard knowledge graph question answering. First, the output is a *pair* rather than an answer alone: the supporting triples belong to the contract, which makes the reasoning auditable instead of merely asserted. Second, that support must be *traversed* and not simply retrievable after the fact, which forecloses justifying post hoc an answer the system reached some other way. Third, the system checks the justification *before* it emits the answer, so it can remove or hedge unsupported content rather than merely flag it.

1.5 Research Questions

1.5.1 RQ1: Does Agentic Navigation Improve Multi-Hop Factual Accuracy?

Does interleaving retrieval with reasoning over a knowledge graph produce more accurate answers than static retrieval, and does the advantage grow with the number of hops a question requires? Answering this needs a no-retrieval control, because the standard benchmarks predate current models and may be partly memorised.

1.5.2 RQ2: What Does Pre-Generation Verification Contribute Beyond Graph Navigation?

Given a system that already navigates the graph, what does a claim-level check against the traversed triples add? We pose the question as *what does it contribute* rather than *does it reduce hallucination*, because the answer turns out to have several parts that a yes-or-no framing would obscure (Section 8.9).

1.5.3 RQ3: Which Components Contribute What, at What Token Cost?

Decomposition, backtracking, verification, and learned candidate scoring each cost model calls. Which of them measurably improve accuracy or groundedness, by how much, and at what price in tokens and latency?

1.6 Our Contribution

1.6.1 The AGR Framework and Its Verification Layer

We present Agentic Graph Reasoning (AGR), a knowledge-graph question-answering framework built as an explicit state machine over a constrained, deterministic graph tool API. It carries a *Structural Verification Layer* that decomposes the draft answer into atomic claims and checks each one against the traversed subgraph before the system emits anything. Unsupported claims trigger targeted re-exploration, or drop out of the answer, and AGR returns what survives paired with its supporting triples.

1.6.2 A Component-Level Ablation of Agentic Mechanisms

We ablate four components — the planner, the backtracking mechanism, the verification layer, and the learned component of candidate scoring — and report accuracy, groundedness, and token cost for each condition, with paired significance testing. This closes, for one architecture, the attribution gap we identify in Section 1.3.

1.6.3 Stratum-Dependent Decomposition: When Planning Hurts

The literature treats question decomposition as straightforwardly beneficial. We find that it is not: removing the planner *improves* accuracy on the shallower benchmark while trending toward harming it on the deeper one. The asymmetry, rather than the pooled average, is the result.

1.6.4 The Echo Attractor as a Named Failure Mode

We identify and characterise a recurring failure in which several independent systems converge on the same wrong answer — typically the question's own topic entity, an intermediate node from the reasoning chain, or a coarser-granularity container. Because the systems share it, no evaluation that treats them as independent can see it.

1.6.5 Quantified Benchmark Defect Rates for WebQSP and CWQ

We quantify label defects in the two standard benchmarks through a consensus pre-pass followed by per-item adjudication, separating two kinds of label error because they carry opposite evidence signatures, and we report results both with and without the affected items.

1.6.6 An Evaluation Protocol with Pre-Registered Thresholds

We fixed four decisions before touching any test data: the scoring hyperparameters, the sample size, the baseline-certification diagnostic, and the judge-agreement bar. We report the outcome against all four, including the one we narrowly missed.

1.7 Scope and Delimitations

Our knowledge environment is a static, curated subset of Freebase [18] — specifically, the union of per-question subgraphs, each pre-extracted so as to contain its own question's answer, which makes it a friendlier place to search than the full knowledge base and puts answer reachability above 97% by construction (Section 4.7). The absolute accuracies we report are therefore not comparable to evaluations over full Freebase, though the comparisons between systems are, because every system navigates the identical environment; Section 9.9.3 states the limit in full. We do not address graph construction from unstructured text, knowledge-base completion, or the temporal evolution of facts, and we draw only English factoid questions, from WebQSP [19] and ComplexWebQuestions [20]. Long-form generation and conversational settings lie outside our scope.

All systems share one backbone model, which we hold constant so that no difference we report can be credited to model capability rather than to architecture. We therefore establish how these architectures compare on a single backbone, not how they scale across backbones. Finally, AGR is a research prototype that we evaluate offline, and we report latency as a cost measure rather than as evidence of production readiness.

1.8 Thesis Organization

Chapter 2 establishes the background the technical chapters assume. It covers knowledge graphs and their Freebase-specific idiosyncrasies, knowledge graph question answering and its evaluation conventions, language models as reasoning engines, retrieval augmentation, graph search, and the statistical tools we use throughout. Chapter 3 surveys prior work and positions AGR against it, culminating in a comparison of agentic knowledge graph question answering systems that identifies the gap this thesis addresses and justifies our choice of baselines.

Chapter 4 describes how we build and validate the knowledge environment, including the answer-reachability gate that bounds the accuracy attainable in it. Chapter 5 specifies the AGR framework: the tool API, the state machine, the planner, the scoring function, the backtracking policy, and the budgets that guarantee termination. Chapter 6 treats the Structural Verification Layer in detail, as the thesis's principal architectural contribution.

Chapter 7 sets out the experimental design: test sets, baselines, metrics, judge validation, gold-answer quality control, and the pre-registration policy. Chapter 8 reports what we measured and answers the three research questions. Chapter 9 analyses the failures that remain, including the benchmark defects and the shared-attractor pattern, and discusses threats to validity. Chapter 10 summarises the contributions, states the limitations, and identifies future work.

Chapter 2

Background and Preliminaries

This chapter establishes the concepts and notation the technical chapters assume. It is deliberately compact: only material actually used later is included, and standard results are stated rather than derived.

2.1 Knowledge Graphs

2.1.1 Triples, Entities, and Relations

A *knowledge graph* $G = (E, R, T)$ consists of a set of entities E, a set of relation types R, and a set of triples $T \subseteq E \times R \times E$. A triple (h, r, t) asserts that relation r holds from head entity h to tail entity t — for example, (Ada Lovelace, place_of_birth, London). Relations are directed and typed, and both directions carry meaning: the inverse of place_of_birth is a legitimate but distinct relation.

Two properties matter for what follows. Triples are *atomic*: each asserts exactly one fact, so a claim can be checked against the graph by testing for the existence of specific edges. And they are *composable*: a chain $(e_0, r_1, e_1), (e_1, r_2, e_2), \ldots$ expresses a multi-step relationship that no single triple states, which is what makes a knowledge graph the natural substrate for multi-hop questions.

2.1.2 Freebase: MIDs, Schema, and Mediator Nodes

Freebase [18] is a large collaboratively-built knowledge base, frozen since 2016, and the substrate for the standard knowledge graph question answering benchmarks. Three of its characteristics affect any system built on it.

Entities are identified by opaque *machine identifiers* (MIDs) such as m.02mjmr, with human-readable names attached as a separate property. Names are neither unique nor stable, so entity

identity and entity surface form must be handled separately.

Relations are namespaced paths such as people.person.place_of_birth, encoding domain, type, and property. The vocabulary is large — tens of thousands of relation types — and includes many that carry no factual content, such as internal type and key predicates.

Most consequentially, Freebase represents n-ary facts through *mediator* or *compound value type* (CVT) nodes. A fact with more than two participants — an actor playing a particular character in a particular film — is not a single edge but an anonymous intermediate node linking the participants. A relationship a user would describe as one hop is therefore two edges in the graph. Any system reporting hop counts over Freebase must state how it treats these nodes, since the choice changes every depth measurement; systems that ignore them lose the facts they encode [11].

2.1.3 Property Graphs, Neo4j, and Cypher

The triple model above is the RDF view. An alternative is the *property graph* model, in which nodes and edges are both typed and may carry arbitrary key–value properties. This is the model implemented by Neo4j [21] and queried by Cypher [22], a declarative language whose MATCH clause expresses graph patterns directly. A triple store maps onto it naturally: entities become nodes carrying their MID and name, and relations become typed edges. Traversal patterns that would require recursive joins in a relational schema are expressed as literal path patterns, which is the practical reason for choosing a graph store here.

2.2 Knowledge Graph Question Answering

2.2.1 Multi-Hop Questions and Constraint Satisfaction

Knowledge graph question answering (KGQA) takes a natural-language question and a graph, and returns the entity or entities that answer it. A question is k*-hop* if answering it requires traversing k edges from the entities named in the question — its *topic entities* — to the answer.

Beyond hop count, questions vary in *constraints*: conditions that filter candidates rather than extending the chain. “Which of Tolstoy’s novels was published after 1875?” is one hop plus a temporal constraint. Constraints must be satisfied jointly with the chain, and a system that traverses correctly while silently dropping a constraint returns a superset of the answer — a failure that hop-count analysis alone does not expose.

Two families of approach have historically divided this field. *Semantic parsing* methods translate the question into a formal query (SPARQL or equivalent) and execute it, giving exact and interpretable results but requiring the parse to be exactly right [19, 23]. *Information retrieval* methods retrieve a relevant subgraph and rank candidate answers within it, degrading more

gracefully but sacrificing transparency. The agentic systems considered in this thesis are closer to the second family, but recover much of the first family's transparency by logging the traversal itself.

2.2.2 Evaluation Conventions: Hits@1 and F1

Two metrics are conventional. *Hits@1* scores a question 1 if any entity the system asserts is in the gold answer set, and 0 otherwise. It is a per-question binary correctness measure and the headline number in most of this literature.

The name is historical and slightly misleading. The metric was defined for systems that emitted a *ranked* list, where only the top entry counted; it has been carried over to systems that emit an unordered set, where it becomes the set-intersection test just stated. This thesis uses the set-intersection form throughout, because that is what the systems it compares against report, and Section 7.5 fixes the string-matching rule exactly. The looseness is not incidental — it is precisely why F1 is reported alongside.

Because many questions admit multiple correct answers, Hits@1 alone rewards a system that names one correct entity and ignores the rest. *F1* therefore complements it, computed per question between the predicted answer set A and the gold set A^*:

$$P = \frac{|A \cap A^*|}{|A|}, \qquad R = \frac{|A \cap A^*|}{|A^*|}, \qquad F_1 = \frac{2PR}{P+R},$$

and averaged over questions. The distinction between P and R is load-bearing in this thesis: a system that asserts many entities can achieve high recall while its precision falls, and Section 1.3 argues that precisely this failure goes unmeasured when only Hits@1 is reported.

2.3 Large Language Models as Reasoning Engines

2.3.1 In-Context Learning and Chain-of-Thought Prompting

Sufficiently large language models perform tasks specified entirely in their prompt, without gradient updates — *in-context learning* [24]. Prompting the model to produce intermediate reasoning steps before its final answer, *chain-of-thought* prompting [25], substantially improves performance on multi-step problems. Both are used here: the agent's planner, scorer, evaluator, and verifier are all prompted components of a single frozen model, and no component is fine-tuned. This matters for comparability, since several systems surveyed in Chapter 3 are fine-tuned and therefore not directly comparable to prompting methods.

2.3.2 Tool Use and the ReAct Loop

A language model can be given *tools*: named operations it may invoke, whose results re-enter its context [26]. The *ReAct* pattern [27] interleaves reasoning and action in a loop — the model reasons about what to do, acts, observes the result, and reasons again — which is the control structure underlying agentic graph navigation. Where the tools are graph operations, each iteration extends a traversal, and the sequence of tool calls is a complete record of how the answer was reached.

2.3.3 A Working Definition of Hallucination for This Thesis

"Hallucination" is used loosely in the literature. This thesis adopts a narrow, operational definition tied to the knowledge environment: an asserted entity is *ungrounded* if it does not exist in the graph, or exists but is not reachable from the question's topic entities within the traversal budget. A claim is *unsupported* if the traversed triples do not entail it.

Two consequences of this definition should be stated plainly. It is relative to the environment: an entity absent from the graph is ungrounded even if true in the world. And groundedness is not correctness — an answer can be perfectly grounded, every entity real and reachable, and still be the wrong answer to the question. Chapter 8 shows that this gap is not hypothetical, and Section 1.3 is built on it.

2.4 Retrieval-Augmented Generation

2.4.1 Dense Retrieval and Sentence Embeddings

Dense retrieval encodes queries and documents into a shared vector space and retrieves by nearest neighbour, capturing semantic similarity where lexical matching fails [28]. Sentence-level encoders such as Sentence-BERT [29] produce embeddings suitable for direct cosine comparison. In this thesis, embeddings serve two bounded purposes — linking question phrases to graph entities, and pre-ranking relation candidates during exploration. They never establish that a fact is true; that role belongs exclusively to the symbolic triples.

2.4.2 Graph-Based Retrieval

Graph-based retrieval replaces flat passages with structured context [4, 5]. The retrieved unit is a subgraph rather than a document, which preserves relationships explicitly. The distinction that matters for this thesis is not graph versus vector but *static versus interleaved*: whether retrieval happens once before reasoning begins, or repeatedly as reasoning proceeds.

2.5 Search over Graphs

2.5.1 Best-First and Beam Search

Best-first search maintains a frontier of candidate nodes ordered by a heuristic estimate of promise and expands the most promising first [30, 31]. *Beam search* bounds the frontier to the b highest-scoring candidates at each depth, trading completeness for a fixed memory and computation budget. With an expensive scoring function — here, a language model call — the beam width is also the principal cost control.

2.5.2 Backtracking and Dead-End Detection

Bounded search commits to choices that may prove wrong. *Backtracking* recovers by abandoning the current branch and resuming from a previously unexplored alternative, which requires maintaining a stack of such alternatives and a trigger condition for when to pop it. Trigger conditions and the stack discipline used here are specified in Section 5.6.

2.5.3 Relation to Monte Carlo Tree Search: A Terminological Clarification

The original proposal for this work described its navigation strategy as "inspired by Monte Carlo Tree Search". That description is withdrawn here, and the reason is worth stating explicitly.

Monte Carlo Tree Search [32, 33] is defined by four phases: selection, expansion, *simulation* (a rollout to a terminal state), and *backpropagation* of the rollout's value to the ancestors of the expanded node. The two phases that give MCTS its statistical character are simulation and backpropagation, which together let node values be estimated from sampled outcomes rather than from a fixed heuristic.

The method used in this thesis has neither. There are no rollouts, and no value is propagated back up the search tree; each candidate is scored once, by a fixed function combining embedding similarity with a language-model judgement, and that score is never revised in light of what is found later. What the method is, is *guided best-first search with a bounded beam and backtracking*. It is described that way throughout.

The contrast is not hypothetical. Genuine tree search has been applied to this exact task: KBQA-o1 [34] drives an agentic knowledge base question answering process with full MCTS — policy and reward models, rollouts, and value backpropagation — over Freebase, and Reasoning with Trees [35] likewise builds an explicit search tree. That those systems exist is precisely why the terminology matters here: describing a single-pass scoring heuristic as MCTS-inspired would invite comparison with machinery this framework does not implement.

2.6 Statistical Tools Used in This Thesis

Three tools recur, each chosen for a specific property of the data.

Bootstrap confidence intervals [36] quantify sampling uncertainty by resampling the evaluated questions with replacement and recomputing the metric; the 2.5th and 97.5th percentiles of the resulting distribution give a 95% interval. They are used because the test sets are samples, and because they require no distributional assumption about metrics such as per-question F1.

McNemar's test [37] assesses whether two systems differ in accuracy when both are evaluated on the *same* questions. It considers only the discordant pairs — questions one system answers correctly and the other does not — and tests whether the split between the two discordant counts departs from chance. Paired testing is essential here because all systems are run on identical question samples, so an unpaired test would discard the pairing and lose power.

Cohen's κ [38] measures agreement between two annotators on a categorical judgement, correcting for agreement expected by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is observed agreement and p_e is the agreement expected if both annotators labelled independently at their observed marginal rates. It is used to validate the automatic groundedness judge against human annotation. Conventional interpretation follows Landis and Koch [39], on whose scale values between 0.61 and 0.80 denote substantial agreement.

Chapter 3

Related Work

This chapter surveys the work AGR builds on and competes with. It proceeds from static graph-augmented generation, through path-retrieval and agentic exploration, to verification and factuality measurement, and closes with a structured comparison of the seven closest systems. That comparison serves two purposes: it locates the gap this thesis addresses, and it justifies the choice of baselines in Section 7.4.

3.1 Static Graph-Augmented Generation

3.1.1 GraphRAG and Query-Focused Summarization

GraphRAG [4] constructs a knowledge graph from a text corpus using a language model, partitions it into communities, and pre-computes summaries at each level, so a query can be answered from structured context rather than retrieved passages. Its strength is global questions — those whose answer depends on the corpus as a whole rather than any single passage — which flat retrieval handles poorly.

The relevant limitation is that retrieval remains a single step preceding generation. The context is assembled before the model has begun reasoning, so it cannot be conditioned on intermediate findings. GraphRAG also builds its graph by language-model extraction, which is appropriate for unstructured corpora but introduces extraction error into the reference itself — a consideration that shapes the environment design in Section 4.1.

3.1.2 HippoRAG and Memory-Structured Retrieval

HippoRAG [5] draws on the hippocampal indexing theory of human memory, combining a knowledge graph with a personalised PageRank retrieval step so that a single query can integrate evidence distributed across many passages. It substantially improves multi-hop retrieval over flat baselines at low cost.

Its retrieval is nonetheless a single graph-propagation step, not an agent-directed traversal: the propagation is fixed and the model does not choose where to look next. Temporal extensions such as T-GRAG [6] address a different axis — conflicting and redundant facts across time — which is orthogonal to the navigation question and is noted here for completeness rather than compared against.

3.2 Path-Retrieval and Semantic-Parsing KGQA

3.2.1 Reasoning-on-Graphs

Reasoning-on-Graphs (RoG) [9] adopts a planning–retrieval–reasoning pipeline: a fine-tuned model generates relation paths grounded in the graph schema, those paths are used to retrieve matching instantiations, and the model reasons over the retrieved paths. Because answers are produced from retrieved paths, RoG is faithful by construction and can present the path as an explanation.

Two properties distinguish it from the present work. It is *fine-tuned*, which places its reported numbers outside direct comparison with prompting methods. And its plan is generated in one shot before retrieval: if the generated relation path does not instantiate in the graph, there is no mechanism to revise it mid-course. The pre-generation verification question does not arise, because grounding is guaranteed structurally rather than checked.

3.2.2 StructGPT and KG-Agent

StructGPT [12] provides a general interface for reasoning over structured data — knowledge graphs, tables, and databases — through an invoke–linearise–generate loop, in which specialised interfaces extract evidence that is linearised into text for the model. It is deliberately general rather than graph-specific, and performs no open-ended graph search.

KG-Agent [10] is closer to the agentic setting: a multifunctional toolbox, a knowledge-graph executor, and a dynamic memory, driven by an instruction-tuned model that autonomously selects tools across iterations. It demonstrates that a small fine-tuned model with good tool abstractions is competitive with much larger prompted ones. It plans implicitly through tool selection rather than committing to explicit sub-objectives, provides no backtracking mechanism, and performs no verification of the final answer.

3.3 Agentic Graph Exploration

3.3.1 Think-on-Graph

Think-on-Graph (ToG) [7] established the template for training-free agentic KGQA. The model performs iterative beam search over the graph: at each step it is shown candidate relations and entities and prunes them itself, with beam width and maximum depth both set to three. Exploration terminates when the model judges the retrieved paths sufficient, at which point it answers directly.

ToG is the most directly comparable prior system to AGR — training-free, prompting-based, operating on Freebase, evaluated on the same benchmarks — and is therefore the agentic baseline in Section 7.4. Its limitations are instructive. There is no explicit decomposition, so multi-constraint questions are handled implicitly. Low-scoring branches are pruned within the beam, but there is no mechanism to return to an abandoned node once the beam has moved on. And the transition from exploration to answering is a single model judgement: nothing checks the answer that follows.

3.3.2 Plan-on-Graph

Plan-on-Graph (PoG) [8] is the closest system to AGR in ambition. It decomposes the question into sub-objectives including constraint conditions, explores adaptively with a breadth that varies by sub-objective, and adds Guidance, Memory, and Reflection mechanisms. Reflection decides both whether to self-correct and which entity to backtrack to, making PoG the one surveyed system with genuine backtracking.

PoG's self-correction operates on the *search process*: it reconsiders where to explore. It does not decompose the draft answer into claims and check each against the retrieved triples before responding. The distinction is the gap this thesis occupies, and it is stated precisely in Section 3.7.

A naming hazard must be recorded here. A second, unrelated system is also abbreviated PoG — Paths-over-Graph [40] — and reports considerably higher figures. Survey tables that cite "PoG" without disambiguation are a known source of contaminated comparisons, and this thesis always writes Plan-on-Graph or Paths-over-Graph in full.

3.3.3 Generate-on-Graph and Reasoning with Trees

Generate-on-Graph (GoG) [11] targets the incomplete-graph setting through a thinking–searching–generating loop: when the graph lacks a required fact, the model generates candidate triples from its parametric knowledge and continues. It also expands subgraphs dynamically to handle the mediator nodes described in Section 2.1, which ToG tends to skip. The trade-off

is directly opposed to this thesis's: GoG deliberately admits model-generated facts into the evidence, whereas AGR admits only graph triples.

Reasoning with Trees [35] applies explicit tree search to knowledge graph question answering. It is cited in Section 2.5.3 as an example of genuine tree-search machinery, in contrast with the guided best-first search implemented here.

3.3.4 KBQA-o1 and Genuine Tree Search

KBQA-o1 [34] is the closest system to the search strategy the present work's proposal originally described, and therefore the most important one to distinguish from it. It performs agentic knowledge base question answering with full Monte Carlo Tree Search — a policy model proposing actions, a reward model scoring them, and rollouts whose values propagate back through the tree — wrapped around a ReAct-style agent that generates logical forms step by step against a Freebase environment. The exploration additionally generates annotations used for incremental fine-tuning, so the search improves the model that drives it.

Two aspects bear on this thesis. First, KBQA-o1 is what the withdrawn "MCTS-inspired" description of Section 2.5.3 would have claimed: it has the rollouts and value backpropagation that AGR does not, and naming the difference honestly is what keeps the two separable.

Second, its ablation is directly relevant to the methodological argument of Section 3.7. Removing MCTS from KBQA-o1 while leaving every other component intact drops overall GrailQA F1 from 78.5 to 48.5 — a thirty-point attribution to a single mechanism, obtained precisely because the authors ablated components rather than reporting the assembly as a whole. That is the same instrument this thesis applies in Section 8.8, and it is evidence that the instrument is informative.

KBQA-o1 is not a baseline here. It is fine-tuned rather than prompted, evaluates under a low-resource few-shot protocol, and does not report ComplexWebQuestions, so no controlled comparison against it is available in this environment.

3.4 Verification and Self-Correction in Language Models

3.4.1 Chain-of-Verification

Chain-of-Verification (CoVe) [15] has the model draft a response, plan verification questions about it, answer those independently, and regenerate a corrected response. It establishes the core premise of this thesis's verification layer — that decomposing a draft and checking its parts reduces factual error — but performs the checks against the model's own knowledge rather than an external symbolic source. Multi-agent debate [16] achieves a related effect by having several model instances critique one another.

This distinction matters given evidence that language models cannot reliably self-correct reasoning without external feedback [41]. A verification step whose ground truth is the same model that produced the error is limited by construction. AGR's verification is grounded in traversed triples, making the check external to the generator.

3.4.2 Ontology-Guided and Self-Correcting Graph RAG

A self-correcting agentic graph RAG system for clinical decision support in hepatology [17] demonstrates the combination of agentic retrieval with a correction loop in a high-stakes domain, and ontology-guided reasoning constrains an agent to logically valid relation use. These substantiate the practical motivation for verification, though in domain-specific settings rather than on the open-domain benchmarks used here. Reflexion-style verbal self-improvement [42] provides the general agentic pattern of which these are graph-specific instances.

3.5 Measuring Factuality

3.5.1 Claim-Decomposition Metrics

FActScore [43] evaluates long-form generation by decomposing text into atomic facts and verifying each against a reference source, reporting the supported proportion. This decomposition-then-verify protocol is the basis for the groundedness metric in Section 7.5.3, with the knowledge graph serving as the reference. The essential methodological requirement, adopted here, is that the verifying judge must be independent of the system that produced the text.

3.5.2 Factuality Benchmarks and Their Limits

FactBench [44] provides a dynamic benchmark for in-the-wild factuality evaluation. The original proposal for this thesis intended to use it both as a source for the knowledge environment and as the hallucination evaluator. That plan was abandoned, and the reason is methodological rather than practical: if the evaluator's facts are contained in the graph the system retrieves from, a low hallucination rate follows by construction rather than by merit. The evaluation in Chapter 7 therefore measures claim grounding against the graph with a human-validated judge, keeping the evaluator's ground truth outside the system's evidence.

3.6 Comparative Summary of Agentic KGQA Systems

Table 3.1 compares the seven closest systems on the mechanisms relevant to this thesis, and Table 3.2 records their evaluation settings and published accuracy.

Table 3.1: Mechanism comparison of the seven closest prior systems. No system performs claim-level verification of the final answer against retrieved triples before responding.

System	**Exploration strategy**	**Decomposes query?**	**Backtracks?**	**Verifies before answering?**
ToG [7]	Training-free LLM-guided iterative beam search; beam width and depth both 3	No	No — prunes within beam, cannot return	No
PoG [8]	Self-correcting adaptive planning with Guidance, Memory, Reflection	Yes, into sub-objectives with constraints	Yes — Reflection selects the entity to return to	Partial — corrects the search, not the answer
RoG [9]	Planning–retrieval–reasoning over generated relation paths	Partial — relation-path plans	No	Faithful by construction; no explicit check
KG-Agent [10]	Toolbox with KG executor and dynamic memory; autonomous tool selection	Implicit, via tool calls	No	No
GoG [11]	Thinking–searching–generating; generates triples when the KG is incomplete	Yes	No — expands subgraph instead	No
StructGPT [12]	Iterative reading-then-reasoning via invoke–linearise–generate interfaces	No	No	No
KBQA-o1 [34]	Monte Carlo Tree Search with policy and reward models over a ReAct agent; fine-tuned	Implicit, stepwise logical form	Via MCTS tree revisiting	No
AGR (this work)	Guided best-first beam search with explicit backtracking	Yes, into ordered sub-objectives	Yes — threshold, dead end, or evaluator veto	**Yes — claim-level, pre-generation**

Three caveats attach to Table 3.2 and are carried into every comparison in this thesis.

First, the figures are drawn from the original publications and reflect different backbones. RoG and KG-Agent are fine-tuned, and their numbers are not comparable with prompting methods at all. Among the prompting methods, the backbone accounts for much of the spread — the same methods drop sharply when run on smaller open-weight models. This is the direct motivation for holding the backbone constant across every system in Chapter 7: the literature does not, so a cross-paper table cannot settle which architecture is better.

Second, evaluation protocols differ. Some papers evaluate on sampled test subsets, and answer-matching conventions are not uniform across implementations.

Third, and most concretely, two papers disagree about a third. Plan-on-Graph's own comparison table measures Think-on-Graph at 57.1 and 67.6 on CWQ, while Think-on-Graph reports 58.9 and 69.5 for itself. Neither is wrong: one is an original result and the other a reimplementation,

Table 3.2: Reported evaluation settings and Hits@1 as published. These figures are *not* directly comparable to one another: backbones differ, some papers evaluate on sampled subsets, and the backbone accounts for much of the variation.

System	Backbone	Datasets	WebQSP	CWQ
ToG	GPT-3.5	CWQ, WebQSP, GrailQA [45], QALD10-en, SimpleQuestions, WebQuestions, T-REx, Zero-Shot RE, Creak	76.2	58.9
	GPT-4		82.6	69.5
PoG	GPT-3.5	CWQ, WebQSP, GrailQA	82.0	63.2
	GPT-4		87.3	75.0
RoG	Fine-tuned LLaMA2-7B	WebQSP, CWQ	85.7	62.6
KG-Agent	Fine-tuned 7B	WebQSP, CWQ, plus other KBs	83.3	72.2
GoG	GPT-3.5	WebQSP, CWQ (complete and incomplete KG)	78.7	55.7
	GPT-4		84.4	75.2
StructGPT	GPT-3.5	WebQSP, MetaQA, TableQA, Text-to-SQL	72.6	—
KBQA-o1	Llama-3.1-8B	GrailQA, WebQSP, GraphQ; 100-shot low-resource protocol	59.8†	—
	Llama-3.3-70B		67.0†	—

† KBQA-o1 reports **F1**, not Hits@1, and evaluates under a 100-shot low-resource protocol; these cells are therefore *not* comparable with the rest of the column and are shown only to situate the system. It does not evaluate CWQ.

Every figure is the one its own authors published. That this is not the same as a comparison is visible inside the table: PoG's paper reproduces Think-on-Graph and measures it at 57.1 and 67.6 on CWQ, against the 58.9 and 69.5 Think-on-Graph reports for itself. Two published papers disagree by 1.8 and 1.9 points on the same system and benchmark, which is the order of many claimed improvements in this literature and the reason Chapter 7 reimplements its baseline rather than citing a number.

which is exactly the situation this thesis is in with respect to its own agentic baseline (Section 7.4.4). The gap is 1.8 and 1.9 points — the same order as many improvements claimed in this literature. A cross-paper table cannot distinguish a real advance from a reproduction difference of that size, which is why no claim in this thesis rests on one.

3.7 Positioning of This Work

Table 3.1 makes the gap explicit. Decomposition is well established — PoG and GoG both do it. Backtracking exists, in PoG. Faithful grounding by construction exists, in RoG. But in the final column, *no surveyed system decomposes its draft answer into atomic claims and checks each against the retrieved triples before responding*. PoG's Reflection corrects the search while it is running; RoG's answers are grounded because of how they are produced, not because anything examined them. The slot is open.

AGR's contribution is therefore positioned as follows.

The **primary** contribution is the Structural Verification Layer: a claim-level check against traversed triples, executed before the answer is emitted, with repair policies that either resume

exploration or remove unsupported content. Chapter 6 specifies it.

The **secondary** contribution is a systematic component ablation quantifying what each agentic mechanism actually contributes, in accuracy and in tokens. As Table 3.1 shows, these mechanisms are typically reported as bundles; which of them earns its cost is not established in this literature.

A deliberate consequence of this positioning is that the thesis does not depend on beating prior systems on raw Hits@1. The comparison that matters is against the same architecture with the verification layer removed, on the same graph, with the same backbone — which is why the ablation in Section 8.8, rather than Table 3.2, carries the argument. Where AGR is compared against a prior system, that comparison is run under controlled conditions in this thesis's own environment rather than quoted across papers.

Chapter 4

The Knowledge Environment

Everything the agent can know, it knows from the graph. The environment therefore bounds the accuracy any system in this thesis can attain, and a result reported without that bound is uninterpretable. This chapter describes how the environment was built, what it contains, and — equally important — what it cannot express.

4.1 Design Goals

4.1.1 Decoupling the Graph Source from the Question Sets

The original proposal for this work described ingesting WebQSP and FactBench into a graph database. That description conflated two distinct roles, and the correction is worth stating because it shaped everything downstream.

WebQSP [19] is not a knowledge graph. It is a set of questions annotated with semantic parses over Freebase [18]; the graph is Freebase, and WebQSP supplies questions and gold answers against it. The two roles are kept strictly separate here: Freebase-derived triples constitute the environment, while WebQSP and ComplexWebQuestions [20] supply questions and gold answers and contribute no facts to it.

FactBench [44] was removed entirely. It had been intended both as a source of graph content and as the hallucination evaluator, which is circular: if the evaluator's facts are inside the graph the system retrieves from, a low hallucination rate follows by construction. Groundedness is instead measured against the graph by an independent judge, as described in Section 7.6.

4.1.2 Why a Curated Subgraph Rather Than LLM-Extracted Triples

The proposal also specified a language-model entity-relationship extractor to build the graph. This is appropriate when the source is unstructured text, and it is what GraphRAG does [4]. It is

self-defeating here. The graph is the reference against which model output is verified; building that reference with a language model would introduce the very error class the thesis measures, and no claim about hallucination reduction would survive the observation that the ground truth was itself generated.

The environment is therefore assembled exclusively from curated Freebase triples. Language models are used to *navigate* the graph and to *judge* answers, never to populate it.

4.2 Source Data

4.2.1 The Freebase Snapshot

Freebase has been frozen since 2016, which makes it stable and reproducible while also dating it — a property that cuts both ways and motivates the no-retrieval control in Section 7.4. Two routes to a working Freebase environment were considered.

The first is a full SPARQL endpoint: the community-standard preprocessed Virtuoso image, which ships a database of over 53 GB and whose maintainers recommend on the order of 100 GB of RAM for acceptable query latency. Its advantage is a complete and realistic search space, including the hub entities of very high degree that make naive traversal explode.

The second is the per-question subgraph distribution published alongside Reasoning-on-Graphs [9], in which each record pairs a question with its topic entities, gold answers, and the triples of a multi-hop neighbourhood around those topic entities.

The second route was chosen, for a reason that should be recorded plainly: the hardware for the first was not available. The consequence is a scoped environment rather than all of Freebase, and Section 4.7 quantifies what that costs.

4.2.2 WebQSP and ComplexWebQuestions as Question Sets

WebQSP [19] contains questions whose answers lie predominantly one or two hops from the topic entity; it descends from WebQuestions [23]. ComplexWebQuestions [20] is constructed by composing WebQSP queries into deeper chains with conjunctions, superlatives, and comparatives, and is the genuine multi-hop benchmark of the two. Reporting both is deliberate: WebQSP establishes comparability with prior work, and CWQ is where the multi-hop claim is actually tested.

4.3 Graph Construction

4.3.1 Union of Per-Question Subgraphs from the RoG Distribution

The environment is the *union* of every per-question subgraph in the RoG distribution, across both datasets and all three splits. Each record's triple list is read, whitespace-normalised, and inserted into a global deduplicated set; entity and relation strings are interned to integers so that the union and its adjacency structure fit in memory.

Unioning is not incidental. A per-question subgraph in isolation is a near-oracle: it was extracted to contain the answer, so an agent searching it faces almost no distractors. Merging every question's neighbourhood into one global graph restores distractor mass, since the agent exploring one question's topic entity encounters edges contributed by unrelated questions. WebQSP contributes 3,791,302 unique triples over 1,298,304 entities; adding CWQ brings the union to 8,309,194 triples over 2,592,892 entities.

Two consequences must be disclosed. First, the RoG subgraphs were pre-extracted so as to guarantee answer reachability, which makes the resulting environment friendlier than raw Freebase — fewer irrelevant hub explosions, and a higher answer-reachability rate than a BFS extraction from the full store would produce. This is a threat to external validity, revisited in Section 9.9.3. The mitigating argument is that *every system compared in this thesis navigates this identical environment*, so relative comparisons remain sound even where absolute numbers are optimistic.

Second, the distribution's triples are keyed by human-readable entity names rather than by Freebase machine identifiers. The node identifier is therefore the name string, which means distinct real-world entities sharing a surface form are merged into a single node. This is inherent to name-keying and is also revisited in Section 9.9.3.

4.3.2 Relation Filtering and Noise Removal

An extraction from raw Freebase would require aggressive predicate filtering: administrative namespaces, type and key predicates, and above all type.type.instance, which contributes millions of edges per type node. Degree capping would likewise be mandatory, since uncapped breadth-first expansion from a hub entity explodes at the second hop.

No such filtering was applied here, and the reason is structural rather than an oversight: the RoG distribution ships subgraphs that have already been scoped around topic entities, so the administrative predicates and hub explosions that filtering exists to remove are largely absent from the source. The only transformation applied to relation names is sanitisation into Cypher-legal identifiers, described in Section 4.4. What survives is a vocabulary of 7,058 distinct relation types, which is the relation search space the agent faces at every exploration step.

4.3.3 Treatment of Mediator Nodes and Its Effect on Hop Counts

Freebase reifies n-ary facts through mediator nodes, as described in Section 2.1. In the name-keyed distribution these surface distinctively: named entities appear as their names, while unnamed mediators survive as raw machine identifiers of the form m.0d05w3 or g.11b62t84jq. Nodes are flagged as mediators by matching that pattern, and the flag is stored as an is_cvt property.

Mediators are *retained*, not collapsed into hyper-edges. Retaining them preserves the facts they encode — systems that skip mediators lose exactly those facts [11] — at the cost of inflating raw hop counts, since one logical hop through a mediator is two graph edges. The conversion is roughly two raw hops per logical hop.

Two conventions follow, and they are not the same one. Every statement about *distance in the graph* — the reachability gate of Section 4.7, the hop strata of Section 7.2, the path-length bounds quoted anywhere in this thesis — is in *raw* hops, because that is what the graph actually contains. The agent's depth budget is not: max_depth counts explorer expansions, and one expansion crosses a mediator transparently, so it may consume two raw edges while costing one unit of depth. This is deliberate — the agent should not be charged twice for an artifact of the encoding (Section 5.2.3) — but it means max_depth $= 4$ does not bound the search at four raw hops, and the two numbers should not be compared without the conversion.

The scale of this decision is easy to understand: **1,652,618 of the 2,592,892 nodes — 63.7% — are mediator-form nodes carrying no human-readable name.** Nearly two thirds of the environment is connective tissue rather than nameable entities, which is why the tool layer in Section 5.2 must traverse through mediators transparently rather than present them to the agent as candidate answers.

4.4 Graph Store Construction

4.4.1 Property-Graph Schema

Entities become :Entity nodes carrying an integer id, the name string, and the boolean is_cvt flag. Triples become typed relationships whose type is the sanitised relation name — non-alphanumeric characters replaced by underscores, since Freebase predicates contain dots that are illegal in unquoted Cypher relationship types — with the original dotted string retained as an fb_name property.

Native relationship types were chosen over a single generic type carrying a name property. The alternative simplifies import but slows filtered expansion, and filtered expansion is the agent's hottest operation: every exploration step retrieves the relations of an entity and then the neighbours along one of them.

Table 4.1: The knowledge environment as constructed and imported.

Property	Value
Entities (nodes)	2,592,892
Triples (relationships)	8,309,194
Node and relationship properties	16,087,870
Distinct relation types	7,058
Mediator-form nodes (is_cvt)	1,652,618 (63.7%)
Triples contributed by WebQSP subgraphs	3,791,302
Entities after WebQSP alone	1,298,304
Rows skipped during import	0
Import wall-clock time	36.4 s
Peak import memory	1.09 GiB
Neo4j version	Community 5.26.28

Integer identifiers were chosen over names as the import key. Entity names contain commas, quotation marks, and unicode, all of which invite escaping bugs and silent collisions in bulk-import CSVs. The name survives as an ordinary property and is what the entity resolver matches against.

4.4.2 Bulk Import into Neo4j

Import used the offline bulk importer of Neo4j Community 5.26.28, which operates on a stopped database and is orders of magnitude faster than transactional loading at this scale. Duplicate nodes and bad relationships were configured to be skipped rather than to abort the import.

After import, a uniqueness constraint on Entity.id and a full-text index on Entity.name were created; the latter is what the resolver's lexical tier queries.

4.4.3 Graph Statistics

Table 4.1 records the environment as built. These figures, together with the dataset versions and the hop cap, constitute the reproducibility record for the environment.

The zero skipped rows deserve emphasis, because the import was deliberately configured to tolerate skips. Node and relationship counts in the database match the CSV row counts exactly, so the count-reconciliation check of Section 4.7 passes without any unexplained gap to account for.

4.5 The Vector Index

4.5.1 Entity Embeddings

Entity names were embedded with all-MiniLM-L6-v2, a 384-dimensional sentence encoder [29] chosen for CPU throughput at this node count. Mediator nodes were skipped: embedding a string such as m.0y5k79m produces noise, and by Section 4.3 that is almost two thirds of the graph, so skipping them is also the difference between a tractable and an intractable embedding job. Vectors were written back into Neo4j and indexed with a cosine-similarity vector index.

4.5.2 Relation-Vocabulary Embeddings

The scoring function of Section 5.5 compares candidate relations against the current sub-objective in embedding space, so the 7,058 relation names are embedded once and held in memory rather than embedded per step.

Two details make this work. Relation names are *verbalised* before encoding: people.person.place_of_birth becomes "people person place of birth", with consecutive duplicate tokens removed. The encoder was trained on natural language, and the verbalised phrase lands near a question like "where was X born" in embedding space where the raw dotted path does not. And relations are embedded with the *same* model as entities and sub-objectives, since vectors from different models occupy unrelated spaces and their cosine similarity is meaningless.

The resulting artefact is a 7058×384 float32 matrix with a parallel name list. Because vectors are L2-normalised at encoding time, similarity against the whole vocabulary is a single matrix–vector product.

A functional check confirms the verbalisation is doing its job. Encoding the probe "where was the person born" and ranking the vocabulary against it returns people.person.place_of_birth first at cosine 0.709 and location.location.people_born_here — the inverse relation — second at 0.675, with the remaining top-five entries plausible near-misses drawn from birth-related and fictional-character namespaces. A degenerate verbaliser would have produced an unordered top five here.

4.5.3 Scope: Linking and Candidate Ranking, Never Ground Truth

The role of embeddings in this thesis is deliberately narrow. They resolve question phrases to graph entities, and they pre-rank relation candidates so the language-model scorer can be applied to a short list rather than to thousands of relations. They never establish that a fact holds. Every factual check — whether a triple exists, whether an entity is reachable, whether a claim is supported — is answered by symbolic graph queries. Nothing in the verification layer of Chapter 6 consults an embedding.

Table 4.2: Entity resolution over all topic-entity mentions in both test sets. A mention counts as resolved only when the returned node's name equals the mention exactly.

Dataset	Mentions	Exact stage	Unresolved
WebQSP	1,661	1,661 (100%)	0
CWQ	5,488	5,478 (99.8%)	10
Total	**7,149**	**7,139 (99.86%)**	**10 (0.14%)**

4.6 Entity Resolution and Surface-Form Linking

4.6.1 The Exact, Lexical, and Vector Cascade

Mapping a question's surface mention to a graph node uses a three-stage cascade, each stage attempted only if the previous one fails.

Exact matching queries for a node whose name equals the normalised mention. Because the environment is name-keyed and the benchmarks supply topic entities as names, this resolves nearly everything.

Lexical matching queries the full-text index, handling typographical variation and partial names. Mentions are escaped for Lucene syntax before the query: benchmark entity names contain parentheses, colons, and slashes, all of which are Lucene operators that would otherwise crash or silently distort the query.

Vector matching encodes the mention and queries the entity vector index, handling paraphrases and misspellings that survive the first two stages.

The cascade is referred to by these names throughout, never as "tiers", which this thesis reserves for the two-tier groundedness metric of Section 7.5.3.

4.6.2 Threshold Selection on Development Data

The lexical stage accepts its top hit only when the full-text score exceeds 1.0, falling through to vector matching otherwise. The threshold exists because the full-text index returns a ranked list for almost any input, including inputs that match nothing meaningfully; without a floor, the lexical stage would confidently return the least-bad token overlap and the vector stage would never be reached.

Resolution was measured over every topic-entity mention in both test sets, and Table 4.2 reports the outcome.

Surface forms supplied by the benchmarks therefore resolve to nodes in this environment essentially without loss. The limits of that measurement should be read with it: the graph is keyed on the same name strings the benchmarks supply, both coming from the same RoG distribution, so this establishes that the *resolver* does not lose entities the environment contains

Table 4.3: Environment coverage over the full test sets, at a bound of four raw hops. The reachability column is the Hits@1 ceiling for every system evaluated in this thesis.

Dataset	Test questions	All answers exist	≥1 exists	≥1 reachable
WebQSP	1,628	94.7%	97.3%	**97.3%**
CWQ	3,531	99.4%	99.7%	**99.7%**

— not that entity linking would be easy against an independently-built graph, and not that linking from free text would work at all. Topic entities are given rather than linked throughout (Section 9.9.3). What follows for the error analysis of Chapter 9 is the narrow version: a failure in this environment cannot be attributed to the resolver having failed to find the question's topic.

4.7 Environment Validation Gate

No experiment was run until the environment was certified. The gate has three parts of increasing strength.

4.7.1 Answer-Reachability Protocol

For every test question, two properties are computed over the union graph. *Existence* asks whether the gold answer entities appear as nodes at all. *Reachability* asks whether at least one gold answer is connected to at least one topic entity within a bounded number of hops.

Reachability is computed by breadth-first search treating edges as *undirected*, because the agent's neighbour-retrieval tool looks in both directions and the gate must model the agent's actual traversal semantics. The bound is four raw hops, which by Section 4.3 is approximately two logical hops through mediators — enough for all of WebQSP and most of CWQ. A question whose topic entity is itself a gold answer counts as reachable at zero hops.

This number is the **accuracy ceiling**: if a gold answer is unreachable, no navigating system can find it, however well designed.

4.7.2 Coverage Results and the Induced Accuracy Ceiling

Table 4.3 reports the gate.

Three observations follow. Every question in both datasets has at least one topic entity present in the graph, so no question is lost at the entry point.

More strikingly, *existence and reachability are identical* — 1,584 of 1,628 for WebQSP and 3,519 of 3,531 for CWQ. Every gold answer that exists in this environment is reachable within four hops. Reachability is therefore never the binding constraint; when an answer is unavailable

it is because the entity is absent altogether. This is a direct consequence of the construction in Section 4.3: the subgraphs were extracted around topic entities precisely to guarantee reachability, and the measurement confirms that the property survived unioning.

Finally, CWQ's coverage exceeds WebQSP's, which inverts the expected ordering — the deeper benchmark has the better-covered answers. The explanation is again constructional rather than semantic: CWQ contributes far more questions and therefore far more subgraphs to the union, so its own answers are better represented. The residual WebQSP shortfall is dominated by answers that are not entities at all, which Section 4.7.4 takes up.

4.7.3 Analysis of Linking Misses and Gate Failures

The ten unresolved mentions of Table 4.2 are not a random residue. Every one is a bare machine identifier of the form g.123852dg appearing as a CWQ topic entity — an unnamed mediator node that the benchmark supplies where a named entity was expected. Exact matching fails because no node carries that string as a *name*, and the lexical stage then returns a meaningless single-character match. The correct characterisation is that these questions are anchored on nodes with no surface form, not that resolution is unreliable.

The strongest gate check re-verifies the Python-computed reachability against the live database, confirming that nothing was lost between the CSV files and the imported graph. A random sample of 400 questions flagged reachable, drawn under a fixed seed, was re-tested with a shortest-path query in Neo4j. The pass criterion is strict — these were already known to be reachable, so anything below 100% would indicate dropped edges. **All 400 verified; the failure set is empty.** Together with the zero skipped import rows and the exact count reconciliation, the environment is certified.

4.7.4 What the Environment Cannot Express

Certifying reachability establishes that answers which *are* entities can be found. It says nothing about answers that are not entities, and this is where the environment's real limits lie.

The import schema of Section 4.4 gives every node a single :Entity label. There is no typed literal node, no datatype, and no distinction between an entity and a value. A date or a number can therefore exist in the graph only incidentally, as an entity whose *name* happens to be a numeric string. Measured over all 2,592,892 nodes by scripts/count_literals.py, which writes results/phase1/literal_census.json:

- **Dates** are effectively absent. Only 77 nodes carry a full YYYY-MM-DD name and 175 a bare four-digit year — together under 0.01% of the graph, and untyped, so they cannot be compared, ordered, or filtered by range.

- **Numeric values** appear on 12,799 nodes (0.49%) under the strict rule that the entire name is a single number, but the population is dominated by identifiers such as ISBNs rather than by measurable quantities. That count is sensitive to its own definition, which is why the definition is committed alongside it: admitting names that merely carry no letters ("(1955)", "12-4") raises it to 13,888. The argument does not turn on which is used — both are under 0.6% — but the figure should not be read as a property of the graph independent of how it was counted.

- **Ordinals and superlatives** are not expressible at all. This is a property of the tool layer rather than the data: Section 5.2 exposes no aggregation, sorting, or comparison operator, so "the smallest" or "the earliest" cannot be computed even when every candidate is present in the graph.

This defines a class of questions that is *unanswerable in this environment*: any question whose gold answer is a date, a measured quantity, or a superlative selection over candidates. Such a question is not a reasoning failure when the system misses it — the environment could not have supplied the answer. The class is referenced by name in Section 9.6, where it accounts for a substantial share of the residual failures, and it is a known consequence of a design decision: modelling literals as typed nodes was considered and not adopted, in exchange for a uniform schema and a simpler tool API.

Chapter 5

The AGR Framework: Architecture and Navigation

This chapter specifies the Agentic Graph Reasoning framework: how the agent is structured, what operations it may perform on the graph, how it decides where to look next, how it recovers from wrong turns, and what guarantees bound its cost. The verification layer, which is the thesis's principal contribution, is separated into Chapter 6; everything here is the machinery that produces the traversal it verifies.

5.1 Architectural Overview

5.1.1 The Controlled-Agent Pattern

There are two ways to build a graph-navigating agent. The first hands the model a set of tools through native function-calling and lets it drive the loop itself, ReAct-style [27]. The second inverts control: the *program* orchestrates the loop, calls the graph tools deterministically, and consults the model only at decision points, through constrained prompts that must return JSON. AGR uses the second, and the choice is deliberate enough to be worth defending, because it determines what the word "agentic" means in this thesis. Four consequences follow:

- **Every graph operation is logged by construction.** The traversal is not reconstructed from the model's narration of what it did; it is the literal sequence of tool calls the program made.
- **Budgets are enforced in code, not requested in prompts.** A prompt asking the model to stop after four hops is a suggestion. A counter checked in a router is a guarantee (Section 5.8).

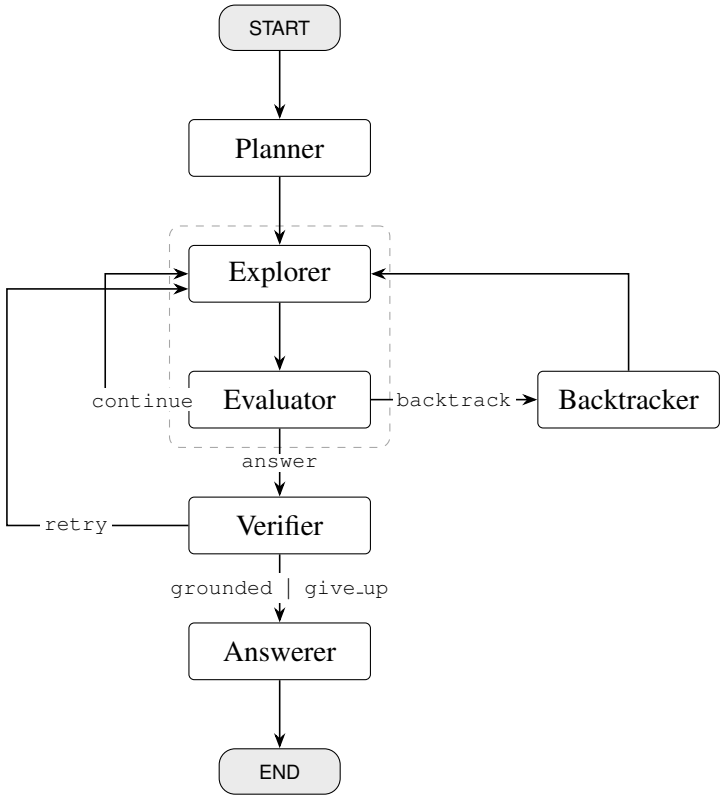

Figure 5.1: The AGR state machine. Two cycles exist: the dashed region marks the Explorer ↔ Evaluator search loop, and Verifier → Explorer is verification-driven re-exploration. Both are bounded by the budgets of Section 5.8 rather than by the graph structure. Every path from Evaluator to Answerer passes through the Verifier, which is the structural expression of this thesis's contribution.

- **Failures are reproducible.** A malformed model response degrades a decision; it cannot corrupt the control flow.
- **Each model call is small, single-purpose, and cacheable**, which is what makes the response cache of Section 5.9 viable.

The agency in AGR therefore lives in the *decision policy* — which sub-objectives to pursue, which edges look promising, whether the current path is a dead end, whether the draft answer is supported — and not in unconstrained tool invocation. Prior systems in this space make the same choice [7, 8]; stating it explicitly forecloses the objection that AGR is merely a prompt around a function-calling loop.

5.1.2 The State Machine

The framework is a state machine: a directed graph of six nodes over a shared typed state, built with LangGraph. Figure 5.1 gives the topology.

Two properties of this topology matter. It contains *cycles*, which is why a state-machine framework is used rather than a linear pipeline: exploration loops until the evaluator is satisfied, and verification can send the agent back to explore. And the verifier sits *between* the decision to answer and the answer itself, which is the structural expression of this thesis's contribution — there is no path from evaluator to answerer that does not pass through verification.

5.2 The Graph Tool API

The five operations are specified in full, with the Cypher each executes, in Chapter B.

5.2.1 Rationale: Constrained Tools Instead of Free-Form Cypher

An obvious alternative is to let the model write Cypher directly. This was rejected. Generated-query approaches introduce a failure mode that is indistinguishable, in the results, from a reasoning failure: when a question is answered incorrectly, one cannot tell whether the agent reasoned badly or merely emitted invalid syntax. Since the thesis's central claims are about reasoning and verification, a confound that contaminates every error category is unacceptable.

The tools are therefore parameterised Cypher templates (Section 7.10), written once and unit-tested. They accept entity identifiers and return names alongside them, so the model always sees human-readable text while the program tracks identity. Every invocation is appended to a JSONL log with its arguments, result, and latency.

5.2.2 The Five Operations, of Which Four Are Live

Table 5.1 lists the operations. There are five, not the four originally planned, and only four of the five are reachable from a node in the final design — both facts are consequences of the same episode and are recorded rather than tidied away.

verify_triple was built first, as the claim checker: given a claim $\langle h, r, t \rangle$ it asked whether that relation held between those entities. It did not survive contact with generated text. Claim decomposition produces relations phrased in English — "played for", "is the capital of" — and Freebase predicates are neither phrased that way nor reliably one edge long, so requiring the relation to match rejected claims that were true. verify_connection was added in response: it drops the relation entirely and asks only about adjacency. It answered the question the verifier actually needed answered, and it superseded rather than complemented its predecessor. Section 6.3.1 describes the two routes the verifier ended up with, and verify_triple is not one of them.

The operation is left in the API because it is exercised by the tool integration tests and remains the natural primitive for a relation-aware check, which Section 10.3 proposes. But nothing in the running loop calls it, and the table says so.

Table 5.1: The graph tool API. All operations are deterministic parameterised Cypher; none consults an embedding except search_entity, which delegates to the resolver of Section 4.6.

Operation	Returns	Role
search_entity(*s*, k)	up to k candidate entities with id, name, score, and resolver stage	anchor seeding; the only entry point from text to graph
get_relations(*e*)	relation types incident to e in *both* directions, each with its fanout count	candidate generation for the scorer
get_neighbors(*e*, *r*, *d*)	entities reached from e along r in direction d, each with the via chain that reached it	frontier expansion
verify_triple(*h*, *r*, *t*)	{direct, via_cvt, supported}	*not called by any node*; built for claim checking and superseded by verify_connection (Section 5.2.2)
verify_connection(*a*, *b*)	{direct, via_cvt, connected}	relation-agnostic claim checking — the check the verifier actually uses (Section 5.2.3)

Three design decisions inside these operations are load-bearing.

Relations are returned with fanout counts, in both directions. The count is free signal: a relation with a fanout of forty thousand is almost never the edge that answers a specific question, and the scorer can use this without an extra query. Retrieving incoming as well as outgoing edges matters because Freebase's relation direction is a modelling artefact — a question about where someone was born may be answerable through people.person.place_of_birth outward or location.location.people_born_here inward.

Administrative predicates are filtered at the tool layer. Relations whose names begin with common., freebase., type., kg., user., dataworld., rdf-schema#, or owl# are dropped before the candidate list is built. These carry no factual content, and Section 5.10 shows what happens when they are not filtered: they consume beam slots and drag the agent into the schema rather than the data.

The relation list is capped at 300 entries, sorted by fanout. An earlier cap of 80 was raised after it was found to hide precisely the low-fanout, high-precision relations that answer specific questions — again, Section 5.10.

5.2.3 Mediator Traversal and Relation-Agnostic Adjacency

Section 4.3 established that 63.7% of the environment's nodes are unnamed mediators. If the agent saw these as candidate entities, most of its frontier would consist of nodes with no surface form and no meaning to a language model.

get_neighbors therefore hops through them transparently. When an expansion reaches a node flagged is_cvt, the tool immediately expands that node one further step, excluding the origin and excluding other mediators, and returns the entities on the far side. Each returned neighbour

Table 5.2: The agent state. Fields marked *append-only* accumulate across node executions through reducers; all others are overwritten by whichever node returns them.

Group	Fields
Question	qid, question, gold_q_entities
Plan	plan: ordered sub-objectives, each with a status (pending/active/done) and its resolved entities
Search	anchors (current frontier), traversed (*append-only* triple log), backtrack_stack, banned, last_expanded
Answer	candidate_answers, draft, answer, answer_entities, supporting_triples
Verification	supported_claims, unsupported_claims, unsupported_claim_objs
Control	budget (meter, mutated in place), trace (*append-only*), and the router scratch keys _eval_decision, _last_n_new, _frontier_max_score

carries a via chain — one relation for a direct edge, two for a mediator-mediated one. The model reasons about logical neighbours; the log retains the full raw chain, so hop accounting stays exact.

verify_triple applied the same principle to checking: it matched *undirected* and accepted a mediator in the middle. Direction-strictness would have generated false rejections, because the claims it was given came from generated text that has no notion of which way a Freebase edge points. That relaxation was necessary but not sufficient, for the reason Section 5.2.2 records: relaxing direction does not help when the relation itself has no Freebase counterpart.

verify_connection goes further and drops the relation entirely, asking only whether two entities are adjacent directly or through one mediator. It exists because claim decomposition frequently produces a claim whose relation does not correspond to any single Freebase predicate — "X played for Y" may be realised through a roster mediator with no edge literally named for playing. Requiring an exact relation match in such cases rejects true claims. This is the adjacency test the verification layer uses, and Section 6.3.1 gives the two routes it sits in. The cost of dropping the relation is stated there and not here: an adjacency test cannot tell a true claim from a false one that happens to join the same two entities, which is the boundary case Section 9.5 analyses.

5.3 Agent State

The state is a typed dictionary threaded through every node. Table 5.2 groups its fields by purpose.

Two disciplines govern this structure and are worth stating because violating either produces bugs that are extremely hard to localise.

Nodes write, routers read. A router is a pure function that inspects state and returns an edge label; it never mutates. Any signal a router needs must therefore be deposited in state by the node

preceding it — which is what the three underscore-prefixed scratch keys are for. The explorer records how many triples it added and the best score it saw; the evaluator records its decision; the routers read those and decide.

The budget meter is mutated in place and never returned as a delta. It is a mutable object shared by reference across all nodes, because counters that are merged rather than mutated can be silently lost when two nodes return overlapping state. Enforcement must not be able to drift.

5.4 The Planner

5.4.1 Decomposition into Ordered Sub-Objectives

The planner makes one model call, converting the question into an ordered chain of sub-objectives plus the entity mentions to seed the search. The chain is strictly sequential; later objectives reference earlier results with a #N back-reference. A general dependency-graph planner was not built, because a sequential chain covers the composition structure that dominates CWQ and a DAG planner is a research problem in its own right.

The prompt is reproduced in full as Section A.1. It is few-shot with one example per question archetype — single hop, composition, conjunction, superlative — and the single-hop example exists to demonstrate that *not* decomposing is a valid output. Over-decomposition of simple questions is a real failure mode, and Section 8.8 shows it is not hypothetical: on the shallower benchmark, removing the planner improves accuracy.

The prompt also instructs the model to extract only specific named entities as topic mentions, and explicitly not generic type words such as “celebrity”, “university”, “country”, or “senator”. That instruction was added in response to evidence discussed in Section 5.10: type words resolve successfully to type nodes in the graph, consuming anchor slots with entities that lead nowhere.

5.4.2 Plan Validation and Degenerate-Plan Handling

The planner’s output is validated, not trusted. Four checks run: the plan must contain a non-empty list of sub-objectives; it must contain no more than four; every #N reference must point strictly backwards; and topic mentions must be present. The back-reference check catches the planner’s most common structural error, a step that references itself or a future step.

If any check fails — or if the model call itself raises — the planner falls back to a single sub-objective consisting of the whole question, seeded with the dataset’s topic entities. This is a deliberate design principle: **planning can degrade but can never abort a run.** A degraded plan is a data point that appears in the results; a crash is a lost question that biases them.

Anchor seeding has two modes. In the default benchmark configuration, anchors are seeded from the dataset’s annotated topic entities, which isolates reasoning from entity linking and is the

standard protocol in this literature [7, 9]. The alternative seeds from the planner's own extracted mentions, giving the end-to-end number. Because Section 4.6 showed resolution succeeds on 99.86% of mentions, the two modes differ very little in this environment.

Setting the planner ablation flag bypasses the model call entirely and installs the fallback plan directly, which is what makes "no planner" a one-configuration-line change rather than a separate code path.

5.5 The Explorer

The explorer performs one expansion of the frontier. It is the only node that touches the graph in bulk, and its structure is the heart of the navigation policy.

5.5.1 Frontier Construction and the Single Scoring Pass

The node proceeds in five steps:

1. **Checkpoint.** The current frontier, the current depth, and the best score seen at this point are pushed onto the backtrack stack.
2. **Gather.** For every anchor, get_relations is called and the results are collected into a *single* candidate list, each row tagged with the anchor it came from.
3. **Score.** The whole list is scored in one pass (Section 5.5.2). Pooling candidates across anchors before scoring is what allows the hybrid scorer to consult the model *once* per expansion rather than once per anchor.
4. **Beam.** Candidates are taken in descending score order into per-anchor buckets of width three, skipping any (anchor, relation, direction) triple on the ban list (Section 5.6.3).
5. **Expand.** Each selected edge is expanded with get_neighbors; the resulting triples are appended to the traversal log, and the neighbours become frontier candidates.

The new frontier is deduplicated by entity identifier, sorted by the score of the edge that produced each entity, and truncated to the anchor cap. Sorting by score rather than by insertion order matters: an insertion-ordered frontier keeps whichever entities happened to be produced first, which causes the agent to drift away from the anchor that mattered.

5.5.2 The Hybrid Scoring Function

Each candidate edge receives a hybrid score blending semantic similarity with a model judgement; the prompt carrying the second is Section A.2:

$$\text{score}(c) \;=\; \alpha \cdot \cos\big(\mathbf{v}_{\text{rel}(c)},\, \mathbf{v}_{\text{obj}}\big) \;+\; (1-\alpha)\cdot \text{LLM}(c),$$

where $\mathbf{v}_{\text{rel}(c)}$ is the precomputed embedding of the candidate's relation name (Section 4.5), $\mathbf{v}_{\text{obj}}$ is the embedding of the *rendered active sub-objective*, and $\text{LLM}(c) \in [0, 1]$ rates how likely following that edge from that anchor is to advance the sub-objective.

Three properties of this design deserve emphasis.

The objective, not the question, is embedded. Hop two of a two-hop question is scored against "find the director of Titanic", with the back-reference already substituted by the resolved entity name, rather than against the original question text. This is the concrete mechanism by which decomposition is supposed to help, and isolating it is what makes the planner ablation interpretable.

The model scores only a shortlist. Candidates are pre-ranked by embedding similarity and only the top twenty are shown to the model, in one batched call. Without the pre-filter, an expansion from five anchors with hundreds of relations each would be unaffordable.

The equation above understates what that shortlist does, and the gap is worth closing here. A candidate outside the top twenty is not scored with the model term *omitted*; it is scored with $\text{LLM}(c) = 0$, which is the lowest rating the model could have given it. The embedding pre-filter is therefore a hard gate rather than a soft ranking aid: at $\alpha = 0.7$ an unrated candidate is penalised by up to 0.30 against an identically-similar rated one, so it can only re-enter the beam if its embedding similarity exceeds the rated candidate's by that margin. This is defensible — the pre-filter is an embedding ranking and a candidate below the cut is one the embedding already judged worse — but it means the twenty-candidate cut interacts with α rather than sitting orthogonally to it, and a larger α softens the gate as a side effect of weighting the model less.

The blend is applied after the call, not inside it. The scoring prompt contains no configuration value — no α, no τ. This is not cosmetic: because the response cache (Section 5.9) keys on prompt content, a prompt free of configuration means every condition in an α sweep shares the same cached model responses, and the sweep costs one set of calls instead of one per condition. Embedding a configuration value in that prompt would multiply sweep cost by the number of conditions without changing any result.

If the model call fails or the budget is exhausted, the scorer treats every model score as zero rather than aborting the expansion, so each candidate scores $\alpha \cdot \text{emb}$. That is the embedding ordering scaled by α, not the embedding score itself, and the distinction is worth stating precisely because it does not matter and could be assumed to: $\alpha > 0$ is a positive scaling, so the ranking is

identical, and the backtracking trigger compares τ against σ, which takes the unscaled embedding maximum as one of its terms (Section 5.6.3). No reported number changes on this path. Setting $\alpha = 1$ disables the model term entirely, which is the embedding-only ablation.

5.6 The Evaluator and Routing Policy

5.6.1 Sub-Objective Completion Judgment

The evaluator makes one model call per expansion. It is shown the active sub-objective, a window of retrieved facts, and the remaining budget, and returns a decision (continue, backtrack, or answer), whether the objective is complete, and which entities satisfy it.

The fact window is *relevance-ranked, not recent*. Traversed triples are verbalised, embedded, and the thirty most similar to the current objective are shown, in traversal order. The distinction is not incidental: a large expansion can add hundreds of triples, and a window showing the most recent thirty will push the answer out of view in exactly the situations where the expansion succeeded. Section 5.10 documents the case that forced this change.

5.6.2 The Groundedness Filter on Resolutions

The evaluator's resolved entity names are not accepted as given. Each is looked up against a table built from the tail entities of the *traversed* triples, and any name that does not appear there is discarded.

This filter is small in code and significant in effect. A language model asked which entities satisfy an objective will readily list entities it knows from training but never retrieved. The filter refuses to launder that parametric recall into the agent's evidence. Section 5.10 records a case in which the evaluator confidently resolved nine correct countries that appeared in no retrieved triple, and the filter rejected all nine — costing a point on that question while preserving the property that every entity the system proposes came from the graph.

The table is built from tail names only, and the verifier's is not. A traversed triple contributes its tail to this lookup and not its head, so an entity the search *departed from* — a topic entity, or any intermediate the agent has since moved past — cannot be resolved as an answer by the evaluator, however the model phrases it. The structural check of Section 6.3.1 takes the opposite convention: it collapses traversed triples into undirected endpoint pairs, so heads and tails are interchangeable there.

The asymmetry is deliberate in the evaluator and worth naming because it is not obviously so. A topic entity is exactly what an echoing model reaches for (Section 9.7), and refusing to resolve it is a cheap structural defence against the most common degenerate answer. The cost is that a question whose gold answer genuinely *is* a topic entity, or an intermediate the plan has passed

through, cannot be answered through this path at all. Those cases exist — Section 9.7 counts 13 echo failures, and the filter is what prevents them from being far more numerous — but the two layers disagreeing about what counts as a reachable entity is a design seam, not a proved optimum, and no experiment in this thesis isolates it.

Resolutions that survive the filter are accumulated into candidate_answers on *every* evaluation, whether or not the objective was judged complete. An earlier design discarded resolutions when the objective was incomplete, which threw away correct answers found mid-run.

When an objective completes and further objectives remain, the plan advances, the decision is forced to continue, and the resolved entities become the next frontier. When the last objective completes, the decision is forced to answer.

5.6.3 Backtracking Triggers and the Backtrack Stack

Backtracking fires on the first of three conditions, evaluated in this order:

1. **Dead end** — the last expansion produced no new triples.
2. **Low score** — the frontier's *signal maximum* fell below the threshold τ.
3. **Evaluator veto** — the model judged the direction a dead end.

The second trigger deserves a precise statement, because it is easy to assume it tests the blended score and it does not. The quantity compared against τ is

$$\sigma \;=\; \max\Big(\; \underbrace{\max_{c\in E} \operatorname{score}(c)}_{\text{blended, expanded edges}} \;,\; \underbrace{\max_{c\in R} \cos(v(r_c), v(o))}_{\text{embedding, all candidates}},\; \underbrace{\max_{c\in R_K} s_{\mathrm{LLM}}(c, o)}_{\text{model, rated candidates}} \Big),$$

the largest of three signals: the best blended score among the edges actually expanded, the best raw embedding similarity over every candidate relation the anchors offered, and the best model score over the top-K candidates the model was asked to rate. The trigger therefore fires only when *all three* signals are weak. A frontier where the blend is depressed — because α happens to discount whichever signal is confident here — does not backtrack if either underlying signal still points somewhere. This makes the trigger deliberately conservative: it abandons a branch only on unanimous evidence of hopelessness, and Section 8.1 shows the consequence, which is that τ barely fires at the values swept.

The implementation departs from that formula in two places, and both are recorded rather than repaired, under the standing rule of Section 5.10. Repairing them would change the frozen results this thesis reports, which is a larger claim than either defect warrants.

First, $\sigma = 0$ is read as "no signal recorded". The comparison is written so that a missing signal defaults to a value above any usable τ, which is right when no expansion has run yet. But 0.0

is falsy in Python, so a genuine signal maximum of exactly zero takes the same branch, and the trigger does not fire at the one point where the formula says it most clearly should. This is the same falsy-empty trap that Section 6.6 records in the answerer, found in a second place. Its practical reach is narrow: a signal maximum of exactly zero means nothing was expanded at all, and the **dead end** trigger — evaluated first — fires on that same condition. The order of the triggers is what masks the defect, which is a weak reason for the behaviour to be correct.

Second, at $\alpha \geq 1$ only the first term is computed. The scorer takes an early return when the model term is disabled, and the two auxiliary maxima are never recorded, so σ collapses to the blended maximum over expanded edges. Since $\alpha = 1$ makes the blend the embedding score, the effect is not that σ is measured on a different scale but that it is measured over a *smaller set*: the beam-selected, non-banned edges rather than every candidate the anchors offered. After a backtrack has banned the best relation, σ falls in this condition where it would have held steady in the reference. That is exactly the embedding-only ablation, and Section 8.8.4 reports what it does to that condition's backtrack count instead of attributing the difference wholly to τ.

The order matters. The first two are deterministic and cost nothing; checking them before deferring to the model's judgement means a mechanically hopeless frontier is abandoned without an argument about it.

The backtracker does three things. It pops the *highest-scoring* snapshot from the stack — not the most recent one, which would make backtracking a simple undo. It restores that snapshot's anchors and depth. And, critically, it adds every (anchor, relation, direction) triple expanded since that snapshot to a **ban list** that the explorer consults on its next pass.

The ban list is what makes backtracking a search operation rather than a loop. Without it, restoring a frontier is pointless: a deterministic scorer facing an identical frontier selects identical edges, producing identical facts and an identical decision to backtrack, until the backtrack budget is exhausted. With it, a backtrack forces the scorer to its next-best alternatives, which is what "exploring an alternative branch" actually requires. Section 5.10 shows both behaviours in the logs.

5.6.4 Routing

route_after_eval is pure Python and enforces budgets before preferences. Wall-clock and depth are checked first; if either is exhausted the agent is routed to answer with whatever it has. If a backtrack trigger fired, the agent backtracks if the backtrack budget permits and otherwise answers. Only then is the evaluator's own decision honoured. Routers are deterministic by construction: no adversarial model output can change what they do.

5.7 The Answerer

The answerer is a finalizer rather than a generator. Which of its three paths runs is decided entirely by what the verification layer left in state, and two of the three are specified in Chapter 6: the *verified* path, where no claim was unsupported and the draft is emitted with no further model call, and the *repair* path, where unsupported claims remain and one rewrite call restates the answer around a deterministically filtered entity list (Section 6.5). On the development set the verified path handled 77 of 80 questions at zero additional cost.

The third path exists for the case where the verifier raised before producing a draft at all. It makes one model call over the traversed facts and the accumulated candidates. If the model budget is exhausted it falls back without a call to the resolved entities of completed plan steps, or failing that to the accumulated candidates, producing a degraded but non-empty answer. If the call raises for any other reason, the error is recorded in the trace and the answer is an explicit failure string.

That last distinction is deliberate. An earlier version caught all exceptions as budget exhaustion, which silently misreported genuine errors as expected behaviour — exactly the kind of defect that inflates apparent robustness.

5.8 Budgets and Termination Guarantees

Budgets do three jobs: they bound cost, they guarantee termination, and they produce the efficiency data of Section 8.7. Table 5.3 lists them.

The two tool-layer caps deserve a sentence of their own, because they bound recall rather than cost and a reader could otherwise take the budget table for a complete account of what limits the search. max_fanout caps neighbours returned from a single expansion at 200. On a high-degree entity — a country, a large film franchise, a prolific author — the true neighbour set exceeds that, so the frontier sees a truncated view and an answer sitting outside the first 200 is unreachable through that edge no matter how well the scorer ranks the relation. The hub-noise failures of Section 9.7 are the place this would be expected to matter.

The tool returns a truncated flag alongside the neighbours, and two defects in it are recorded rather than repaired, under the rule of Section 5.10. First, the flag does not reach the tool log: the logger records the returned neighbour count and not the flag, so the condition is in fact *not* visible in the log, and the count alone cannot distinguish a capped expansion from one that happened to return exactly 200. Second, the flag is computed before mediator traversal, on the raw row count, while the final list is truncated after mediators have been expanded into it. An expansion that returns few raw rows but many entities through those rows' mediators is therefore cut without the flag being set. Both mean the flag is a lower bound on truncation, not a record of it, and no

Table 5.3: Budget configuration, and the two tool-layer caps that bound retrieval alongside it. The rows above the rule are the budget proper: they are hashed and the hash is recorded in every run record, so results can always be attributed to the configuration that produced them. The two rows below the rule are constructor arguments of KGTools and are *not* in that hash — they are listed here because they bound what the agent can see, which makes them budgets in effect if not in implementation.

Parameter	Value	Enforced in
max_depth	4	route_after_eval
beam_width	3	explorer (slice)
max_anchors	5	explorer (slice)
max_backtracks	3	route_after_eval
max_llm_calls	25	LLM client
max_verify_iters	2	route_after_verify
max_seconds	300	route_after_eval
max_fanout	200	get_neighbors — neighbours returned per expansion; sets truncated when it binds
max_relations	300	get_relations — candidate relations offered to the scorer

analysis in this thesis reads it — the cap's effect is measured instead from the returned counts, as in Section 7.4.4.

Each budget has exactly one enforcement site, which is what keeps enforcement auditable. The model-call budget is checked inside the client before every call, so no node can bypass it; exceeding it raises an exception that each node catches into a degraded path rather than a crash. Depth, backtracks, and wall-clock are checked only in the post-evaluation router. Verification iterations are checked only in the post-verification router. Beam width and anchor cap are not checks at all but slices inside the explorer.

Together these give a termination guarantee that does not depend on model behaviour: every cycle in Figure 5.1 passes through a router that decrements or checks a monotone counter, so no sequence of model outputs, however adversarial, can produce a non-terminating run. This property is verified directly by the mechanics tests of Section 7.10, which script an evaluator that always says continue and assert that the depth budget halts it.

5.9 Instrumentation and Reproducibility

Two logs are written per run. The *tool log* records every graph operation with its arguments, result size, and latency. The *run log* records, per question: the answer and its extracted entities, the draft, the verification outcome and any unsupported claims, the count of supporting triples, every backtrack with its trigger reason and restored depth, the full decision trace, the budget snapshot,

wall-clock time, and — for provenance — the backbone description, the budget configuration hash, the run configuration, and the current git commit.

The response cache deserves separate description: it underwrites the thesis's reproducibility claims. Every model call is keyed by a hash over the model identifier, temperature, reasoning-effort setting, and the full prompt text; hits are served from disk. The subtle part is that a cache hit must be *indistinguishable* from a live call inside the agent: the budget check still runs, the call counter still increments, and the original token counts are replayed into the meter from the stored record. A fully cached rerun therefore reproduces the original run's budget snapshots exactly, rather than reporting zero tokens. A separate counter records cache hits so that live and replayed runs can still be told apart in analysis.

This has three consequences. Repeated experiments cost nothing. A code change that is not supposed to alter behaviour can be verified by confirming that a rerun is served entirely from cache. And because the key includes the dated model snapshot, a silent server-side change to a model alias cannot poison previously cached results.

5.10 Design Validation on the Development Set

The first end-to-end run of the assembled framework, over a twenty-question development set drawn from the training splits, answered roughly three to four questions correctly. The final configuration answers thirteen to fourteen. The interval between those numbers is not incidental tuning; it is where several of the mechanisms described above were shown to be necessary, and it is documented here because each fix is evidence that a component earns its place.

The development set was constructed to exercise specific machinery rather than to sample the benchmark distribution: six single-hop questions, eight two-hop compositions, three conjunctions, two whose shortest path traverses a mediator, and one question about a deliberately non-existent entity whose correct answer is a refusal. Questions were drawn from the training and validation splits only, never from test, so that everything tuned against them remains outside the evaluation.

The first run produced no crashes, terminated cleanly on every question, and abstained rather than confabulated in almost every failure — which was the actual exit criterion for the framework's mechanics. But its sixteen or seventeen failures collapsed into just five systematic causes, summarised in Table 5.4.

Three observations from this exercise are worth carrying forward, because they bear directly on later chapters.

The two confabulations in the first run were caused by a retrieval defect, not a generation defect. Two questions about national currencies produced a confident "United States Dollar". The relation cap had hidden the currency_used relation; the beam surfaced triples about GNI per

Table 5.4: Root causes of failure in the first end-to-end run, and the mechanism each one motivated. Every fix is described in the section indicated.

Observed failure	Mechanism introduced	Section
Backtracking was a deterministic no-op: the restored frontier produced identical scores, facts, and decisions until the budget drained	Ban list on abandoned (anchor, relation, direction) triples	Section 5.6.3
The evaluator's fact window showed the most recent thirty triples, so a large successful expansion pushed the answer out of view	Relevance-ranked fact window	Section 5.6
The relation list, capped at 80 by descending fanout, hid low-fanout precise relations behind hub relations	Cap raised to 300, plus the administrative-predicate blocklist	Section 5.2
Entities resolved while an objective was still incomplete were discarded	candidate_answers accumulated on every evaluation	Section 5.6
The frontier was truncated in insertion order, so the agent drifted away from the anchor that mattered	Frontier sorted by edge score before truncation	Section 5.5

capita denominated in dollars; and the answerer pattern-matched. No retrieved triple supported the answer. After the cap was raised both questions resolved correctly — and at roughly a third of the token cost. These two traces are the motivating example for Chapter 6: a claim-level check against the traversed triples would have caught them regardless of the retrieval defect, which is precisely the argument for verifying before emitting.

The groundedness filter on resolutions was validated by a failure. On one question the evaluator listed nine correct countries; none appeared in any retrieved triple, because the beam had gone down an unrelated branch. The filter rejected all nine, and the system abstained. This cost a point. It also demonstrated, one layer earlier than intended, that a language model inside an agentic scaffold will assert entities it never retrieved — and that catching this requires an explicit check, because nothing about the model's confidence distinguishes the nine countries it recalled from entities it actually found.

One instrumentation defect was found and removed. A counter reporting how many banned expansions had been skipped was computing the number of anchors instead. It was reporting a plausible-looking number that was unrelated to what it claimed to measure. It was corrected rather than deleted, but the episode motivates a standing rule adopted for the remainder of the work: a metric that has never been checked against a case where its correct value is known independently should not be trusted, and should not be analysed.

5.11 The Complete Navigation Algorithm

Algorithm 1 states the loop.

Algorithm 1 AGR navigation (verification layer abstracted; see Chapter 6)

Require: question q, graph G, budgets B, blend α, threshold τ
Ensure: answer A with supporting triples S

$P \leftarrow \textsc{Plan}(q)$; **if** invalid **then** $P \leftarrow [\,q\,]$
$F \leftarrow \textsc{SeedAnchors}(P)$; $T \leftarrow \emptyset$; $C \leftarrow \emptyset$; $stack \leftarrow \emptyset$; $ban \leftarrow \emptyset$
while budgets permit **do**
  push $(F, depth, \sigma)$ onto $stack$ {σ from the previous pass; 0 initially}
  $R \leftarrow \bigcup_{e \in F} \textsc{GetRelations}(e)$
  $scored \leftarrow \alpha \cos(\cdot) + (1-\alpha)\mathrm{LLM}(\cdot)$ over R against the active objective
  $E \leftarrow$ top-β of $scored$ per anchor, excluding ban
  $T \leftarrow T \cup \textsc{GetNeighbors}(E)$; $F \leftarrow$ top-m new entities by score
  $\sigma \leftarrow \max(\max_{c \in E} scored(c),\ \max_{c \in R} \cos(\cdot),\ \max_{c \in R_K} \mathrm{LLM}(\cdot))$ {signal maximum}
  $(d, done, \rho) \leftarrow \textsc{Evaluate}(\text{objective}, \textsc{TopFacts}(T))$
  $\rho \leftarrow \{x \in \rho : x \text{ appears in } T\}$; $C \leftarrow C \cup \rho$ {groundedness filter}
  if $done$ **then**
    advance P; **if** no objective remains **then** $d \leftarrow$ answer **else** $F \leftarrow \rho$
  end if
  if $|T_{\text{new}}| = 0$ **or** $\sigma < \tau$ **or** $d =$ backtrack **then**
    $ban \leftarrow ban \cup E$; $(F, depth) \leftarrow$ pop highest-scoring snapshot
  else if $d =$ answer **then**
    break
  end if
end while
$(A, S) \leftarrow \textsc{VerifyAndAnswer}(q, T, C)$ {Chapter 6}
return (A, S)

Chapter 6

The Structural Verification Layer

Chapter 5 described a navigation loop that ends with a set of traversed triples and a set of accumulated candidate entities. Nothing in that loop constrains what the final answer text *asserts*. This chapter specifies the layer that does, together with the repair, filtering, and evidence-reporting policies built on it. Every quantity reported here is recomputed from the archived development-set run at the frozen configuration ($\alpha = 0.7$, $\tau = 0.20$, verify_claims = true, $n = 80$), and where a behaviour is established on a handful of questions rather than a distribution, the count is given in the sentence that makes the claim.

6.1 Motivation: Verifying Before Emitting, Not After

Section 1.3 argued that agentic KGQA systems are typically evaluated on whether the correct answer is *among* what they assert, leaving the precision of the assertion unmeasured and unconstrained. The navigation loop makes that concrete: its final act is a language-model call over a fact window and a candidate list, the model is free to name an entity that appeared in neither, and no downstream component objects.

Two episodes from the first end-to-end development run (Section 5.10) sharpen the point. In one, a retrieval defect hid the currency_used relation, the beam surfaced triples about income denominated in dollars, and the answerer confidently asserted "United States Dollar" for two different countries — an assertion no retrieved triple supported. In another, the evaluator listed nine countries that appeared in no retrieved triple at all. The first was caught only by fixing the retrieval defect; the second by the groundedness filter on resolutions (Section 5.6.2), one layer earlier than intended. Neither was caught by anything examining the answer.

The design response is to check the draft against the environment *before* it is emitted rather than to fact-check an emitted answer afterwards. This is where the layer parts company with the post-hoc verification literature. Chain-of-Verification [15] improves a response by drafting it, planning verification questions, answering them, and revising; FActScore [43] decomposes

generated text into atomic facts and scores each against a corpus. Both are placed after generation, and both take the model's own answers to the verification questions as evidence. The layer specified here borrows their decomposition step — Section 6.2 extracts atomic claims in much the same spirit — and rejects their placement and their evidence source: the check runs before emission and adjudicates against traversed triples rather than against further generation. That matters because a model asked to check its own output is a weak detector of its own errors [41], which is precisely the failure mode a symbolic environment can avoid.

Three properties follow from placing the check inside the loop. The checker sees the same retrieved evidence the drafter saw, so a disagreement between them is informative rather than an artifact of differing context. The check happens while budget remains, so a rejected claim can trigger further exploration rather than only a retraction. And the check is symbolic where the graph can answer, so the arbiter of whether a fact holds is the environment and not a second opinion from the model that produced the draft.

What the layer promises should be stated precisely, because the terms are easy to inflate. It certifies that the atomic claims a draft asserts are *supported* by the environment in the sense fixed in Section 2.3.3: the traversed triples, or the graph in the retrieved neighbourhood, exhibit a connection consistent with the claim. It does not certify that the answer is correct. An answer every one of whose claims is supported can still be the wrong answer to the question, and Section 6.8 shows that this is a structural gap rather than a hypothetical one.

6.2 Claim Decomposition of the Draft Answer

Verification begins with a single model call that produces three things at once: a prose draft, the list of entities that draft asserts as the answer, and a decomposition of the draft into atomic claims.

The call sees a fact window of at most sixty traversed triples — ranked by embedding similarity between the triple's verbalisation and the *question*, with traversal order preserved among the survivors — together with up to twenty-five accumulated candidate entities. Each claim is emitted as a triple of strings $\langle h, r, t \rangle$ over entity *names*, not identifiers, since the model has never seen an identifier.

Four provisions in the prompt are load-bearing, and each exists because its absence produced a specific defect.

Names are to be copied exactly as they appear in the facts. The graph's surface forms are frequently not the ones a language model would generate unprompted — Freebase renders Yale University as University Yale — and every downstream check is keyed on the name string.

answer_entities names what the draft asserts as the answer, not every entity it mentions. Without this the list drifts toward a bag of mentioned names, including the question's own

subject.

A hedged draft has zero claims. This makes abstention explicit rather than something inferred from the absence of an answer.

Answer entities must be specific entities of the kind the question asks for, never a restatement of the question or an entity type. This provision was added after development runs produced answers of the form "the senators from New Jersey are the United States Senate members associated with New Jersey" — grammatically an answer, semantically empty, and, as Section 6.8 explains, entirely invisible to a claim-level check. It is a drafting-side mitigation of a failure the checking side cannot address.

The relation slot r is deliberately free text. The model is not asked to name a Freebase predicate, because it has no reliable way to do so and a wrong guess would be indistinguishable from a wrong claim. The consequence is that the structural checks below cannot match on the relation at all; the cost of that choice is analysed in Section 6.3.1 and again in Section 6.8.

Decomposition on the development set was sparse. Across 80 questions the drafter emitted 121 claims — a mean of 1.51 and a median of 1 per question, with a maximum of 5. Ten questions produced no claims at all. For those ten the layer certified nothing, and Section 6.8 returns to what that means for the outcome label they received.

6.3 Two-Stage Claim Checking

Each claim is routed through at most three tests: two structural, one model-based. Algorithm 2 states the procedure and Figure 6.1 draws the same routing as a single claim's path through it; the subsections that follow explain each test and the reason for their order.

6.3.1 The Structural Check

The structural check operates on entity identifiers, so it first needs a mapping from the names the model produced back to graph nodes. That mapping, μ, is built from the traversed triples alone: every head and tail name encountered during navigation is registered against its identifier, first occurrence winning. Its scope is worth stating explicitly, because it bounds everything that follows. **A claim naming an entity the agent never traversed cannot reach either structural route**, regardless of whether that entity exists in the graph. Such claims fall directly to the entailment check.

For claims whose endpoints both resolve, the first route is *traversed adjacency*. The traversed triples are collapsed into an undirected set of endpoint pairs, and a claim is supported if its endpoint pair is in that set. This route is exact, costs nothing beyond a set lookup, and — uniquely among the three — produces citable evidence: every traversed triple joining the pair is added to the supporting-triple set.

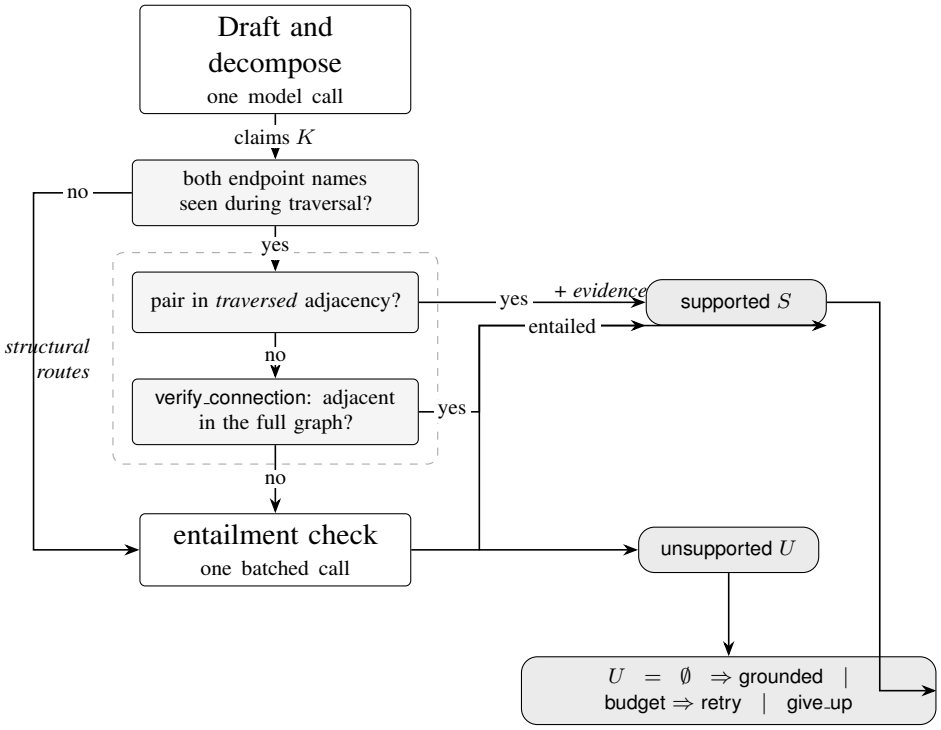

Figure 6.1: The path of a single atomic claim through the layer, mirroring Algorithm 2. Three tests are tried in a fixed order and a claim leaves at the first one that settles it. The two structural routes are boxed; only the first of them adds to the supporting-triple set, which is why Section 6.6 can report evidence for some grounded answers and not others. A claim naming an entity the agent never traversed skips both structural routes entirely, since the name-to-identifier map is built from traversed triples alone.

Algorithm 2 Structural verification of a draft

Require: question q, traversed triples T, candidates C, tools, budget
Ensure: draft, answer entities, supported and unsupported claim sets, supporting triples

$W \leftarrow \textsc{TopFacts}(q, T, 60)$; $\mu \leftarrow \{\text{ name} \mapsto \text{id} \}$ over T {first occurrence wins}
$(\mathit{draft}, A, K) \leftarrow \textsc{DraftAndDecompose}(q, W, C)$ {one call}
if $\neg$ verify_claims **then**
  return $(\mathit{draft}, A, \emptyset, \emptyset, \emptyset)$ {ablation condition}
end if
$\mathit{adj} \leftarrow \{(h,t),(t,h) : (h,r,t) \in T\}$; $S \leftarrow \emptyset$; $P \leftarrow \emptyset$; $E \leftarrow \emptyset$
for all $c = \langle h, r, t \rangle \in K$ **do**
  if $h \notin \mu$ **or** $t \notin \mu$ **then**
    $P \leftarrow P \cup \{c\}$ {endpoint never traversed}
  else if $(\mu[h], \mu[t]) \in \mathit{adj}$ **then**
    $S \leftarrow S \cup \{c\}$; $E \leftarrow E \cup \{\, \theta \in T : \{\theta_h, \theta_t\} = \{\mu[h], \mu[t]\} \,\}$
  else if $\textsc{VerifyConnection}(\mu[h], \mu[t])$ **then**
    $S \leftarrow S \cup \{c\}$ {no evidence recorded}
  else
    $P \leftarrow P \cup \{c\}$
  end if
end for
$U \leftarrow \emptyset$
if $P \neq \emptyset$ **then**
  $V \leftarrow \textsc{Entail}(W, P)$ {one batched call; on any failure $V \equiv$ false}
  partition P into S and U by V
end if
if $U \neq \emptyset$ **and** verify budget remains **then**
  $\mathit{iters} \leftarrow \mathit{iters} + 1$; append repair objective for the first $c \in U$; re-anchor on $\mu[c_h]$
end if
route: $U = \emptyset \Rightarrow$ grounded; else verify, depth, and time budget all remain $\Rightarrow$ retry (**goto** explorer); else give_up
return $(\mathit{draft}, A, S, U, E)$

The second route is the verify_connection tool (Section 5.2.3), a single Cypher query asking whether the two entities are adjacent in the *full* graph, either directly or through one mediator node. It exists because the first route tests the *traversal*, not the graph: the agent may have reached both endpoints along different branches without ever walking the edge between them.

Both routes are relation-agnostic. Since r is free text with no correspondence to a Freebase predicate, matching on it would reject true claims wholesale — "X played for Y" is realised through a roster mediator carrying no predicate named for playing. The design intent is a division of labour: the structural routes provide high recall for "some supporting edge exists," and semantic precision is delegated to the entailment check. Section 6.8 examines what that division actually buys and what it lets through.

On the development set the second route was consulted five times, across five distinct questions — four per cent of the 121 claims. All five returned connected, and all five had a mediator path;

in three of them there was no direct edge at all, so a mediator-blind adjacency test would have rejected them. Two were direct edges the traversal had simply never walked. Query latency ranged from 1.9 to 75.7 ms. The route is therefore cheap and, on this evidence, correct — but it is also rare, and the structural check is in practice overwhelmingly the traversed-adjacency test.

6.3.2 The Entailment Check

Claims that fail both structural routes — together with those whose endpoints were never traversed — accumulate into a pending set and are checked in one batched model call. The claims are numbered, rendered as h – r – t, and presented against a fact block, with the model asked for a boolean verdict per claim. The prompt defines unsupported explicitly, to include the three cases that would otherwise be resolved charitably: plausible but absent, contradicted, and inferable only through outside knowledge.

Three properties of this check must be stated, because each bounds what its verdicts mean.

The fact block is the same window the drafter saw. It is the same sixty triples, ranked by relevance to the *question* rather than to the individual claim under test. A claim about an entity peripheral to the question may therefore be judged against facts that never mention it. The bias this introduces is directional: it disfavours peripheral claims and favours claims about material the drafter had already been shown.

Failure is conservative and lossy. If the call raises for any reason — budget exhaustion, transport error, malformed output — every pending claim is marked unsupported. This is the right default for a system whose purpose is to avoid unsupported assertions, but it means an infrastructure failure and a genuine rejection produce the same record except in the trace.

The check can only add support, never remove it. Claims accepted structurally never reach it. A claim whose endpoints are adjacent for a reason unrelated to the asserted relation is accepted without any semantic examination.

6.3.3 Why the Structural Check Runs First

Three reasons, in descending order of importance. *Authority*: the thesis's position, fixed in Section 4.5.3, is that embeddings and language models rank while symbols decide, so the model must be consulted only where the graph is silent — it fills gaps and never overrides a finding of the environment. *Evidence*: only traversed adjacency yields triples that can be attached to the answer, so the order does not change which claims are accepted but does change how many can be justified. *Cost*, least importantly: a set lookup is free, a verify_connection probe is one indexed query, and the entailment call is the layer's only mandatory expense beyond drafting, so ordering cheap-to-expensive runs it on a residue rather than on every claim.

6.4 Repair Policies for Unsupported Claims

The router following the verifier is pure Python and has three outcomes: no unsupported claims routes to grounded; unsupported claims with budget remaining route to retry; otherwise give_up.

6.4.1 Targeted Re-Exploration Under Remaining Budget

On retry, the verifier has already re-targeted the agent before the router runs. It takes the *first* unsupported claim, synthesises a repair objective of the form verify whether $h\ r\ t$, marks every active plan step done, appends the repair step as the new active objective, and — if the claim's head name resolves — resets the frontier to that single entity. Control returns to the explorer, which now searches for the missing evidence specifically.

Two limitations follow. Only one claim is repaired per iteration, and the one chosen is first by position in the draft's decomposition rather than by importance to the answer. Both keep the repair path cheap, and neither was revisited because the path was rarely exercised.

The retry route requires verify, depth, *and* time budget. The depth condition was added after a development trace showed a question completing with depth 5 against a maximum of 4: the retry edge re-enters the explorer without passing through the post-evaluation router, which is where the depth check lives, so each retry silently granted one expansion beyond budget. Since Section 5.8 claims budgets are hard guarantees, the hole was closed in the router rather than documented as an exception.

That condition turns out to be decisive, and the honest statement of what this route contributed is that **it never executed**. On all three development questions that produced an unsupported claim, the depth budget was already exhausted when the verifier ran, so the router took give_up in every case and no repair exploration occurred. This is not a coincidence of small numbers so much as a structural correlation — the verifier is reached either because the plan completed or because a budget ran out, and all three firings came from the fourteen questions that exhausted depth, none from the sixty-six that finished with search budget in hand. With $n = 3$ that is an observation and not a finding, but it means the targeted re-exploration described above is specified and unit-tested rather than empirically characterised, and no claim in this thesis depends on it.

One instrumentation consequence follows, and matters when reading run records. The verifier increments verify_iters and appends the repair objective *before* the router decides, so a record showing verify_iters: 1 records the verifier's intent to repair, not an executed repair. Whether exploration actually resumed is visible only in the node sequence of the trace.

6.4.2 Answer Rewriting and Hedging

On give_up the answerer makes one rewrite call. It receives the draft, the unsupported claims, and — crucially — the list of entities already decided to survive, with an instruction to keep them named exactly as written. It is asked to remove or hedge the unsupported material and to respond with prose only. That last phrase is the whole of the fix described in Section 6.5: in the original design this call also regenerated answer_entities, which turned out to be the single largest defect in the layer.

6.4.3 Iteration Cap and the Draft-Only Fallback

Repair is capped at two iterations, which together with the depth and time conditions guarantees the verify–explore cycle terminates. On the development set the cap was never reached, for the reason just given: the cycle never began.

The grounded path costs nothing. When no claim is unsupported the answerer emits the draft verbatim with no further model call — 77 of 80 development questions took this path. The layer's aggregate cost is correspondingly small: mean tokens per question rise from 4,858 without claim checking to 4,922 with it, about 64 tokens or 1.3%, and mean model calls from 7.2 to 7.3. Whatever the case for or against the layer, it is not a cost argument.

A distinct path exists when verify_claims is false: the verifier returns the draft and its entity list untouched with outcome draft_only. This is the ablation condition of Section 7.8, not a runtime fallback — the system never selects it dynamically. A third path, the legacy answerer of Section 5.7, runs only when the verifier raised before producing a draft at all.

6.5 Entity Filtering of the Final Answer

The rule is one line and its justification is the most instructive result in this chapter.

The rule. On the give-up path the final entity list is computed deterministically from the draft's own list: an entity is kept if it appears in at least one supported claim, or if it appears in no unsupported claim. Strings are copied verbatim. The function can only remove entities, never add one. The disjunction's second branch is deliberately permissive — an entity the decomposer never mentioned in any claim is retained, on the principle that the layer should punish unsupported assertions and not imperfect decomposition recall.

The scope. This filter applies to the give-up path only. On the grounded path the draft's entity list passes through unchanged, because there are no unsupported claims to filter against. On the development set the filter therefore ran on 3 of 80 questions.

Table 6.1: The complete attribution census: every development-set question whose answer differed between claim checking and its ablation, at $\alpha = 0.7$, $\tau = 0.20$. Three questions, one run per condition. "Pre-fix" is the original design in which the rewrite call regenerated the entity list; "post-fix" is deterministic filtering. Per Section 7.5.2, the three questions the environment cannot answer hedge in every condition and are counted as hedges rather than excluded, so the totals are over all 80.

Question	**Draft-only**	**Verified, pre-fix**	**Verified, post-fix**
Harbaugh's teams founded before 1996 *gold:* 4 teams	4 gold teams *plus* Baltimore Ravens; scores as a hit	[] — hedge	[] — hedge
Party of Hitler and Carl Voss *gold:* Nazi Party	Nazi Party — hit	Adolf Hitler, Nazi Party; hit, but asserts the question's subject	Nazi Party — hit
Franco's university, founded 1701 *gold:* University Yale	University Yale — hit	Yale University — **wrong**	University Yale — hit
Whole run — all 80 questions; the three counts partition the set			
hits / hedges / wrong	66 / 10 / 4	64 / 11 / 5	65 / 11 / 4

The Attribution Census

The filter exists because of a paired comparison between the verified condition and its verify_claims = false twin at the frozen α. In aggregate, claim checking *cost* accuracy: 64 hits against draft-only's 66. Aggregate counts cannot distinguish a mechanism working as intended — turning an unsupported lucky hit into an honest hedge — from a mechanism misfiring, so the two runs were diffed per question. Exactly three questions differed. Table 6.1 gives all three.

The classification is unambiguous. *No entailment verdict was wrong*, and no repair exploration was involved — as Section 6.4 records, none ran. All three differences trace to one component: the rewrite call, failing three different ways. On Harbaugh it over-deleted, discarding four grounded-correct team names along with the unsupported one. On Hitler/Voss it laundered the question's subject into the answer list — an entity the draft never asserted. On Franco it silently respelled the graph's University Yale as the parametric Yale University, converting a defensible qualified answer into a scored error.

The finding is worth stating in one sentence, because it is the conclusion this chapter's design rests on:

> The verification logic was sound; its repair path free-generates entities from a language model, which reintroduces exactly the ungroundedness the layer exists to prevent.

Replacing generation with deterministic filtering removed two of the three regressions, exactly as predicted: Hitler/Voss and Franco left the diff.

What the Census Does and Does Not Establish

Three qualifications belong here rather than in a limitations chapter, because without them the numbers above would be read as more than they are.

The census is $n = 3$. It is a complete enumeration of the questions on which the two conditions differed, which is exactly why it could be classified by inspection — but three questions support a mechanism claim, not a rate.

Claim checking did not reduce wrong answers on this set. Post-fix verified scores 65 / 11 / 4 against draft-only's 66 / 10 / 4. The wrong count is identical; the layer cost one correct answer and removed none. Its case therefore cannot be made on development-set accuracy, and this thesis does not attempt to make it there. The case rests on the groundedness measurements of Section 8.6, which are the metric the layer was built to move, and on the precision effects that Hits@1 is structurally unable to see.

The residual difference is a single question, and its interpretation involves a label changed after the fact. The sole remaining diff is Harbaugh, where draft-only's "hit" asserts Baltimore Ravens — supported by no traversed triple, and not a team Jim Harbaugh played for — alongside an unverifiable founding-date filter. Set-intersection Hits@1 scores that as a clean hit; per-question F1 in Chapter 8 does not. The question was additionally relabelled from cvt_heavy to unanswerable_env after this analysis, on the grounds that it carries a date constraint the environment cannot express (Section 4.7.4). The relabelling is defensible on its own terms and was recorded as a decision, but it was made after seeing the result it affects, and is flagged as such here and in Section 9.9.3. The aggregate figures reported above are not rescored on account of it.

6.6 Output Contract: Answer Paired with Supporting Triples

The system pairs each answer with the triples that support it. This is the explainability deliverable named in Section 1.6, and its actual coverage is narrower than the phrase suggests in two independent ways: one route of three attaches no evidence, and what the run record keeps is a count rather than the triples.

Only one of the three routes records evidence. Supporting triples are collected in the traversed-adjacency branch alone. A claim certified by verify_connection is accepted with nothing attached, and so is a claim certified by entailment. The output contract is therefore "answer paired with whatever traversed triples happened to justify it", not "answer paired with justification for every claim".

What survives in the log is the count, not the evidence. The pairing holds inside the run: the answerer receives the verifier's triple list and emits against it. But RunLogger writes n_supporting_triples, an integer, and drops the list, so no committed artifact in this work contains

a single supporting triple, and every statistic in this section is computed from that counter. An examiner can confirm that these answers came from a system which tracked its evidence; they cannot inspect the evidence. This bounds the auditability the section claims, and it is the same defect approached from the other side by Section 10.3.2, which asks for accepted claims to be persisted with their matching triples — one logging change repairs both. It is recorded here rather than repaired late, under the standing rule of Section 5.10, because repairing it now would separate the code from the results already frozen against it.

The measured consequence on the development set: 13 of 80 answers carry no supporting triples at all. Ten of those asserted no claims — they are hedges, and zero is the honest number. Of the remaining three, one had every claim rejected, so there was nothing to record; the other two had their single claim certified by verify_connection, which records no evidence. Across all 80 records the median count is 3 and the mean 4.1, with a maximum of 16.

Two instrumentation defects in this area are recorded rather than quietly fixed, in keeping with the standing rule adopted in Section 5.10.

The first was a Python falsy-empty-list trap. The answerer substituted the entire traversed set for the supporting triples whenever the verifier's list was empty — treating "the verifier ran and matched nothing" identically to "the verifier never ran". A development trace consequently recorded 1,927 supporting triples for an answer that asserted nothing whatsoever. The fix tests for None rather than for falsity, so an empty list is reported as the zero support it is.

The repair is effective but its stated rationale was wider than the code. The intent was a three-way distinction — absent means the verifier never ran, empty means it ran and matched nothing — and the second half is what the system delivers. The first half is unreachable: the state initialiser seeds supporting_triples to an empty list, so the key is never absent and the fallback branch never executes. Every answer therefore reports the verifier's own list, which is the correct behaviour and the one the results depend on; what does not exist is the legacy path the description implies. The branch is dead code guarding a state the initialiser rules out, and saying so is cheaper than leaving a reader to reconcile the two files.

The second is unrepaired and therefore fenced off. A counter in the verifier's trace named n_structural counts every supported claim, including those certified by the entailment call, so its name misdescribes it. Under the standing rule — a metric that has never been checked against a case whose correct value is known independently should not be analysed — it is not used anywhere in this thesis, and no table or figure derives from it. The route-by-route counts reported in Section 6.3.1 come from the tool invocation log, which records each verify_connection call with its arguments and result.

6.7 A Worked Example

Which university, founded in 1701, did actor James Franco attend? Gold: University Yale.

Navigation resolves James Franco as the anchor and expands along education.education.student and people.person.education. The plan's first objective — find the university founded in 1701 — is never satisfied, because the environment holds essentially no date literals (Section 4.7.4). Three evaluator-triggered backtracks follow, and the run terminates on the depth budget with 18 model calls, 17 of them cache replays.

The verifier drafts: *"James Franco attended University Yale, which was founded in 1701."* Its entity list is [University Yale] — copied from the fact window, in the graph's spelling. It decomposes the draft into an attendance claim and a founding claim.

The attendance claim resolves both endpoints in μ and its endpoint pair is in the traversed adjacency set. It is supported, and four traversed triples joining the pair are recorded as evidence. The founding claim names 1701, which no traversed triple mentions, so it cannot resolve and falls to the entailment check — which correctly reports it unsupported, because the fact is genuinely absent from the environment rather than merely unretrieved. The verifier prepares a repair objective anchored on University Yale, but the depth budget is already spent, so the router takes give_up and the exploration never resumes; there was in any case nothing to find.

The answerer now filters. University Yale appears in a supported claim, so it is kept, verbatim. The rewrite call produces prose that asserts the attendance and hedges the founding date, and its own opinion about entities is discarded. Final answer entities: [University Yale] — a hit.

The instructive part is the counterfactual. Before deterministic filtering, this identical trace — same draft, same claims, same verdicts — ended with [Yale University]. The rewrite call, asked to restate the answer, restated it in the model's own spelling rather than the graph's, and the answer was scored wrong. Nothing about the verification was different. The claim-level machinery had worked perfectly and the layer still emitted an entity string that appears nowhere in the environment, which is the precise thing it exists to prevent.

6.8 Failure Modes by Construction: Wrongful Rejection and Wrongful Acceptance

This section is analytical. It enumerates the ways the layer can be wrong that follow from its design, independently of how often they occur. Which of these mechanisms actually fired, and how often, is reported in Section 9.5; the vocabulary for the two polarities is fixed in Section 9.5.1 and used here. A claim is *wrongly rejected* when the layer withholds support the environment would in fact provide, and *wrongly accepted* when it certifies a claim the environment does not actually support.

Mechanisms of Wrongful Rejection

Composite claims. The decomposer emits claims corresponding to what the draft asserts, not to every relationship the draft relies on. When a draft asserts a filtered set — "these teams were founded before 1996" — the decomposition is about the founding, and the membership relation that makes each team a candidate may never be claimed at all. Every team's only claim is then the unverifiable one, the filter drops every entity, and a draft containing four correct answers becomes a hedge. This is the Harbaugh question of Table 6.1, and the mechanism is worth naming: *a composite assertion whose unverifiable component drags down its verifiable component*. The conservative rule is defensible for a hallucination-mitigation system, and it is also the layer's most expensive single behaviour on the development set.

Endpoints outside the traversal. μ is built from traversed triples only, so a claim about an entity the agent never walked to skips both structural routes regardless of what the graph contains. It is judged solely by a model, against a fact window that by construction does not mention it.

A question-ranked window for a claim-level judgement. The entailment check's evidence is ranked for relevance to the question, not to the claim under test. Peripheral claims are systematically disadvantaged.

Conservative failure. Any exception in the entailment call marks all pending claims unsupported. Infrastructure failure is recorded as rejection.

Name collisions in μ. The mapping keeps the first identifier seen for a name. Two distinct graph entities sharing a surface form collapse to one, and adjacency is then tested for the wrong node.

Mechanisms of Wrongful Acceptance

Relation-agnostic adjacency. Both structural routes ignore r and ignore direction. Any edge between the endpoints certifies any claimed relation between them. A claim that X is Y's mother is accepted on an edge recording that X is Y's child; a claim that a film was directed by a person is accepted on an acting credit. This is the direct cost of allowing free-text relations, and it is not a small one — it is the layer's principal acceptance risk.

The mediator hop. verify_connection accepts a path through one mediator node. Because a Freebase mediator encodes a single n-ary fact, the endpoints of such a path are genuinely co-participants in one fact — but which role each plays is exactly what the mediator encodes and the test discards. An actor, a film, a character, and a period are mutually "connected" through one performance node, and any claim relating any two of them is certified.

Vacuous certification of empty decompositions. A draft with no claims has no unsupported claims, so it is routed grounded. On the development set this describes 10 of 80 questions. The label is not false — an answer that asserts nothing asserts nothing unsupported — but it must not be read as a positive verification, and any statistic computed over grounded outcomes inherits

this.

Entities that appear in no claim are never checked. This is the broadest of the mechanisms listed here and deserves to be stated without softening. Verification operates on claims; the answer's entity list is a separate output of the same call. An entity the drafter named but the decomposer never wrote into a claim passes through untouched on the grounded path, and is deliberately retained by the permissive branch of the filter on the repair path (Section 6.5). Since 77 of 80 development questions took the grounded path, the layer's guarantee over emitted entities is in practice mediated entirely by decomposition recall — and decomposition recall is itself a language-model output, subject to no check.

Grounded-but-vacuous answers. This is the boundary of what triple-level verification can do, and it deserves to be stated as such. An answer whose every claim is supported can still fail to answer the question: "the senators from New Jersey are the United States Senate members associated with New Jersey" decomposes into claims the graph genuinely supports, and is certified. No claim-level check can catch this, because the defect is not in any claim — it is in the relationship between the claims and the question. The anti-vacuity provision of Section 6.2 addresses it at drafting, which is mitigation, not verification.

Structural acceptance is final. A claim accepted by adjacency is never examined semantically. The entailment check, which is the only component that looks at meaning, sees only what the structural routes could not resolve.

What the Layer Is

Stated at the level of precision the preceding list requires: the layer establishes that the entity pairs a draft's claims relate are connected in the environment, and on the repair path it makes the emitted entity list a deterministic function of those verdicts rather than a second generation. It does not establish that the claimed relation is the one the edge records, that an entity absent from every claim was grounded at all, that the answer is responsive to the question, or that a claim it could not check is false. Those boundaries are properties of the design rather than defects discovered in it, and Chapter 9 reports how much of the residual error they account for.

Chapter 7

Experimental Setup

This chapter fixes everything that had to be decided before any test question was answered: which questions, which systems, which numbers, and which rules for reading them. It is written to be sufficient for replication, and it records the decisions that went the other way as well as the ones that stood — a backbone choice that was made, reversed, and then remade under a pre-registered rule; a test-set build that read the wrong split and was caught before harm; a validation threshold that was missed by half a thousandth. Each of those is reported where it belongs rather than collected into a confessional at the end.

Every figure quoted in this chapter and the two that follow is regenerated by scripts/build_thesis_numbers.py into a single file, results/phase4/thesis_numbers.json, from the scoring, groundedness, judge, and census artifacts. No number in the thesis is transcribed by hand from a console log.

That claim is about *provenance*, and its limit should be stated with it. The numbers come from the generated file, but most of them reach the page by being typed into a table body from that file rather than emitted into it: of the twenty-eight tables in this thesis, two — Table 9.2 and Table 8.4 — are generated directly, as are three of the five figures. The state machine of Figure 5.1 and the claim path of Figure 6.1 are drawn by hand, since neither carries a measurement. Every other table is a hand copy that agrees with the JSON but is not mechanically tied to it, which is a weaker guarantee than the sentence above may suggest. It is also not a hypothetical one: Table 9.2 is generated because its hand-typed predecessor had drifted from the file it was copied from, and converting it was the fix. Where a table's body is generated, its caption says so; the rest were checked against the JSON by hand.

7.1 Mapping Experiments to Research Questions

Each of the three research questions of Section 1.5 is answered by a distinct experiment rather than by one table read three ways. **RQ1** — whether agentic navigation improves multi-hop

accuracy — is answered by the main matrix of five systems on two datasets, scored with Hits@1 and F1, broken down by hop stratum, with paired significance tests (Sections 8.2, 8.3 and 8.5); the per-stratum breakdown, not the aggregate, is what makes the claim about *multi-hop* accuracy. **RQ2** — what pre-generation verification contributes — is answered by three measurements that do not agree with one another, which is itself the finding: the two-tier groundedness metric (Section 8.6), the precision and F1 columns of the main matrix, and the verifier ablation (Section 8.8). It was posed as *what does it contribute* rather than *does it reduce hallucination* for the reason Section 8.9 gives. **RQ3** — component attribution at a token price — is answered by four single-deviation ablations on stratified half-samples, each compared against the unmodified system restricted to the same questions (Sections 7.8 and 8.8).

7.1.1 Pre-Registration and the No-Tuning Policy

One rule governed everything from the moment the test files were built: *nothing gets tuned*. No prompt was edited, no configuration value changed, and no threshold adjusted after test data was touched. Defects discovered mid-matrix were documented rather than fixed, and any fix that did become unavoidable would have been applied identically to every system with all affected files re-run — a situation that did not arise.

Four decisions were fixed in writing before the measurements they govern:

- The frozen configuration $\alpha = 0.7$, $\tau = 0.20$, claim verification enabled, tuned on the development set (Section 7.9) and never revisited.
- The sample size: stratified samples of 400 questions per dataset, named as the fallback before the API budget was known and adopted once it was (Section 7.2).
- The baseline certification threshold: a retrieval baseline is certified if it hedges while the gold answer is visible in its own retrieved facts on fewer than 15% of its hedges (Section 7.4).
- The judge validation threshold: Cohen's $\kappa \geq 0.7$ against blind human annotation before the semantic-support numbers may be cited as validated (Section 7.6).

The last of these was missed, by 0.0005. It is reported as missed in Section 7.6, because a threshold that is only pre-registered when it is met is not pre-registered at all.

One further discipline, cheaper than it sounds and load-bearing for Section 8.8: the single code change Phase 4 required — a `use_planner` flag to disable decomposition — was certified as a no-op on the frozen path by re-running an already-answered development question and confirming that every language-model call was a cache hit. Byte-identical prompts are byte-identical behaviour, so the reference condition in the ablation table is provably the same system that produced the main matrix.

7.1.2 Metrics Proposed but Not Run

The approved proposal named three evaluation criteria: answer accuracy, groundedness, and *path fidelity* — agreement between the relation chain the system traversed and the chain the benchmark's gold SPARQL query specifies. Path fidelity is not reported. Computing it requires the gold relation chains, and the RoG distribution used to build the knowledge environment (Section 4.2) publishes name-keyed triples without the originating SPARQL, so the chains would have to be reconstructed from the upstream releases and re-aligned to this environment's relation vocabulary. That was outside the project's remaining budget. Section 10.3.7 treats it as future work; the point is not relitigated here.

7.2 Test Sets

7.2.1 WebQSP and ComplexWebQuestions

Two benchmarks are used, in the RoG distribution [9] whose per-question subgraphs also form the knowledge environment (Section 4.2). **WebQSP** [19] is the semantic parses of WebQuestions: natural questions from search logs, mostly one or two hops, with multi-valued answers common. **ComplexWebQuestions** (CWQ) [20] is built by composing WebQSP questions with additional constraints through SPARQL templates, giving deeper chains, conjunctions, superlatives, and comparisons — and, as Section 7.7 documents, a higher rate of machine-generated label defects.

The distinction that matters operationally is visible in the gold answer sets. WebQSP averages 7.98 gold entities per question, but its median is 1.5: exactly 200 of its 400 questions carry a single gold answer, and the mean is pulled up by a long tail whose largest question lists 382. CWQ averages 1.78 with a median of 1. So the two benchmarks differ less in the typical question than the means suggest, and more in the tail — and it is the tail that decides how a metric rewarding one correct entity behaves. This is one of the reasons F1 is reported alongside Hits@1 (Section 7.5).

7.2.2 Stratified Sampling, Seeds, and Published Question IDs

Only the *test* splits are used. The development set of Section 7.9 was mined from train and validation, and the builder asserts zero overlap between the two pools.

Sampling was a budget decision. The full test splits hold $1{,}628$ and $3{,}531$ questions; five systems over both, at the measured token rates, projected to roughly \$45–55, against a stated ceiling of \$15–20. The pre-registered fallback — stratified samples of 400 per dataset (Table C.1), giving bootstrap confidence intervals of roughly ± 5 points on Hits@1 — was therefore adopted, with the arithmetic recorded before the runs fired. The whole benchmark, including the ablations,

Table 7.1: The two certified test samples. The unreachable stratum is retained deliberately; the ceiling is the share of questions with at least one gold answer reachable within four hops, and is the upper bound on Hits@1 for every system evaluated here.

Dataset	n	h1	h2	h3plus	unreach.	Ceiling
WebQSP	400	256	128	4	12	97.0%
CWQ	400	137	211	49	3	99.2%

ultimately cost $11.13 in API spend. That figure is read from the provider's billing record rather than derived from the run artifacts, and it is the one number in this thesis a reader cannot recompute from the repository. The reproducible proxy is the token and call accounting of Section 8.7, which is recorded per question in every run record.

Sampling is proportional within hop strata at seed 42, and the resulting question sets are committed as immutable artifacts (results/phase4/test_webqsp.json, results/phase4/test_cwq.json); every question identifier is listed in Chapter C. They were never regenerated after any system ran.

A near miss worth recording. The first build read train-*.parquet rather than test-*.parquet — a glob copied from the development-set miner, which legitimately reads train. Pool sizes matching the train splits and eighteen resolved shards were suggestive; the conclusive signal was the development-overlap guard reporting 32 and 45 overlapping questions where a genuine test pool must report zero. The contamination guard, built for hygiene, is what caught the contamination. No system had run against the files; they were deleted and rebuilt, and the rebuilt pools report 1,628 and 3,531 with skipped={} — the same evidence now testifying the other way, and the reason a cheap overlap assertion belongs in every benchmark builder.

7.2.3 Hop-Count Stratification and the Accuracy Ceiling

The stratification variable is the length of the shortest path in *this* environment between any question topic entity and any gold answer entity, bounded at four raw hops — the same machinery used to certify the development set, so that mediator nodes are counted consistently as the two edges they are (Section 5.2.3). Questions fall into h1, h2, h3plus, or unreachable.

Table 7.1 carries two facts the results chapter leans on repeatedly. WebQSP is 64% one-hop with a four-question h3plus tail — too small to interpret, and labelled as such wherever it appears. CWQ is majority two-hop with a genuine deep tail of 49. The samples preserve the strata proportions of their parent pools, so per-stratum numbers describe the benchmark rather than the sample.

The unreachable stratum was kept rather than dropped. Dropping it would raise every system's score by the same amount and misdescribe the environment; keeping it means the ceiling is

reported honestly and, as Section 8.6 shows, that these questions become the sharpest available probe of parametric leakage — any "hit" on a question with no graph path to gold is an assertion the environment cannot support.

7.3 Backbone Model Selection and Qualification

All systems share one frozen backbone: gpt-5.4-mini-2026-03-17 at temperature 0.0, with reasoning_effort explicitly set to "none" and the response cache enabled. The model string, both sampling parameters, and the cache mode are stamped into every one of the $4{,}000$ test-matrix records and every ablation record, so any record found anywhere states what produced it.

Two properties motivated a non-reasoning model: sampling parameters remain available, so temperature can be pinned; and no hidden reasoning tokens contaminate the cost metric. The second was verified rather than assumed — a defensive check logs the reasoning-token count on every call, and it read zero across all runs in this thesis.

Selection between gpt-4.1-mini-2025-04-14 and gpt-5.4-mini-2026-03-17 was made by a qualification script that runs both candidates over the same twenty development questions and reports tokens, calls, latency, reasoning tokens, errors, and a determinism probe. The probe re-runs a subset of questions and counts how often the *decision signature* — a hash of the plan, the expanded relations, the evaluator verdicts, and the answer, not of timings or scores — repeats exactly.

The first round reported $3/3$ signature matches for 4.1-mini against $1/3$ for 5.4-mini, and 4.1-mini was chosen on that basis. A second round with the cache disabled reversed it: $2/3$ against $3/3$. The reversal, not either number, is the finding.

7.3.1 Trajectory Stability Under Temperature-Zero Decoding

Temperature-zero decoding on a hosted API is greedy, not deterministic. The logits are not bit-stable across calls, because floating-point reduction order depends on how a request is batched with other traffic, and when the top two tokens are near-tied a wobble of order 10^{-6} flips the argmax. A sequential agent then *amplifies* that: one divergent token in a planner call changes a sub-objective's wording, which changes its embedding, which changes the beam, which changes everything downstream. A model that is 98% stable per call can be 70–90% stable per trajectory.

With three probes per round, an estimator of a rate in that band bounces between $1/3$ and $3/3$ by sampling variance alone. That is what happened. A tiebreaker was therefore pre-registered before rounds three and four — *if the two candidates finish within three matches of each other, take gpt-5.4-mini on longevity grounds, the 4.1 line being deprecation-adjacent* — and both rounds returned $2/3$ for each candidate. Pooled over all four rounds the score is $9/12$ against $8/12$: a tie, the rule fired, and the backbone was frozen.

The reported measure of trajectory stability is therefore $\approx 75\%$, and the thesis makes no claim of determinism. One provenance limit belongs with that number: only the first round survives as per-question artifacts, because the qualification script writes its detail file in truncating mode and later rounds overwrote it; the tool logs accumulated across all four rounds, and the console tables for rounds two through four exist in the project record rather than as committed artifacts.

Two observations from those four runs matter more to this thesis than the choice they settled. First, each candidate carried a *persistent* ungrounded answer: 4.1-mini asserted "United States Dollar" for a Dominican-peso question in two of four runs and a wrong European Union president in one; 5.4-mini named the wrong Beckham child in all four and a different wrong president in one. Second, both carried *intermittent* ones — the same question, the same graph, the same budget, with the answer moving between grounded-correct and ungrounded-wrong as the sampling weather changed. Backbone selection does not remove this class of error; it relocates it. That is the premise of this thesis, measured over four runs and two frontier-family models, and it is the reason Chapter 6 exists.

7.3.2 Response Caching as the Reproducibility Backstop

Reproducibility here rests on the cache, not on the model. Every completion is stored keyed on model string, temperature, reasoning effort, prompt, and schema hint, and — the detail that makes it work — the original token usage is stored alongside and *replayed into the budget meter* on a hit. The budget check still runs and the call counter still increments, so a fully cached rerun reproduces the original run's budget snapshot rather than approximating it — with one field's worth of care. The replay restores prompt and completion tokens but not the reasoning_tokens counter, which the cache stores and does not add back. The backbone runs at reasoning effort none, and that field is zero in all $6{,}060$ records this thesis reports, so the snapshots do reproduce exactly here; under a reasoning-effort setting they would not. The defect is recorded rather than repaired, for the reason Section 7.1.1 gives. Every table in Chapter 8 comes from one recorded run per condition, and that run replays byte-identically for as long as the cache survives, independent of API weather or model retirement.

Fresh-run variance is a different problem and is handled the standard way: bootstrap confidence intervals over questions and paired significance testing, both specified in Section 7.5.

The cache leaves one artifact in the record. The WebQSP run of the full system was started, interrupted after seventeen questions, and restarted; the restart replayed those seventeen from cache. Answers, entities, tokens, calls, and traces are unaffected — that is what byte-exact replay means — but their wall-clock readings measure a disk, not a system. The original cold log was recovered and merged back by a script that asserts every field except the timing and cache counters is identical, so the archived file carries true cold latencies throughout; no record in it is a full cache replay. Latency is in any case computed over cold-cache records only, which protects

the column regardless.

7.4 Baseline Systems

Four baselines span the space between no retrieval and agentic navigation. All were implemented against the same tools, the same graph, and the same backbone.

7.4.1 No-Retrieval LLM (Parametric-Memory Control)

One call, the question, no facts, with an instruction to return an empty entity list if the answer is unknown. WebQSP and CWQ predate current models and are plausibly memorised, so this system measures the floor above which any retrieval result must be read. Its entity strings come from parametric memory rather than the graph, so surface forms mismatch gold more often than for graph-grounded systems; its score is consequently a *lower bound* on parametric knowledge. That asymmetry is reported rather than corrected, because grading one system leniently and another strictly would corrupt the comparison.

7.4.2 Vector-RAG over Verbalised Triples

Every triple is verbalised into a sentence, embedded with MiniLM [29], and indexed; the question retrieves the top 30 by exact inner product and one call answers from them — dense retrieval [28] applied to a graph. Its limitation is structural rather than incidental: a mediator-reified fact, a marriage or a film role, has no single triple expressing it, so no single retrieved sentence can express it either. Triples with a mediator endpoint are filtered at build time, since a verbalised half-edge is meaningless and leaks raw machine identifiers into answers; filtering the noise does not repair the paradigm, and both facts are reported.

7.4.3 Static GraphRAG

The structure-without-navigation comparison [4]: resolve the question's topic entities, expand a *one-logical-hop* neighbourhood under the same relation blocklist and the same mediator pass-through contract every other graph-touching system uses, rerank by embedding similarity to the question, and answer from the top 30 facts in one call. It shares Vector-RAG's answering prompt and cutoff exactly, so the difference between the two isolates retrieval strategy and nothing else.

The radius deserves a precise statement, because an earlier draft of this section and the implementation's own docstring both described it as two hops, and it is not. Expansion runs once from each topic entity. A mediator node is hopped through and its two relations concatenated, so a mediator path costs two graph edges while remaining one logical hop; an ordinary neighbour is

emitted and never expanded again. The consequence is checkable from the run records without re-reading the code: every fact this baseline emits is rendered as topic relation endpoint, so a genuine second hop would produce facts anchored at an intermediate entity. Across all 7,431 logged facts — the top ten of the thirty retrieved per question, which is what the run records keep — 7,431 begin with a topic entity. There are no exceptions, and the sample is the highest-ranked tenth of retrieval rather than an arbitrary slice, which is what a one-hop radius looks like in the data.

The radius is not the only thing that bounds this baseline; the fanout is, and it binds harder. Expansion retrieves at most 100 edges per topic entity and at most 20 per mediator hop, and the Cypher that fetches them carries no ORDER BY — so on any entity with more than 100 post-blocklist edges, the 100 retrieved are an arbitrary sample of the neighbourhood rather than its best or its first, and the embedding rerank then picks the top 30 facts out of that sample. This matters at scale rather than at the margin. This baseline seeds from the dataset's annotated topic entities and expands nothing else, so the population its cap acts on is exactly identifiable: the 1,032 topic mentions across the 800 test questions resolve to 711 distinct entities, every one of which the full system also expanded and measured. Their median post-blocklist degree is 131, the ninetieth percentile is 1,623, the largest is 136,742, and 55.1% of them carry more edges than the cap retains. Since a question carries 1.29 topic entities on average, that entity share does not give the share of questions affected, which is measured separately: on 72.5% of the 800 questions at least one topic entity is truncated, and on 59.0% every one of them is. For a hub topic the baseline therefore answers from a small fraction of the available neighbourhood — on the largest, well under one percent — chosen without a stated criterion.

Identifying that set costs nothing, because the resolution is deterministic. The baseline resolves each topic name through the same entity resolver the full system uses, and every one of the 711 names matches exactly one node at the resolver's exact-match tier (Section 4.6), so the two systems necessarily seed the same nodes and the tool log records which. The generator asserts that property rather than assuming it: if any topic name ever resolved ambiguously, the build would fail rather than quietly substitute a different population. The consequence for Section 8.5's diagnosis is direct and is stated there: the answer may not merely be outranked in the reranked window, it may never have entered the sample the reranker saw. Because the ordering is unspecified rather than seeded, this is also the one place in the harness where a re-run is not guaranteed to reproduce the same retrieved set, even though it did across the runs recorded here.

Two implementation details fixed on development data are worth naming, because both were comparability repairs rather than improvements: neighbourhood expansion initially passed *through* ordinary entities and stopped at mediator nodes, exactly inverted for Freebase and the reason machine identifiers were reaching answers; and retrieved facts are deduplicated by endpoint pair before reranking, because cosine similarity favours redundant phrasings of one

connection and was crowding the informative row out of the top 30. That dedupe has a cost, stated rather than hidden: two distinct facts sharing both endpoints collapse to one, observed but rare.

What this baseline licenses, and what it does not. A one-hop retriever cannot contain a two-hop chain, so its decay across hop strata (Section 8.5) is partly a property of the radius and not only of the paradigm. That distinction has to be drawn, because the alternative is to read an implementation limit as a result. The evidence that it is a radius effect is visible in the strata themselves: this baseline scores 0.44 on WebQSP's h2, where two-hop questions are largely mediator paths that a one-hop expansion *does* reach, against 0.16 on CWQ's h2, where the chains are genuine compositions it cannot reach at all. A retriever failing uniformly on depth would not show that gap.

The claim this thesis therefore makes on the strength of this baseline is the narrow one: *retrieval fixed before reasoning must commit to a radius in advance, and any fixed radius is wrong for some question in the set.* Widening it to two hops moves the boundary without removing it, and costs neighbourhood size at every step. The claim it does *not* make is that a two-hop static retriever would score as this one does — that experiment was not run, and the honest expectation is that it would recover a substantial part of the CWQ h2 gap while losing precision to a larger and noisier window. Re-running it is the first item in Section 10.3.8. Where the per-stratum argument in Section 8.5 needs a static comparator that is definitely not radius-limited, it uses Vector-RAG, whose one-triple context cannot represent a chain at any radius.

7.4.4 Think-on-Graph (Agentic Baseline)

The credibility-critical comparison. Think-on-Graph [7] is reimplemented on this environment's tools: beam width 3, depth 3, language-model relation pruning, language-model entity pruning, and a sufficiency check at each depth. It is a reimplementation, not the authors' code, and it runs on this graph with this backbone — which makes it a *stronger* comparison than citing published numbers, because the graph, the model, the budget, and the scoring rule are all held constant and the algorithm is what varies. Its numbers are therefore not comparable to the published ones and are not presented as such.

Holding the budget constant is not the same as making it irrelevant, and the distinction is load-bearing rather than pedantic. A fixed cap binds harder on a method whose cost grows multiplicatively, so the comparison measures search quality and affordability together. Section 8.4 separates them by splitting the results on whether the cap actually truncated the baseline, and the answer changes what the aggregate margin should be taken to mean.

One shared-budget consequence is reported as a finding rather than a defect. The cost of beam search is multiplicative in width and depth, so under the same 25-call cap the full system operates under, Think-on-Graph is clipped on $117/400$ WebQSP questions (29.2%) and $176/400$ CWQ

questions (44.0%). A clipped run still answers, from whatever it gathered, by the same graceful degradation the full system uses — symmetry that fairness requires.

One asymmetry is *not* symmetric, and it is disclosed here because it bears on how Section 8.4 should be read. Both systems call the same tools, but they truncate the returned candidate sets at different widths. Think-on-Graph keeps the first 40 relations per entity and the first 20 neighbours per relation, the pruning widths of the reimplemented algorithm; AGR keeps 300 and 200 (Table 5.3). Because get_relations returns rows sorted by *descending* fanout (Section 5.2), a binding cut discards the low-fanout tail — which is precisely the material Section 5.2 reports raising AGR's own cap from 80 to 300 to stop discarding. Think-on-Graph therefore runs at half the cap this thesis found harmful for itself. Measured over the committed tool logs, the 40-relation cut binds on 31.6% of the 1,651 entities it expanded and the 20-neighbour cut on 32.8% of its 7,992 neighbour calls, against 1 of AGR's 3,097 expanded entities and 3.3% of its neighbour calls. Beam width and depth follow the original paper at 3 and 3; the candidate widths behind them do not come from it and are a property of this reimplementation.

The consequence is a bound on one specific claim. Section 8.4 concludes that the margin over Think-on-Graph is affordability rather than search quality. That conclusion is safe in the direction it is stated — a narrower candidate set cannot explain why the baseline runs *out of calls* — but the residual gap on unclipped questions is measured against a baseline searching a thinner pool than AGR, so it is a lower bound on that baseline's attainable quality, not an estimate of it. Equalising the widths is the first item in Section 10.3.8.

7.4.5 Variables Held Constant Across All Systems

Every system shares the backbone and its sampling parameters, the knowledge environment and its relation blocklist, the entity resolver, the question files, the budget meter and its 25-call cap, the output schema so that scoring reads one field for all systems, and the record schema so that any record self-describes with backbone, budget hash, run configuration, and source commit. Both graph-navigating systems seed from the dataset's annotated topic entities, which removes entity linking as a confound and is stated as a limitation in Section 9.9.3.

Two things are *not* held constant, and both are named rather than buried. The first is deliberate: the baselines get a plain answering prompt while the full system's drafting prompt carries the anti-vacuity and grounding provisions of Section 6.2, which are part of the system under test rather than of the shared harness. The second is a property of the algorithms and is where a reader should look first for a confound — the width of the candidate set each system considers per step, 40 relations and 20 neighbours for Think-on-Graph against 300 and 200 for AGR, with the measured binding rates and the bound they place on Section 8.4 given in Section 7.4.4. The static graph baseline's own fanout cap is stated in Section 7.4.3.

7.4.6 Baseline Certification Protocol

Baselines were debugged exclusively on development data and certified before any test question ran. The pre-registered diagnostic targets the one failure mode that would be an implementation defect rather than a paradigm limit: *retrieval found the gold answer and the system hedged anyway*, checkable because both retrieval baselines log a sample of what they retrieved. The rule, written before the numbers: under 15% of hedges certifies the baseline as-is; over 30% means an over-abstinent prompt, to be adjusted identically for both baselines with both files re-run.

Measured: Vector-RAG 0% on both datasets, Static GraphRAG 10% and 14%. Both certified, with two caveats recorded rather than smoothed over. The diagnostic samples the top ten of thirty retrieved facts, so it is a lower bound; and Think-on-Graph's 0% is *vacuous* rather than clean, since it logs no retrieved-fact sample and the diagnostic therefore compared against an empty string. As a multi-step navigator it was not the diagnostic's target, but the reading is a null, not a pass. All ten test files completed at exactly 400 records with zero failures and zero duplicate question identifiers.

7.5 Metrics

7.5.1 Answer Accuracy: Hits@1 and F1

Both metrics are computed over *entity sets* after Unicode NFKC normalisation, case folding, and whitespace stripping — no fuzzy matching, no substring containment. *Hits@1* is 1 if any predicted entity matches any gold entity, which is the convention in this literature and is what makes published numbers comparable at all. *F1* is computed per question between the predicted and gold sets as defined in Section 2.2, and averaged over questions.

Reporting both is not redundancy. Hits@1 rewards a system that names one correct entity among several wrong ones. On WebQSP the loophole is wide, though not on the typical question: the median question has just 1.5 gold answers, so half the set admits only one right entity. It is the tail that opens the loophole — the mean is 7.98 and the largest question lists 382 — and a system is free to fish in it. The precision component of F1 is what closes it, and Section 8.2 shows that the gap between the two metrics is exactly where the verification layer's contribution becomes visible.

Strict matching raises an obvious worry — that "St. Petersburg" versus "Saint Petersburg" is manufacturing failures. It was measured rather than assumed. Of the full system's wrong answers, 1 of 65 (2%) on WebQSP and 6 of 99 (6%) on CWQ are surface-form near-misses under aggressive normalisation. Strict matching does not materially understate performance, and the failure census of Chapter 9 is therefore reading real errors.

7.5.2 Accounting for Hedges and Abstentions

A *hedge* is an empty predicted entity set: the system declined to assert. Hedges score zero on both Hits@1 and F1 — an abstention earns nothing on an accuracy metric — and hedge rate is reported as its own column so the abstention–accuracy trade is visible rather than buried. A system can buy precision with abstention, and the reader is given both numbers to see it happen.

Hedges are not epistemically uniform, and Chapter 9 keeps them separate from wrong answers throughout for that reason: a wrong answer is a reasoning error, whereas a hedge is usually a coverage gap that never produced a committal claim.

One counting convention follows from this and is stated here once, because it governs every hits/hedges/wrong triple in the thesis. *The three counts always partition the whole set.* Nothing is dropped from the denominator — in particular, questions the environment provably cannot answer (Section 4.7.4) are counted as hedges rather than excluded, even though the system had no route to a correct answer. Excluding them would flatter every condition equally and make the counts incomparable with the accuracy figures, which do include them. Where the unanswerable questions matter to an argument they are named, not netted out.

7.5.3 Two-Tier Groundedness and Hallucination Rate

Groundedness is measured at two levels, and the word *tier* in this thesis refers to this metric and nothing else.

Tier 1, structural, is deterministic, free, applies to every asserted entity in all $4{,}000$ records, and operationalises Section 2.3.3 directly: *an asserted entity is graph-grounded if and only if it exists as a node in the knowledge environment and is connected to at least one of the question's topic entities within four hops.* The entity-level ungrounded rate is the fraction of asserted entities failing this test; the question-level rate is the fraction of answered questions containing at least one such entity. The test is deliberately strict about surface form — a parametric spelling that does not exist as a node is ungrounded *by definition of the environment*, which is the point of defining hallucination relative to an environment at all.

Tier 2, semantic, asks the harder question Tier 1 cannot: is the entity supported *as the answer*? For a stratified sample of 60 asserted answers per system per dataset (600 judgements), the graph's own shortest connecting paths between topic and asserted entity are retrieved and a language model judges whether any one of them expresses the relationship the question asks about. The governing instruction is that *mere connectedness is not support*, and that sentence is what separates the two tiers.

Two design choices are pre-registered and consequential. Judging is against **the environment**, not against each system's own retrieved context: the logs do not preserve per-system evidence uniformly, and environment-judging asks every system the identical question, so the metric

applies even to the no-retrieval control. And path retrieval takes the three shortest paths, which means a supporting chain can lose to shorter irrelevant ones. Tier 2 rates are therefore *conservative lower bounds*, uniformly depressed across all systems, and are read as a between-system comparison rather than as absolute support rates.

7.5.4 Efficiency: Tokens, LLM Calls, and Wall-Clock Latency

Tokens and calls are counted by the shared budget meter, identically for every system, and are exact — the cache replays original usage, so they are unaffected by which runs were warm. Wall-clock latency is not: it is computed over **cold-cache records only**, and that restriction is repeated wherever latency appears. Where a condition's records are entirely cache replays the latency cell is reported as unmeasured rather than as a number, which happens for one ablation condition in Section 8.8.

7.6 The Groundedness Judge and Its Validation

7.6.1 Independence of the Judge from the Answering System

The Tier-2 judge sees a question, one asserted entity, and up to three connecting paths rendered as relation chains with mediator nodes masked. It does not see which system produced the assertion, whether the assertion was scored a hit, the gold answer, or the system's own reasoning. Its prompt was fixed before any per-system result was inspected. It runs on the same backbone as the systems it judges, which is a limitation stated in Section 9.9.3 rather than a claim of independence from the model family.

7.6.2 Human Annotation Protocol

One hundred judged items were drawn and written to a sheet stripped of the judge's verdicts, and were labelled by the author *blind* — neither the judge's output file nor the answer key was opened until the sheet was saved. Blinding is what makes the resulting statistic a validity check rather than a measure of anchoring.

A written annotation guideline was fixed during calibration and is reproduced in full in Section D.1. Each of its eight rules was forced by a case: judge the asked relationship *under the graph's vocabulary*; treat a mediator-bridged pair of edges as one relation; label supported if *any* single path supports; judge relations by *role, not type*, so a normally-generic relation counts when it is itself what the question asks; for conjunctions require every asked relationship covered by some specific path, jointly rather than in one chain; judge each entity independently of answer-set completeness, which is recall and is measured elsewhere; ignore superlative and temporal filters

the graph cannot express, judging only the base relation; and treat a path expressing only a question's identifying descriptor, rather than its actual ask, as unsupported.

That last rule is the *composition-echo* case, and it is the same phenomenon the failure census later finds at scale (Section 9.7) — the annotation task and the error taxonomy converging on one mechanism from opposite directions.

7.6.3 Agreement Between Judge and Human Annotator

On $n = 100$ items, observed agreement is 85%, with near-symmetric marginals (54 human-supported against 51 judge-supported) and near-symmetric disagreement (9 human-yes/judge-no against 6 human-no/judge-yes), so neither rater is systematically the more lenient. Cohen's κ [38] is

$$\kappa = \frac{0.8500 - 0.5008}{1 - 0.5008} = 0.6995.$$

The pre-registered threshold was $\kappa \geq 0.7$. Rounded to three decimals this value reads 0.700 and appears to clear; it does not. It falls short by 0.0005, and it is reported as short.

What follows from that is a matter of weight, not of discarding. By the conventional bands [39] a κ of 0.70 is substantial agreement, and the task is genuinely hard — whether a relation chain really expresses an asked relationship has real grey area for a careful human, as the guideline's list of decided edge cases attests. So the Tier-2 numbers are reported, with the κ beside them and the miss stated, and no conclusion in this thesis rests on Tier 2 alone. Where Tier 2 is load-bearing — the between-system semantic-support comparison of Section 8.6 — the finding is also supported by the precision column of the main matrix, which is measured without a judge. The alternative course, tightening the judge prompt and re-judging a fresh sample until the bar was cleared, was available and pre-committed; it was not taken, because the honest number was already in hand and re-running until a pre-registered threshold is met is exactly what pre-registration exists to prevent.

7.7 Gold-Answer Quality Control

7.7.1 The Gold-Noise Problem in WebQSP and CWQ

Both benchmarks carry label defects, and CWQ carries more because its questions and answers are generated by SPARQL templates over the knowledge base rather than written by hand. A question whose gold answer is wrong is not a system failure, and counting it as one corrupts both the accuracy tables and the error taxonomy. Some detection procedure was therefore necessary — but a naive one is worse than none, as the results below show.

7.7.2 Consensus Pre-Pass and Per-Item Adjudication

The detection principle is **independent multi-system consensus against gold**: when at least three of the five systems assert the same non-gold answer, the label is worth examining. Independence is what makes the signal meaningful, so the voters are the five *systems* of the main matrix, spanning two knowledge sources (pretraining and this graph) and four retrieval paradigms. The four ablation conditions would not do: they share the full system's scorer, drafter, and traversal, so their agreement would describe an architecture rather than a label.

The pass emits one row per (question, consensus answer) pair, so a question with three different non-gold consensus answers contributes three rows. Rows and questions are therefore reported as separate units throughout, and question counts are always the deduplicated ones.

Adjudication assigns each question one of exactly three verdicts — gold_ok, gold_wrong, ambiguous_question — with a free-text subtype for grouping that drives nothing. Automatic triage cleared 15 of 89 rows on WebQSP and 2 of 59 on CWQ; the remainder were adjudicated by hand against the question text and, where a world fact was at issue, an external check.

7.7.3 Two Evidence Signatures of Label Error

The most useful methodological result of this pass is that label errors come in two kinds with *opposite* evidence signatures, and that separating them is what makes the adjudication reproducible rather than ad hoc.

A **world-fact error** — gold contradicts reality — requires *mixed* consensus: the parametric baseline and the graph-based systems agreeing against the label means two different knowledge sources agree. The San Antonio city council question is the worked specimen: four systems converge on Bexar County against a Comal County gold, with mixed evidence, and Section 9.8.2 traces where that label came from.

An **annotation-pipeline error** — gold contradicts the very graph the labels were generated from — has the opposite signature: *graph-only* consensus, where every graph-reading system finds the same right answer while the parametric baseline misses it for unrelated reasons. The specimen is WebQTest-55: "Who had a concert named Crazy Love Tour?" asks for a person, the graph-reading systems all return Michael Bublé, and gold lists his professions — the annotation query having traversed one relation too far.

This distinction was not designed in advance; it was forced. The adjudication scope had initially been narrowed to mixed-evidence flags only, on the reasoning that graph-only agreement is weaker evidence. The Bublé case is graph-only and is a genuine label error, so the scope was widened to cover every flag whose consensus answer was not simply an echo of the question's own topic. The narrower scope would have missed an entire class.

Table 7.2: Gold-label adjudication. Rows and distinct questions are separate units because the pass emits one row per consensus answer. Exclusions are the union of gold_wrong and ambiguous_question questions.

Dataset	Rows	Questions	gold_wrong	ambiguous	Excluded
WebQSP	89	58	12	10	22 (5.5%)
CWQ	59	47	17	2	19 (4.8%)

7.7.4 Census-Based Exclusions and the Dual-Reporting Policy

Two numbers must not be conflated, and Table 7.2 separates them: 58 and 47 questions were *flagged* by the consensus pass, while 22 and 19 were *adjudicated* as defective. The gap is not noise — most flagged questions are all five systems converging on the same wrong answer, which is a finding about the systems rather than the labels and is treated as one in Section 9.7. A policy of rescoring on consensus alone would have corrupted the results with exactly those cases.

The excluded questions are removed from the failure census of Chapter 9 — they are benchmark artifacts, not system failures — and ambiguous_question is excluded on the same logic, an underdetermined question being no more a system failure than a broken label. Of the excluded questions, 12 of 22 (WebQSP) and 15 of 19 (CWQ) were failures of the full system; those are the ones that shrink the census pool.

The accuracy tables are not rescored. They are reported exactly as measured, with this section as the footnote, for two reasons. Rescoring on self-adjudicated labels would substitute one contestable ground truth for another. And the errors cut both ways: the remaining 10 and 4 excluded questions are ones the full system scored as *hits* against labels now judged broken or ambiguous. That symmetry is what makes the footnote credible rather than defensive, and it means the direction of the net bias is unknown, not conveniently favourable.

Two scope limits belong on the record. The rule that automatically assigned gold_wrong to questions with implausibly long gold lists is a heuristic, not a proof, and its outputs were spot-checked rather than individually verified. And the manual adjudication covered every mixed-evidence flag plus every graph-only flag that was not a topic echo — broad, but not the complete flag population.

7.8 Ablation Conditions

Four conditions, each a single deviation from the frozen configuration, each isolating one mechanism:

- **no-planner** — decomposition disabled; the whole question becomes one objective. Anchor seeding is untouched, so this varies decomposition and nothing else.

- **no-backtrack** — max_backtracks $= 0$; a configuration change, which correctly alters the budget hash stamped on the records.
- **no-verifier** — verify_claims $=$ false; grounded drafting with no claim extraction or checking.
- **embedding-only** — $\alpha = 1.0$, removing the language-model term from the hybrid scorer while leaving τ at its frozen value.

The reference row is the *unmodified* system restricted to the same questions. It costs nothing: the main-matrix records are filtered to the half-set question identifiers rather than re-run, which also guarantees the pairing that the significance tests require. Comparing against the full 400-question score would be invalid, since accuracy on 400 questions and on 198 are different numbers for the same system and McNemar's test needs matched pairs.

Ablations run on stratified halves — 200 WebQSP and 198 CWQ — rather than the full samples. This was the pre-registered response to the benchmark's actual burn coming in above projection, and it was the ablation *sample* that was cut rather than any system or condition. The odd 198 is floor division within each stratum: CWQ's strata of $137/211/49/3$ halve to $68/105/24/1$. Nothing in the analysis assumes a round number, and the consequence — reduced statistical power — is confronted directly in Section 8.8 rather than left implicit.

7.9 Hyperparameter Selection on the Development Set

7.9.1 Development Set Construction and Coverage

The development set is 80 questions mined from the train and validation splits of both benchmarks, selected to span hop counts and to include mediator-heavy and constraint-bearing questions, and certified for answer reachability by the same procedure applied to the test samples. It is disjoint from both test samples by construction, and the disjointness is asserted by the test-set builder rather than assumed.

7.9.2 The α–τ Sweep Protocol

The two free parameters of the navigation loop are the blend weight α in the hybrid scorer (Section 5.5.2) and the low-signal backtracking threshold τ (Section 5.6.3). They were swept jointly over $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$ and $\tau \in \{0.0, 0.20\}$ — eight conditions, each a complete run of all 80 questions, with two additional draft-only conditions to price the verification layer. Each condition is a constructor argument rather than a source edit, and its values are stamped into every record it writes, so no condition can be mislabelled after the fact.

The chosen values, $\alpha = 0.7$ and $\tau = 0.20$, were frozen on the basis of that sweep and never revisited. Section 8.1 reports the sweep, including the non-monotonicity at $\alpha = 0.5$ that a smoothed curve would have hidden.

7.10 Implementation, Environment, and Reproducibility

The system is implemented in Python against LangGraph for the state machine, the Neo4j Python driver for graph access [21], and sentence-transformers for embeddings [29]. The graph runs in a local Neo4j instance; all Cypher [22] is templated inside the tool layer and never composed by a language model (Section 5.2). Randomised operations use seed 42 throughout.

Three practices carry more reproducibility weight than the dependency list, and Chapter E records them as techniques. *Records self-describe*: each carries the backbone description, a budget-configuration hash, the full run configuration including the system name, and the source commit, so any record found anywhere states what produced it — which is what made the cache-replay incident of Section 7.3 diagnosable. *Frozen artifacts are committed and regenerable ones are not*: the certified question sets, their half-splits, and every result file embody spent money and non-repeatable runs, while the graph import, embedding matrices, and triple index are deterministic products of committed scripts. *Code changes are proved harmless by cache identity* rather than by a passing test suite: re-running an answered question and confirming every call is a cache hit proves byte-identical prompts, hence an untouched frozen path, and is the certificate behind the ablation reference row.

Two limits belong here rather than in a footnote. The archived results carry neither per-folder documentation nor generated manifests, so file provenance rests on the self-describing records and on this chapter. And one generated summary in the development-set archive predates a fix applied to the runs it summarises, so rescoring that directory yields different numbers than the committed summary; the discrepancy is stated where those numbers are used (Section 8.1) rather than resolved by editing a generated file.

Chapter 8

Results

This chapter reports the measurements the previous chapter specified. It is organised so that the weakest claims are visible next to the strongest: the headline accuracy result is large and significant on every comparison, the structural groundedness result is emphatic but attributes to a different cause than expected, the semantic groundedness result is underpowered, and three of the four ablations detect nothing. All four outcomes are reported at the same volume.

Every number comes from results/phase4/thesis_numbers.json. Confidence intervals are 95% bootstrap intervals over questions with $10{,}000$ resamples at seed 42 [36]; paired comparisons use the exact-binomial form of McNemar's test [37] on per-question correctness.

8.1 Development-Set Tuning Outcomes

The blend weight α and the low-signal backtracking threshold τ were swept jointly over the 80-question development set (Section 7.9). Table 8.1 reports all eight conditions plus the two draft-only conditions that price the verification layer.

Three readings, of which only the first is the one the sweep was run to obtain.

$\alpha = 0.7$ is the maximum, and the curve is not clean. Hits and F1 rise from $\alpha = 0.3$ to $\alpha = 0.7$ and fall sharply at $\alpha = 1.0$, which is the expected shape: embedding similarity alone ($\alpha = 1.0$) is a weaker guide than the blend, and weighting the language-model judgement too heavily ($\alpha = 0.3$) makes the score noisy. But the wrong-answer column does not follow the hits column. It reads $7, 8, 5, 6$ across the four α values, peaking at $\alpha = 0.5$ — the condition that also has the fewest hedges of the verified rows. At $\alpha = 0.5$ the system commits more often and is wrong more often, and that irregularity is reported rather than smoothed away, because a presented curve that hides a known kink is exactly what a defence finds. It is one sample of 80 questions and the difference is three questions wide; it is noted, not interpreted.

τ does almost nothing on this set, and that is consistent with its design. At $\alpha = 0.7$ and $\alpha = 0.5$ the two τ values produce identical hits, hedges, and wrong counts. At $\alpha = 0.3$ they

Table 8.1: The development-set sweep, $n = 80$. Hits, hedges, and wrong answers partition the set. Verify indicates claim verification enabled; the two draft-only rows disable it. All rows are from the sweep as run, before the answer-entity filter of Section 6.5 was corrected; the frozen condition's post-correction figures are given in the text.

α	τ	Verify	Hits	Hedge	Wrong	F1	Tokens	Calls	Bktrk
0.3	0.00	yes	61	12	7	0.728	4,999	7.4	16
0.3	0.20	yes	61	12	7	0.728	5,030	7.5	18
0.5	0.00	yes	62	10	8	0.743	4,874	7.2	16
0.5	0.20	yes	62	10	8	0.743	4,874	7.2	16
0.7	0.00	yes	64	11	5	0.769	4,925	7.3	17
0.7	**0.20**	**yes**	**64**	11	5	0.769	4,925	7.3	17
1.0	0.00	yes	59	15	6	0.713	3,408	4.9	22
1.0	0.20	yes	59	16	5	0.713	3,400	4.9	26
0.5	0.20	no	64	9	7	0.770	4,807	7.1	16
0.7	0.20	no	66	10	4	0.797	4,858	7.2	17

differ only in backtrack count (16 against 18); at $\alpha = 1.0$, one wrong answer becomes a hedge and backtracks rise from 22 to 26. This is what the signal-maximum formulation of Section 5.6.3 predicts: the trigger fires only when the blended score, the embedding maximum, *and* the language-model maximum are all below threshold, which is rare when the language-model term is contributing. The $\alpha = 1.0$ row is where it has the most to do, and it is exactly there that its effect is visible. The value $\tau = 0.20$ was frozen on the basis of a sweep in which it changed almost nothing; that is stated plainly rather than presented as a tuned choice.

Verification costs accuracy on this set. Against the frozen condition, the draft-only row scores 66 hits to 64 and F1 0.797 to 0.769. After the answer-entity filter of Section 6.5 was corrected — a change made on development data before any test question ran — the frozen condition scores 65 hits, 11 hedges, 4 wrong, F1 0.785. So on the final code the layer costs one correct answer and removes no wrong ones on this set of 80. Section 8.8 finds the same thing at $n \approx 200$ on test data, and Section 8.9 explains why that is the expected result rather than a disappointing one.

One reproducibility note belongs with this table. Only the frozen condition was re-run after the filter correction, so Table 8.1 is the pre-correction sweep throughout — internally consistent, since all eight conditions ran under the same code. The committed summary file for that directory also predates the correction, so rescoring the archived runs yields $65/11/4$ for the frozen condition where the summary records $64/11/5$. The archived per-question records are authoritative.

8.2 Main Results on WebQSP

On WebQSP, AGR answers 302 of 400 questions correctly: 75.5% against a reachability ceiling of 97.0%, +9.5 points over Think-on-Graph and +30.2 over the parametric floor. The floor

Table 8.2: Main results, $n = 400$ per dataset. Brackets are 95% bootstrap intervals. **P** and **R** are mean per-question precision and recall; **Hedge** is the share of questions on which the system asserted nothing. The Hits@1 ceilings are 97.0% and 99.2% (Table 7.1).

Data	System	Hits@1	F1	P	R	Hedge
WebQSP	No-retrieval	0.453 [0.403, 0.500]	0.305 [0.266, 0.345]	0.380	0.303	12.2%
WebQSP	Vector-RAG	0.568 [0.517, 0.618]	0.432 [0.389, 0.475]	0.469	0.460	27.0%
WebQSP	GraphRAG	0.340 [0.292, 0.388]	0.262 [0.224, 0.302]	0.279	0.284	55.8%
WebQSP	Think-on-Graph	0.660 [0.613, 0.705]	0.477 [0.436, 0.518]	0.571	0.480	18.8%
WebQSP	**AGR**	**0.755** [0.713, 0.797]	**0.642** [0.600, 0.684]	**0.697**	**0.653**	**8.2%**
CWQ	No-retrieval	0.307 [0.263, 0.352]	0.279 [0.236, 0.323]	0.285	0.282	38.0%
CWQ	Vector-RAG	0.203 [0.163, 0.242]	0.168 [0.133, 0.204]	0.171	0.189	48.2%
CWQ	GraphRAG	0.205 [0.168, 0.245]	0.183 [0.147, 0.222]	0.187	0.194	60.8%
CWQ	Think-on-Graph	0.435 [0.388, 0.482]	0.378 [0.332, 0.423]	0.403	0.384	22.5%
CWQ	**AGR**	**0.522** [0.475, 0.570]	**0.469** [0.423, 0.514]	**0.484**	**0.481**	23.0%

matters. WebQSP is a decade old and plausibly in the backbone's pretraining data, and the no-retrieval control confirms it: 45.3% of these questions are answerable from memory alone. Any claim about what retrieval contributes has to be read against 0.453, not against zero.

F1 widens the margin that Hits@1 understates. Against Think-on-Graph the Hits@1 gap is +9.5 points; the F1 gap is +16.5 (0.642 against 0.477), and the precision column shows where it comes from (0.697 against 0.571). WebQSP's questions have a mean of eight gold answers, so a system that asserts a long list collects Hits@1 credit for the one entity that matches and pays nothing for the rest. Precision is what charges for it. The two columns together support the sentence this thesis leads with: AGR answers more questions and asserts fewer falsehoods, at an 8.2% hedge rate — the lowest of the five systems, so the precision is not bought with abstention.

Raw hits mislead on the retrieval baselines, and the correction is a finding. GraphRAG scores 0.340, below the no-retrieval control's 0.453, which invites the conclusion that the implementation is broken. Put the wrong-answer counts beside the hits and the picture inverts. No-retrieval asserts on 351 of 400 questions and is wrong on 170 of them — 51.6% assertion precision. GraphRAG asserts on 177 and is wrong on 41 — 76.8%. The mechanism is the prompt: the retrieval baselines are instructed to answer only from the facts given and to return nothing otherwise, so when retrieval misses they abstain, while the no-retrieval control has no such constraint and guesses from memory. On a memorised benchmark, guessing pays. *A grounded system scoring below an ungrounded one on raw hits while committing four times fewer falsehoods* is the contamination story the control was built to expose, and it is reported as a result rather than as an anomaly.

Table 8.3: Paired McNemar tests, AGR against each baseline, on per-question correctness over the same 400 questions. Every comparison rejects.

Dataset	Baseline	AGR only	Baseline only	p
WebQSP	No-retrieval	152	31	2.3×10^{-20}
WebQSP	Vector-RAG	117	42	2.2×10^{-9}
WebQSP	GraphRAG	186	20	6.5×10^{-35}
WebQSP	Think-on-Graph	76	38	4.8×10^{-4}
CWQ	No-retrieval	121	35	2.7×10^{-12}
CWQ	Vector-RAG	151	23	2.8×10^{-24}
CWQ	GraphRAG	148	21	1.0×10^{-24}
CWQ	Think-on-Graph	86	51	3.5×10^{-3}

8.3 Main Results on ComplexWebQuestions

On CWQ, AGR answers 209 of 400: 52.2% against a 99.2% ceiling, +8.7 over Think-on-Graph and +21.5 over the floor. Every system scores lower here, and the ordering changes in one important way: *both* static retrieval baselines fall below the parametric control. Vector-RAG drops from 0.568 on WebQSP to 0.203, GraphRAG from 0.340 to 0.205, while no-retrieval falls only from 0.453 to 0.307.

The reason is structural rather than incidental, and the two baselines earn that description to different degrees. Vector-RAG's context is a single verbalised triple, and one triple cannot contain a chain at any retrieval quality or any radius; that is a paradigm limit Section 7.4 predicted at implementation time and Section 8.5 demonstrates per stratum. GraphRAG's context is a one-logical-hop neighbourhood (Section 7.4.3), so its CWQ number is the weaker evidence of the two: it confounds the paradigm with the radius, and part of the fall is attributable to a boundary chosen in advance rather than to static retrieval as such. The claim rests on the first baseline.

The number worth leading with on this dataset is assertion precision. AGR and Think-on-Graph commit at effectively the same rate — 308 and 310 of 400 questions answered rather than hedged — and out of those near-identical commitments AGR is right 209 times and wrong 99, against 174 and 136. That is 67.9% assertion precision against 56.1%: on the harder, deeper benchmark, the system with an explicit plan, a guided scorer, and a claim check finds 35 more answers and asserts 37 fewer falsehoods than beam search on the same graph with the same model, from the same number of commitments — a margin Section 8.4 attributes to the budget rather than to resolution quality.

Table 8.3 settles the comparisons that matter. All eight reject, including the two against the agentic baseline — the only comparison in the table where both systems navigate the same graph, with the same model, under the same call budget. The discordant counts make the margin concrete rather than distributional: on CWQ, AGR uniquely solves 86 questions Think-on-Graph misses against 51 the other way. What that budget does to the comparison is the subject of the

Table 8.4: The comparison against the agentic baseline, split on whether the shared 25-call cap truncated it. Clipping is read from the per-record budget_exhausted trace flag rather than from a call count, since a run may spend its last permitted call on the step that finishes it. The *combined* rows reproduce the Hits@1 column of Table 8.2, recomputed independently from the raw records. Generated from results/phase4/thesis_numbers.json.

Dataset	Subset	n	ToG	AGR	Δ
WebQSP	ToG ran to completion	283	0.852	0.788	−0.064
	ToG clipped by the cap	117	0.197	0.675	+0.479
	combined	400	*0.660*	*0.755*	**+0.095**
CWQ	ToG ran to completion	224	0.629	0.607	−0.022
	ToG clipped by the cap	176	0.188	0.415	+0.227
	combined	400	*0.435*	*0.522*	**+0.087**

next section, and it changes what the margin means.

8.4 Where the Margin Over the Agentic Baseline Comes From

The shared budget is a control, and like any control it is doing something. The call cap binds on Think-on-Graph far more often than on AGR — beam search costs width times depth, and Section 8.7 records the clip rates — so "same budget" does not mean "same freedom to search". Splitting the 400 questions of each dataset on whether the cap actually cut Think-on-Graph off separates the two things the aggregate confounds: how good its search is, and whether it can afford to run that search to completion.

On the questions it is allowed to finish, Think-on-Graph is ahead on both datasets. It scores 0.852 against AGR's 0.788 on WebQSP and 0.629 against 0.607 on CWQ. The entire aggregate margin comes from the 29.2% and 44.0% of questions where the cap truncates it, and there its accuracy falls to 0.197 and 0.188 while AGR's holds at 0.675 and 0.415. This is stated plainly because it is the first thing an unsympathetic reading of Table 8.3 should look for, and because burying it would misrepresent what was measured.

It does not weaken the result, but it does change its content. The finding is not that guided best-first search resolves entities better than language-model-pruned beam search — on the evidence here it does not, and the unclipped subset is the fairest available estimate of that. The finding is that *beam search cannot afford to finish under a budget a deployed system would actually impose, and guided search can*. Width times depth is multiplicative and paid on every question; a plan plus a scored frontier is not. A comparison run without a cap would measure search quality in the abstract, which is a legitimate experiment and a different one; this thesis fixed the budget in advance (Section 7.4) because cost is a property of the method rather than a

Table 8.5: Hits@1 and F1 per hop stratum. Stratum sizes are those of Table 7.1. WebQSP's h3plus has $n = 4$ and is not interpreted.

Data	System	h1	h2	h3plus	unreach.
WebQSP	No-retrieval	0.43 / 0.29	0.52 / 0.36	0.25 / 0.00	0.17 / 0.17
WebQSP	Vector-RAG	0.82 / 0.63	0.12 / 0.07	0.00 / 0.00	0.08 / 0.08
WebQSP	GraphRAG	0.30 / 0.23	0.44 / 0.35	0.25 / 0.07	0.08 / 0.08
WebQSP	Think-on-Graph	0.63 / 0.45	0.78 / 0.58	0.25 / 0.07	0.17 / 0.17
WebQSP	**AGR**	0.75 / 0.64	**0.84 / 0.73**	0.00 / 0.00	0.08 / 0.08
CWQ	No-retrieval	0.38 / 0.35	0.30 / 0.27	0.14 / 0.14	0.00 / 0.00
CWQ	Vector-RAG	0.33 / 0.27	0.14 / 0.11	0.10 / 0.09	0.67 / 0.67
CWQ	GraphRAG	0.34 / 0.30	0.16 / 0.14	0.02 / 0.02	0.33 / 0.33
CWQ	Think-on-Graph	**0.53** / 0.48	0.37 / 0.32	0.45 / 0.32	0.33 / 0.33
CWQ	**AGR**	0.46 / 0.42	**0.55 / 0.49**	**0.57 / 0.50**	0.33 / 0.33

nuisance to be controlled away. On that reading the clip rate is not an artefact of the setup, it is the result — and Section 8.7 prices the same finding in tokens.

Three qualifications keep the claim honest. The cap of 25 calls is itself a choice, and a larger one would move the split; the clip rates are reported so a reader can see how far. The reimplementation's beam width and depth follow the original paper, so its cost profile is the published algorithm's rather than a weakened variant — which is what makes the cost finding meaningful instead of circular. But the candidate widths *beneath* the beam do not come from the paper, and they are narrower than AGR's by a factor of seven and ten (Section 7.4.4): the baseline prunes from 40 relations and 20 neighbours where AGR prunes from 300 and 200, and those cuts bind on about a third of its expansions against almost none of AGR's. That asymmetry cannot explain the clipping — a thinner candidate set makes a step cheaper, not dearer — so the affordability finding stands. It does bear on the sentence before it: 0.852 and 0.629 on the unclipped subset are a *lower bound* on what this baseline's search could resolve given AGR's candidate widths, not an estimate of it, and the claim that guided search does not out-resolve beam search is correspondingly the weaker one.

The breadth of AGR's answers is checked rather than assumed. On answered WebQSP questions AGR names 3.37 entities against Think-on-Graph's 2.56, and a metric scoring a hit on any match rewards naming more. The check is precision over those same answered questions, where AGR leads 0.760 to 0.703; on CWQ the figures are 1.54 against 1.40 entities and 0.629 against 0.520 precision. Breadth bought without accuracy would show up as precision falling while the entity count rose, and it does not. This is the same reason F1 is reported beside Hits@1 throughout (Section 7.5).

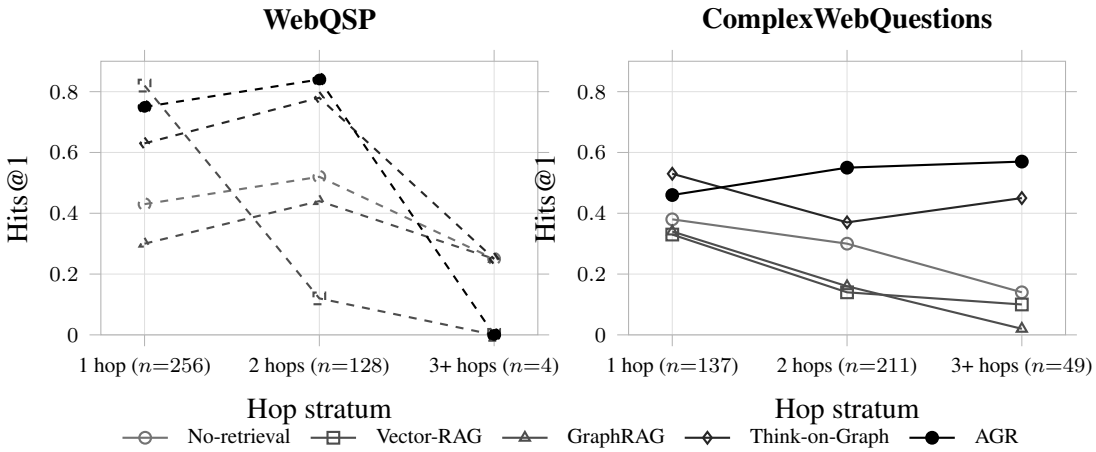

Figure 8.1: Hits@1 across hop strata. The shape is the finding: on ComplexWebQuestions AGR is the only system whose accuracy rises with hop count, while every other profile decays. WebQSP is drawn dashed because its h3plus stratum holds four questions and cannot carry a trend; the stratum sizes are printed on the axis for that reason. The unreachable stratum is omitted here and discussed below, since by construction it measures assertion rather than accuracy. Generated from results/phase4/thesis_numbers.json.

8.5 Accuracy Broken Down by Hop Count

Table 8.5 is the direct answer to RQ1, and it answers it with a shape rather than with a difference of means — Figure 8.1 is that shape.

On CWQ, AGR's accuracy rises with hop count. Hits@1 goes $0.46 \rightarrow 0.55 \rightarrow 0.57$ across h1, h2, h3plus, and F1 goes $0.42 \rightarrow 0.49 \rightarrow 0.50$. Every other system on that dataset decays: Think-on-Graph falls from 0.53 to 0.37 before partially recovering at 0.45; Vector-RAG collapses $0.33 \rightarrow 0.14 \rightarrow 0.10$; GraphRAG $0.34 \rightarrow 0.16 \rightarrow 0.02$, though that last row is radius-limited and is read with the qualification given below. A system that gets *better* as the required chain gets longer is not merely ahead on average — it is the only one in the table whose profile is consistent with actually traversing the chain, and that monotonicity is a stronger form of the multi-hop claim than any aggregate could carry.

Think-on-Graph beats AGR on CWQ's one-hop stratum (0.53 against 0.46, $n = 137$), and this is reported rather than buried because it makes the rest of the story more credible, not less. Section 8.8 identifies the mechanism: decomposition costs accuracy on questions that do not need decomposing, and CWQ's h1 stratum is full of them.

Vector-RAG's WebQSP profile is the paradigm in two cells. It scores 0.82 at one hop — the best single-hop number anywhere in the table, better than AGR's 0.75 — and 0.12 at two. When one triple suffices, dense retrieval over verbalised triples is excellent; when a chain is required, it has nothing to offer. This is the cell the multi-hop argument rests on, because a one-triple context is chain-incapable by construction rather than by configuration.

Table 8.6: Two-tier groundedness. Tier 1 is deterministic over every asserted entity in all 4,000 records; Tier 2 is a language-model entailment judgement over a stratified sample of 60 assertions per system per dataset, validated at $\kappa = 0.6995$ (Section 7.6).

		Tier 1: structural			Tier 2
Data	**System**	**Entities**	**Ungrounded**	**Questions**	**Supported**
WebQSP	No-retrieval	661	179 (27.1%)	37.6%	63.3%
WebQSP	Vector-RAG	1040	38 (3.7%)	6.2%	61.7%
WebQSP	GraphRAG	800	6 (0.8%)	2.8%	55.0%
WebQSP	Think-on-Graph	832	**0 (0.0%)**	**0.0%**	61.7%
WebQSP	AGR	1235	**0 (0.0%)**	**0.0%**	**66.7%**
CWQ	No-retrieval	340	42 (12.4%)	13.7%	36.7%
CWQ	Vector-RAG	289	7 (2.4%)	3.4%	**50.0%**
CWQ	GraphRAG	224	2 (0.9%)	1.3%	36.7%
CWQ	Think-on-Graph	433	**0 (0.0%)**	**0.0%**	43.3%
CWQ	AGR	474	**0 (0.0%)**	**0.0%**	48.3%

GraphRAG's WebQSP inversion has two causes, and only one of them was originally named. It scores 0.30 at h1 against 0.44 at h2. Part of that is hub noise: popular one-hop entities have large neighbourhoods, and the answer is drowned in the reranked window — or, more exactly, drowned *or absent*, since the retrieval step samples at most 100 edges per topic entity without an ordering, and 55.1% of its topic entities carry more edges than that (Section 7.4.3). Which of the two applies to any given question is not recoverable from the run records. But the inversion also marks the edge of the retrieval radius (Section 7.4.3). WebQSP's h2 stratum is largely mediator paths, which a one-hop expansion reaches by hopping through the mediator, whereas CWQ's h2 is genuine composition, which it cannot reach — 0.44 against 0.16 on nominally the same stratum depth. Read GraphRAG's per-stratum row as measuring what its radius covers, not how far static retrieval can see.

The unreachable stratum is a hallucination probe, not an accuracy measurement. By construction no path to gold exists within the hop bound for these questions, so every "hit" is an assertion the environment cannot support that happened to match the label. Vector-RAG scores 0.67 on CWQ's three unreachable questions and no-retrieval 0.17 on WebQSP's twelve. The counts are tiny and no weight is placed on them, but the direction is exactly what Section 8.6 measures at scale: Hits@1 will happily reward what RQ2 is about the absence of.

8.6 Groundedness and Hallucination Rate Across Systems

8.6.1 Tier 1: Structural Hallucination Is a Property of the Paradigm

The differential is emphatic. The parametric control asserts 661 entities on WebQSP of which 179 — 27.1% — have no basis in the knowledge environment, and 37.6% of the questions it

answers contain at least one such entity. Both navigating systems sit at exactly zero, over 1,235 and 832 asserted entities. Parametric answering hallucinates structurally at double-digit rates; graph-navigated answering eliminates structural hallucination outright.

But Think-on-Graph reaches zero as well, and that changes the attribution. Structural groundedness is a property of the *navigation paradigm* — any system that copies its answer entities out of triples it actually traversed inherits it — not of this thesis's verification layer specifically. The honest claim is therefore narrower and more testable than the one the work set out to make: *both navigating systems achieve zero structural hallucination; they differ at the level of semantic support*, where Think-on-Graph's precision of 0.571 and 0.403 against AGR's 0.697 and 0.484 (Table 8.2) shows it asserting substantially more entities that exist, connect to the topic, and were copied from real triples — and are not the answer. Section 1.3 builds on exactly this gap.

Two readings of the middle band belong on the record. The no-retrieval control's CWQ rate (12.4%) is less than half its WebQSP rate (27.1%) not because it fabricates less on harder questions but because it hedges more — 38.0% against 12.2% — and abstention suppresses the measured hallucination rate. That is the abstention trade appearing in a third metric. And Vector-RAG's 38 ungrounded entities are mostly not fabrications: its index is global, so retrieval can surface facts about a similarly-named subject that exists in the graph but has no connection to the question's topics, and the model copies faithfully from them. The metric correctly flags those as unanchored to the question, but the mechanism is retrieval error rather than generation error. Structural ungroundedness conflates the two, and the distinction is recoverable from whether the entity appears in the system's own retrieved facts.

8.6.2 Tier 2: A Narrow Band and an Underpowered Comparison

Tier 2 asks whether the asserted entity is supported *as the answer*, and under the instruction that mere connectedness is not support, every system lands in a 36.7–66.7% band. AGR is highest on WebQSP at 66.7%. On CWQ it is second: 5.0 points ahead of Think-on-Graph, which is the comparison the rest of this chapter makes, but 1.7 behind Vector-RAG's 50.0%. Its margin over the agentic baseline is 5.0 points on both datasets; its lead over *every* system is a WebQSP result only.

Neither gap is resolvable at this sample size. With $n = 60$ per cell the 95% binomial interval has a half-width of roughly 12–13 points, so a 5-point difference is inside the noise — as is Vector-RAG's 1.7 — and no conclusion in this thesis rests on either. What the tier does establish is the level: semantic support is *hard* for every system, including the two that are structurally perfect. Zero structural hallucination coexists with roughly a third to a half of assertions failing a strict semantic-support test — which is the single most useful thing the metric says, and it says it about the paradigm rather than about any one system.

Table 8.7: Cost per question. Latency is computed over cold-cache records only. Clip rate is the share of questions on which the shared 25-call budget was exhausted mid-search.

Data	System	Tokens	Calls	Seconds	Clipped
WebQSP	No-retrieval	113	1.0	1.2	—
WebQSP	Vector-RAG	527	1.0	1.4	—
WebQSP	GraphRAG	531	1.0	2.1	—
WebQSP	Think-on-Graph	3,615	12.8	12.9	29.2%
WebQSP	AGR	4,511	6.2	11.2	0.0%
CWQ	No-retrieval	118	1.0	1.0	—
CWQ	Vector-RAG	535	1.0	1.4	—
CWQ	GraphRAG	470	1.0	2.2	—
CWQ	Think-on-Graph	5,572	18.2	15.2	44.0%
CWQ	AGR	6,818	8.9	17.2	0.0%

Two properties of the measurement bound its interpretation, both pre-registered. Path retrieval takes the three shortest connecting paths, so a genuinely supporting chain can lose to shorter irrelevant ones; the rates are conservative lower bounds, depressed uniformly across systems, which preserves the between-system comparison while making the absolute levels untrustworthy. And the judge was validated at $\kappa = 0.6995$ against a pre-registered bar of 0.7 (Section 7.6) — substantial agreement, marginally short of the threshold, and a further reason to treat this tier as corroborating rather than decisive.

8.7 Efficiency and Cost

8.7.1 Token and Call Budgets per System

Agency costs roughly an order of magnitude over one-shot retrieval: $4{,}511$ and $6{,}818$ tokens per question against ~ 500, and 11–17 seconds against one to two. That is the price of RQ1's accuracy, and it is stated as a price.

The comparison that carries information is against the other agentic system, since the two run under an identical 25-call cap. AGR uses *half* the calls — 6.2 against 12.8 on WebQSP, 8.9 against 18.2 on CWQ — with more tokens per call, because each call carries a richer prompt: a plan, a scored candidate set, a fact window. Fewer, better-informed decisions rather than more, blinder ones.

The clip column is the sharpest form of that. Beam search costs are multiplicative in width and depth, so under the shared budget Think-on-Graph exhausts its calls before finishing on 29.2% of WebQSP questions and 44.0% of CWQ questions, while AGR never does. On nearly half the multi-hop benchmark the baseline is answering from a truncated search. This is a fair constraint — both systems live under the same cap, and a clipped run still answers from what it gathered by

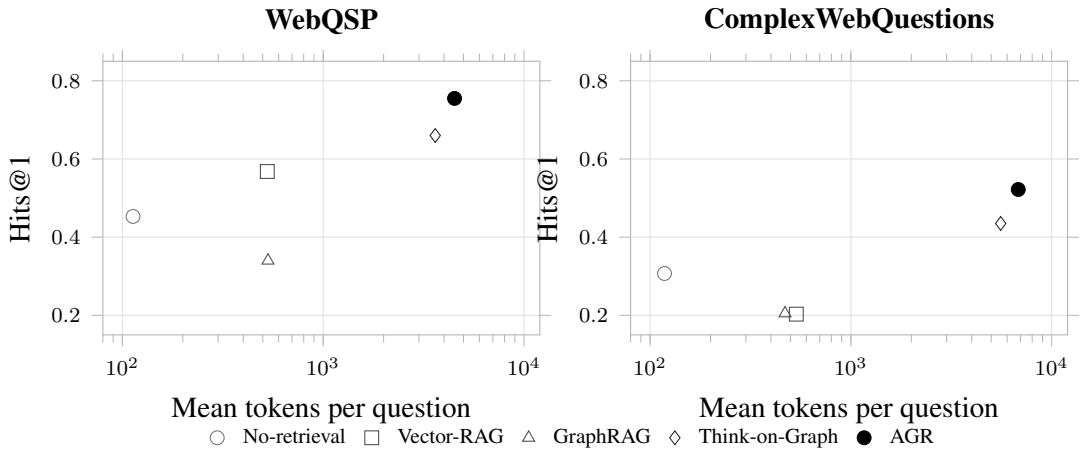

Figure 8.2: Accuracy against token cost, one point per system. The horizontal axis is logarithmic: the agentic systems cost roughly an order of magnitude more than the one-shot ones, and a linear axis would compress every non-agentic system into a single column. Read this figure for the token frontier only. Think-on-Graph sits below *and* to the left of AGR, so it is a frontier point on this axis, not a dominated one; the domination claim holds on calls, which are not plotted and are given in Table 8.7. Generated from results/phase4/thesis_numbers.json.

the same graceful degradation — and it is the efficiency finding rather than a caveat on it: under an identical budget, AGR converts its calls into correct answers at roughly twice the rate.

8.7.2 The Accuracy–Cost Frontier

Reading Table 8.2 against Table 8.7, and plotted as Figure 8.2, the frontier depends on which cost is charged, and the two costs this thesis records do not agree. Stating it on one of them alone would overclaim.

On tokens, the axis of Figure 8.2, Think-on-Graph is not dominated. The Pareto set on WebQSP holds four systems — no-retrieval at 113 tokens, Vector-RAG at 527, Think-on-Graph at 3,615, AGR at 4,511 — and the interior point is static GraphRAG, which the parametric control beats on both axes at once. On CWQ the Pareto set holds three, no-retrieval, Think-on-Graph and AGR, with *both* static retrievers interior for that same reason. Think-on-Graph buys its lower accuracy at a genuinely lower token price on both datasets, and any claim that it is dominated has to name an axis on which that is true.

On calls it is dominated, and calls are what the budget meters (Section 5.8). Think-on-Graph spends 12.8 and 18.2 calls against AGR's 6.2 and 8.9 — lower Hits@1 at roughly double the call cost, on both datasets, with no interior ambiguity. The two costs diverge because AGR spends more than twice as many tokens per call, converting fewer, larger calls into a comparable token total. Which cost binds is a deployment question: token billing separates the two systems very little, whereas a per-call rate or latency budget separates them by a factor of two — and it is the

Table 8.8: Ablation conditions on stratified halves. The reference row is the frozen system restricted to the same questions, at zero additional cost. Cost is reported in tokens and calls only: latency is omitted because every record in the no-verifier condition is a full cache replay, so a seconds column would not be comparable across conditions.

Data	Condition	Hits@1	F1	Hedge	Tok.	Calls
WebQSP	reference	0.755 [0.695, 0.810]	0.659 [0.599, 0.717]	8.5%	4,562	6.2
WebQSP	no-planner	**0.830** [0.775, 0.880]	**0.742** [0.687, 0.794]	5.5%	3,159	4.0
WebQSP	no-backtrack	0.745 [0.685, 0.805]	0.646 [0.586, 0.705]	9.5%	4,261	5.8
WebQSP	no-verifier	0.760 [0.700, 0.820]	0.663 [0.604, 0.721]	8.0%	4,460	6.1
WebQSP	embedding-only	0.740 [0.680, 0.800]	0.643 [0.583, 0.702]	13.5%	3,413	4.4
CWQ	reference	0.495 [0.424, 0.566]	0.445 [0.379, 0.511]	23.2%	7,105	9.1
CWQ	no-planner	0.434 [0.369, 0.505]	0.398 [0.333, 0.464]	26.8%	5,617	6.7
CWQ	no-backtrack	0.490 [0.419, 0.561]	0.445 [0.379, 0.511]	28.3%	5,792	7.4
CWQ	no-verifier	0.490 [0.419, 0.561]	0.436 [0.371, 0.503]	20.2%	6,704	8.7
CWQ	embedding-only	0.475 [0.409, 0.545]	0.437 [0.370, 0.502]	28.3%	4,925	5.9

call budget that clips Think-on-Graph on 29% and 44% of questions (Section 8.4).

Around that, the corners are unchanged. Vector-RAG on WebQSP is the cheap corner: 0.568 Hits@1 for 527 tokens, and 0.82 on the one-hop stratum, which makes it the right choice for shallow questions and useless for deep ones. AGR is the accurate corner on both datasets. The no-retrieval control holds the cheap end of the token frontier while being off any accuracy-worthy frontier once precision is counted — it remains the cheapest way to be right about half the time on WebQSP, which is the contamination point restated as an economic one.

8.8 Ablation Study

Four single-deviation conditions on stratified halves ($n = 200$ WebQSP, $n = 198$ CWQ), each against the unmodified system restricted to the same questions. Table 8.8 reports the conditions and Table 8.9 summarises the component attribution. Throughout, Δ is *ablated minus reference*, so a positive Δ means removing the component *improved* the metric.

8.8.1 Without the Planner: A Stratum-Dependent Effect

This is the one condition that reaches significance, and it points the opposite way to the design intent.

On WebQSP, *removing* the planner improves every axis at once: Hits@1 $0.755 \rightarrow 0.830$, F1 $0.659 \rightarrow 0.742$, hedge rate $8.5\% \rightarrow 5.5\%$, tokens $4{,}562 \rightarrow 3{,}159$, calls $6.2 \rightarrow 4.0$. McNemar gives $p = 5.9 \times 10^{-3}$ on 6 against 21 discordant pairs. The per-stratum breakdown locates the damage precisely: h1 rises $0.75 \rightarrow 0.85$ and h2 rises $0.84 \rightarrow 0.89$, so decomposition is hurting simple *and* two-hop questions on this dataset. The planner prompt's first worked example

instructs the model not to decompose single-hop questions; the ablation proves that instruction under-triggers, with a p-value.

On CWQ the sign reverses and significance is not reached: Hits@1 $0.495 \rightarrow 0.434$, 27 against 15 discordant pairs, $p = 0.088$. The stratum table shows a coherent mechanism rather than noise — h1 *improves* ($0.49 \rightarrow 0.54$) while h2 collapses ($0.48 \rightarrow 0.34$) and h3plus moves little ($0.62 \rightarrow 0.54$) — which is precisely "decomposition helps genuine composition and hurts questions that do not need it". But it does not clear $\alpha = 0.05$ at $n = 198$, and the honest statement carries both halves: *the planner's benefit is stratum-conditional and significant on WebQSP ($p = 0.006$); the CWQ pattern points the same way but does not reach significance at this sample size ($p = 0.088$).*

The finding is therefore not "the planner helps" or "the planner hurts" but that its value is conditional on question structure, and that applying decomposition unconditionally is the defect. Section 9.3 reads the 21 and 15 questions where removing the planner flipped a failure into a success and names the mechanisms; Section 10.3 turns them into an adaptive gating proposal that this ablation earned rather than motivated.

8.8.2 Without Backtracking

Hits@1 moves -0.010 on WebQSP and -0.005 on CWQ; F1 -0.013 and 0.000. Discordant pairs are 5 against 3 and 6 against 5, giving $p = 0.73$ and $p = 1.00$. There is no detectable effect at this sample size.

Two readings are available and the data does not separate them: backtracking's contribution may be small because the scorer and the verifier already prevent most of the excursions it exists to correct, or $n \approx 200$ may simply be too few questions to resolve a small effect. This is reported as "no detectable effect at $n \approx 200$", never as a confirmed null. The mechanism is not idle — the full runs record 63 backtracks on WebQSP and 199 on CWQ, dominated by evaluator-triggered ones (47 and 170) — so the condition removes real activity whose accuracy consequence this experiment cannot measure.

8.8.3 Without the Verification Layer

The cleanest null in the table, and the most informative one. Discordant pairs are 0 against 1 on WebQSP and 1 against 0 on CWQ: across 398 paired questions, removing claim verification changed the correctness of exactly two. Hits@1 moves $+0.005$ and -0.005; F1 $+0.004$ and -0.009. Both $p = 1.00$.

This reproduces the development-set result of Section 8.1 exactly, and it is what Chapter 6 predicted by construction: on 77 of 80 development questions the layer certified the draft and

changed nothing, and the repair path never executed. A mechanism that mostly passes cannot move an accuracy metric.

The conclusion this licenses is narrower and sharper than "the verifier helps", and it is the one Section 8.9 adopts: *claim verification does not measurably change how often AGR is right; it changes how often AGR is right for a reason it can exhibit*. Hits@1 cannot see that distinction — it is what the groundedness metrics were built to see, and what the supporting-triple output contract of Section 6.6 exists to make checkable. The evidence for the layer is Table 8.6, the precision column of Table 8.2, and the specimen analyses of Section 9.5; it is not this row, and this row is reported at full length rather than omitted.

8.8.4 Embedding-Only Scoring

Removing the language-model term from the hybrid scorer costs -0.015 and -0.020 Hits@1 and -0.016 and -0.008 F1 — consistently negative on both datasets and both metrics, and consistently short of significance ($p = 0.66$ and $p = 0.48$).

Two corroborating details make the direction credible without over-claiming from it. The condition is the cheapest of the four (3,413 and 4,925 tokens, 4.4 and 5.9 calls), so the language-model scoring term is buying accuracy with tokens as designed. And it carries by far the most backtracks — 57 and 143 against the reference's 38 and 108.

That second detail was originally read as the τ mechanism of Section 5.6.3 visible in the logs: with the language-model term removed, the signal maximum clears τ less often and the search works harder to compensate. The reading survives, but only partly, and the qualification is recorded because the run logs contain the evidence against the stronger version. Every backtrack stores its trigger, so the increase can be decomposed rather than attributed. On WebQSP the $+19$ additional backtracks are $+14$ low_score, $+5$ evaluator, and no additional dead ends; on CWQ the $+35$ are $+16$ low_score, $+13$ evaluator, and $+6$ dead ends. The low_score trigger — the one τ governs — roughly triples in both datasets, from 7 to 21 and from 8 to 24, and accounts for about three quarters of the WebQSP increase and about half of CWQ's.

So the trigger the explanation names is doing much of the work. But the signal it thresholds is not computed identically in the two conditions, for the reason Section 5.6.3 records: at $\alpha = 1$ the two auxiliary maxima are never taken, so σ is a maximum over beam-selected, non-banned edges rather than over every candidate offered. After a backtrack has banned the best relation, σ falls in this condition and holds steady in the reference, which produces additional low_score backtracks through an implementation asymmetry rather than through the scoring change under test. The two effects push the same way and cannot be separated without re-running the condition with the auxiliary terms restored, which would no longer be the frozen system.

The honest statement is therefore narrower than the original one: removing the model term raises the backtrack count, an unknown but non-trivial share of that rise is an artefact of how σ

Table 8.9: Component attribution, RQ3. ΔF1 is *ablated minus reference*, so positive means removing the component improved F1. Only the planner condition reaches significance.

Component	**WebQSP ΔF1 (p)**	**CWQ ΔF1 (p)**	**Verdict**
Planner	$+0.083$ (0.006)	-0.047 (0.088)	Stratum-dependent
Backtracking	-0.013 (0.73)	0.000 (1.00)	No detectable effect
Verification	$+0.004$ (1.00)	-0.009 (1.00)	No accuracy effect
α-blend	-0.016 (0.66)	-0.008 (0.48)	Directional, underpowered

is computed when the term is disabled, and the backtrack count is consequently corroborating rather than load-bearing. It is worth noting that this cuts against the ablation's own direction — extra backtracks make the embedding-only condition look busier than the scoring change alone would make it — so the defect flatters the component under test rather than the null. Nothing in Table 8.9 depends on it: the accuracy and F1 deltas are computed from answers, not from backtracks, and they are unaffected. The primary evidence for $\alpha = 0.7$ remains the development sweep of Table 8.1, where the $\alpha = 1.0$ condition lost five hits on 80 questions; this ablation corroborates it at test scale rather than independently establishing it.

8.8.5 Paired Significance Testing and Statistical Power

One of four conditions reaches significance. That is the correct outcome to report for four components tested on half-samples, and it is reported as such rather than rewritten into four confident claims.

The power limitation is arithmetic, not rhetorical. McNemar's test reads only discordant pairs, and at $n \approx 200$ with a system near 0.5–0.75 accuracy, a component that changes five questions in a hundred produces on the order of ten discordant pairs — not enough to reject at conventional levels. The planner condition cleared the bar on WebQSP because its effect is large (21 against 6), not because that comparison was better powered than the others. The halves were adopted as the pre-registered response to the benchmark's cost coming in above projection (Section 7.8); doubling them is the straightforward remedy and is named in Section 10.2.

Three conditions are therefore reported as "no detectable effect at this sample size". That phrase is not a hedge for "no effect": for the verifier the point estimates are near zero *and* the mechanism's contribution was never expected to appear in accuracy, whereas for backtracking and the α-blend the point estimates are consistently signed and the mechanism is visibly active in the logs. Those are different epistemic situations, and collapsing them into one word would lose the distinction.

8.9 Findings Against RQ1, RQ2, and RQ3

RQ1 — does agentic navigation improve multi-hop factual accuracy? Yes, and the multi-hop qualifier is doing real work. AGR leads every cell of Table 8.2, and all eight paired comparisons reject, including both against the agentic baseline running on the same graph with the same model under the same budget. The stratum profile is the substance of the answer: on CWQ, AGR's accuracy *rises* with hop count ($0.46 \rightarrow 0.55 \rightarrow 0.57$) while every other system decays, and Vector-RAG collapses from 0.33 to 0.10 on a context that cannot hold a chain at any radius.

Three boundaries are reported with it rather than left for a reader to find. Think-on-Graph is better on CWQ's one-hop stratum. Vector-RAG is better than everything on WebQSP's. And the margin over Think-on-Graph is located in the budget: where the shared cap lets it finish, it is marginally ahead on both datasets, so the claim this thesis defends is that guided search completes under a realistic budget where beam search does not (Section 8.4). The static graph baseline's per-stratum decay is not offered as evidence here, because its retrieval radius confounds it (Section 7.4.3).

RQ2 — what does pre-generation verification contribute beyond navigation? The answer has three parts that a yes-or-no question would have merged, which is why it was not asked as one.

First, navigation eliminates structural hallucination, and the verifier is not what does it. Both navigating systems assert zero ungrounded entities on WebQSP — over $1{,}235$ assertions for AGR and 832 for Think-on-Graph — against 27.1% for parametric answering on the same set. Any system that copies its answer entities out of triples it traversed inherits this; attributing it to the verification layer would be wrong, and Table 8.6 is the evidence that it would be wrong.

Second, the verifier's measurable contribution is precision, not accuracy. AGR's assertion precision exceeds Think-on-Graph's by 12 points on CWQ (67.9% against 56.1%) and its per-question precision by 0.13 and 0.08, at a lower hedge rate on WebQSP — so it is not bought with abstention. The mechanism is visible in Chapter 6: entity-level filtering of the final answer removes assertions the traversal cannot support, which costs recall almost nothing and costs a false assertion every time it fires.

Third, verification shows no measurable Hits@1 effect. Two questions out of 398 change when it is removed. This is not a failure of the layer but a statement about what Hits@1 measures: a metric satisfied by one correct entity among several cannot see a mechanism that operates on the others. The layer's value proposition is that an answer arrives with the triples that support it (Section 6.6) — an auditability claim, not an accuracy claim — and the honest formulation is that *claim verification does not change how often the system is right; it changes how often the system is right for a reason it can exhibit*.

RQ3 — which components contribute what, at what token cost? One of four reaches significance. The planner is stratum-dependent: removing it improves WebQSP by 0.083 F1 at $p = 0.006$ while saving 30% of tokens and 35% of calls, and costs CWQ 0.047 F1 at $p = 0.088$.

That asymmetry, not a pooled average, is the result — and it is a finding about how decomposition should be *applied* rather than about whether it works. Backtracking, verification, and the α-blend show no detectable effect at $n \approx 200$, for the different reasons Section 8.8 separates. On cost, the whole apparatus runs at half the language-model calls of the agentic baseline and never exhausts a budget that clips the baseline on 44% of CWQ questions.

The three answers do not point the same way, and the chapter does not force them to. Navigation is where the accuracy comes from; verification is where the precision and the auditability come from; and decomposition, the component the literature treats as straightforwardly beneficial, is the one component this work can prove is sometimes harmful.

Chapter 9

Error Analysis and Discussion

Chapter 8 measured how often each system fails. This chapter reads the failures and names the mechanism of each one. It is the part of the work no script produced: 259 failure packets — question, gold, answer, plan, explored relations with their scores, backtrack reasons, verifier outcome — read and labelled by hand, one category and one sentence each.

The reading protocol was fixed before it began. For each packet, find where the trajectory *first* left the correct path, and label that point rather than the last visible symptom; when torn, take the earliest plausible cause and record the alternative. That rule is what keeps a run that mis-decomposes, then explores the wrong relation, then hedges from being counted three times under three categories.

A note on the identifiers used throughout this chapter, because they are meant to be looked up. A ComplexWebQuestions identifier is the WebQSP identifier it was derived from followed by a thirty-two character hash, and the stem alone does not identify a question: within these samples one stem covers up to twelve distinct CWQ questions, and thirteen stems are themselves complete WebQSP identifiers naming something else entirely. Cases are therefore cited as stem plus the first eight hash characters, which is unique within both samples; Chapter C lists every identifier in full.

9.1 Annotation Protocol and the Failure Taxonomy

9.1.1 Category and Subtype Schema

Ten categories carry the census, each naming a distinct architectural locus rather than a symptom. That choice is the substantive one. Surveys of hallucination classify by how the output fails — factuality against faithfulness, intrinsic against extrinsic [1] — which is the right cut when the generator is a black box and the output is all there is to read. It is the wrong cut here, because the run record exposes which component produced the failure, and a taxonomy that discarded

that would throw away the only information this system has over a black box. The categories are therefore named for where the trajectory went wrong: relation_selection (the scorer picked the wrong edge), composite_claim (a multi-constraint question handled partially), kg_gap (the environment cannot express what the question needs), decomposition_error (the planner or draft pipeline caused it), answer_selection (the correct candidate was retrieved and the drafter chose another), echo (a real, grounded entity one node away from the answer), premature_termination (the search stopped before the answer), verifier_fn and verifier_fp (a right claim rejected, a wrong one accepted), and gold_noise / ambiguous_question for the benchmark defects that escaped the exclusion pass of Section 7.7. An other category was kept open for mechanisms the schema did not anticipate, and Section 9.3 reports what fell into it.

Subtypes are a second, open vocabulary that drives nothing and exists to group cases when writing. The full schema, with the worked examples that fixed each boundary, is Table D.1 in Section D.2.

One naming decision is recorded here because it was got wrong once and corrected. The verifier's *positive* class is "the claim is supported", so a false positive is a claim wrongly *accepted* and a false negative one wrongly *rejected*. The taxonomy's gloss originally paired those in the opposite order. It was corrected, two mislabelled rows were re-filed (Sections 9.3.2 and 9.5), and the merged histogram was regenerated. Section 9.5.1 states the convention once; everywhere else this thesis says "wrongly rejected" and "wrongly accepted" in plain English, because with outputs labelled supported and unsupported a reader has no reliable way to infer which class is positive.

9.1.2 Population, Not Sample

The census is a *population*, not a sample. Both datasets were read to completion: every wrong answer and every hedge produced by the full system, less three exclusion sets, each removed for a stated reason and each named here rather than folded into one.

The first is the adjudicated benchmark defects of Section 7.7 — 22 WebQSP and 19 CWQ questions where the gold label, not the system, is wrong. The second is the Stage B surface-form near-misses of Section 7.5: 1 WebQSP and 6 CWQ answers that strict string matching scores as failures and aggressive normalisation scores as hits. They are excluded because they are scoring artefacts rather than reasoning failures, and because Section 7.5 already reports them as a measured quantity in their own right. The third is not an exclusion so much as a redirection: the questions read under Stage A, the planner-discordance census, are held out of the Stage D reading so that no question is labelled twice, and they re-enter the histogram through their own row in Table 9.1.

Only the first set removes questions from the analysis altogether. The second is reported elsewhere and the third is counted elsewhere, so the totals of Table 9.1 are what the histogram is

Table 9.1: Census provenance. Stage D is the main reading over the full-matrix failures; Stage A is the separate reading of the questions where removing the planner flipped a failure into a success (Section 8.8); the single migrated row is a Stage D finding later promoted to a formal benchmark-defect exclusion (Section 9.8).

Source	WebQSP	CWQ
Stage D — main census (all remaining wrongs and hedges)	65	157
Stage D — row migrated to a benchmark exclusion	0	1
Stage A — planner-discordance census	21	15
Total failures analysed	**86**	**173**

built over.

The three sets overlap, so their sizes are not additive. This is worth stating because subtracting them in sequence gives the wrong answer, and the generator does not: scripts/dump_failure_packets.py skips their *union*. The full system fails on 98 WebQSP questions and 191 CWQ questions. Of the 22 and 19 adjudicated benchmark defects, only 12 and 15 were failures at all — the rest were questions the system answered correctly despite the label — and of the near-misses, 1 and 6. Those subsets are not disjoint from the Stage A reading: one WebQSP question is both an adjudicated defect and planner-discordant, and on CWQ one defect and one near-miss are. The union skipped is therefore 33 questions on WebQSP and 34 on CWQ, which leaves exactly the 65 and 157 that Stage D read, and the two totals close without remainder. A question that falls in both an exclusion set and Stage A is still analysed, through the Stage A row; net of that re-entry and of the one migrated row, 12 WebQSP and 18 CWQ failures leave the analysis altogether, which is the difference between the failure counts above and the totals in the table.

An earlier plan drew a stratified $40 + 20$ sample from CWQ's failures; the census was read to completion instead, so there is no sampling-bias caveat on any number in this chapter. What does remain is a *composition* caveat: wrong answers and hedges are reported separately throughout and never pooled. They are different failure semantics — a wrong answer is a reasoning error that committed, a hedge is usually a coverage gap that never produced a committal claim — and their proportions differ sharply between the two datasets, so a pooled percentage would describe neither.

9.2 Distribution of Failure Categories Across Datasets

9.2.1 The Shape Flip, and What It Is Not

The two datasets fail differently, but not in the way a first reading of Table 9.2 and Figure 9.1 suggests. relation_selection is the largest category on *both* — 26 of WebQSP's 86 and 39 of CWQ's 173 — so it is not what distinguishes them. Picking the wrong edge is the base rate of

Table 9.2: The merged failure histogram, $n = 259$. Wrong and hedge are kept separate. Counts come from scripts/synthesize_census.py over the three label sources of Table 9.1; the table body is generated from results/phase4/thesis_numbers.json, so it cannot drift out of step with Figure 9.1.

	WebQSP		CWQ		
Category	**wrong**	**hedge**	**wrong**	**hedge**	**Total**
relation_selection	20	6	20	19	**65**
composite_claim	1	—	23	23	**47**
kg_gap	8	4	8	24	**44**
decomposition_error	15	11	10	2	**38**
answer_selection	1	1	8	5	**15**
echo	5	2	4	2	**13**
ambiguous_question	1	1	3	5	**10**
premature_termination	1	4	1	2	**8**
gold_noise	1	—	3	3	**7**
other	—	1	—	5	**6**
verifier_fn	—	3	1	1	**5**
verifier_fp	—	—	—	1	**1**
Total	**53**	**33**	**81**	**92**	**259**

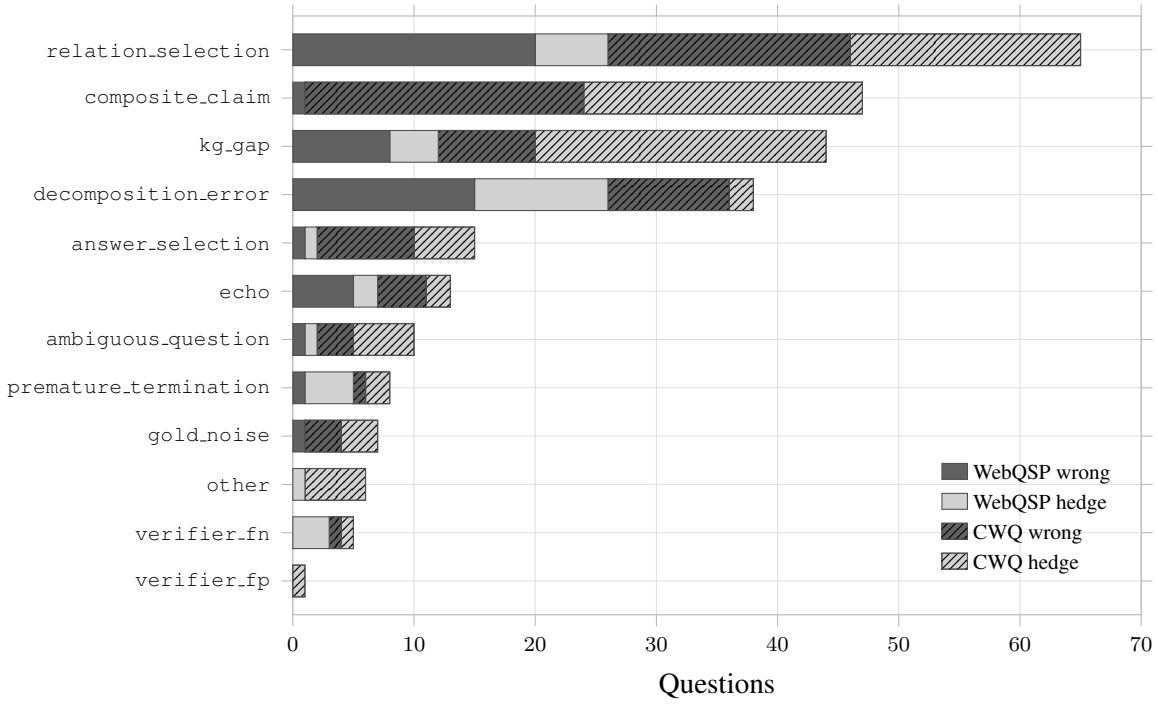

Figure 9.1: The failure histogram of Table 9.2 drawn as stacked bars, ordered by total. Wrong and hedge are shown as separate segments and are never pooled, because they describe different failure semantics. The asymmetry the section below analyses is visible as the length of the CWQ segments on composite_claim and kg_gap against their near-absence on WebQSP. Generated from results/phase4/thesis_numbers.json.

graph navigation, and it is dataset-independent.

What distinguishes them is a pair of categories that are nearly absent from one and dominant in the other. composite_claim is 1 of 86 on WebQSP and 46 of 173 on CWQ; kg_gap is 12 and 32. In the other direction, decomposition_error is 26 on WebQSP — tied with relation_selection, the two together accounting for nearly 60% of its failures — against 12 on CWQ.

That asymmetry is demonstrable rather than assertable, from two facts already established. First, the stratum distribution of Table 7.1: WebQSP is 64% one-hop with a four-question deep tail, so most of its questions have no composition to get wrong and no constraint to drop, while presenting the planner with exactly the questions that do not need decomposing — which is also the significant ablation result of Section 8.8, appearing here as a category count. Second, the expressiveness limits certified in Section 4.7.4: the environment carries no date literals, no numeric literals, and no ordinals, and CWQ's SPARQL templates invoke all three constantly while WebQSP's naturally-occurring questions rarely do. The failure profile of a system on a dataset is, to this extent, a property of the dataset's construction.

9.2.2 Hedging Where It Should

A second reading of Table 9.2 is worth stating as a positive result rather than only as a deficit. On CWQ, kg_gap skews three-to-one toward hedges — 24 against 8 — while on WebQSP it leans the other way, 4 against 8. When CWQ's numeric identifiers, date literals, and ordinal constraints are unreachable, the system much more often declines than asserts something wrong. That is the behaviour the anti-vacuity and grounding provisions of Section 6.2 were written to produce, and it is visible in the one place where the environment guarantees the system cannot succeed.

9.3 Decomposition, Drafting, and Answer-Selection Errors

Three families share this section because they share a locus: everything downstream of retrieval and upstream of the answer. The reasoning is often complete and correct by the time these failures occur, which is what makes them both diagnostic and cheap to fix.

9.3.1 The Entity-Extraction Bug

Nine of the 38 decomposition_error cases carry the subtype extraction_bug, and they are the most actionable finding in the census. On profession-shaped and "what did X do" questions, the evaluator resolves every correct gold value, the drafted text states them verbatim, and answer_entities then collapses to the sentence's grammatical *subject*. Table 9.3 gives three instances read from the raw traces.

Table 9.3: The entity-extraction bug. In each case the drafted answer text names every gold value correctly and the scored entity list holds only the sentence's grammatical subject.

Question ID	Gold	Named in text	Scored entities
WebQTest-1215	6 professions	all 6	[Stephen Covey]
WebQTest-704	6 professions	all 6	[Thor Heyerdahl]
WebQTrn-124_0782789f	4 films	all 4	[Angelina Jolie]

Table 9.3 understates how complete the drafts are, so one is worth quoting whole. For WebQTest-1215, "who was stephen r covey", gold lists Author, Manager, Writer, Consultant, Professor, and Motivational speaker; the drafted answer reads *"Stephen Covey was a Writer, Motivational speaker, Author, Consultant, Professor, and Manager"* — all six, correct, in a different order — and answer_entities comes back as ['Stephen Covey'].

This is not a reasoning failure at all. The navigation found the answers, the verifier certified them, and the prose reports them. What fails is the step that turns answer text into the entity list, which consistently reaches for the subject rather than the predicate values.

The consequence has to be stated as a limitation on Chapter 8, not merely as a bug: scoring reads answer_entities, so every one of these cases is recorded as a wrong answer or a hedge in Table 8.2. The reported accuracy of this system on "list the professions" and "list what X did" question shapes is therefore depressed by an extraction defect independent of any reasoning quality. Nine cases across 259 is not enough to move the headline numbers appreciably, but the correct reading of them is that the measured figure is a floor for that question shape, and Section 10.3 lists the fix among the repairs the census earned.

9.3.2 Right Candidate Retrieved, Wrong One Drafted

answer_selection (15 cases, 13 of them on CWQ) is the extraction bug's more consequential sibling: the exact gold value demonstrably entered the resolved candidate set, and the drafter chose something else. WebQTest-1555 asks what the parliament of Nepal is called; the evaluator resolves Parliament of Nepal itself, and the draft names its two component chambers instead. WebQTrn-2784_abedc8ac asks which Tupac Shakur film Malcolm Campbell edited; the trace records resolved=['Nothing but Trouble'], the literal gold value satisfying both constraints, and the draft substitutes Gang Related and then hedges on it.

One case in this family was originally filed as a verifier error and is worth the correction in full, because it is the kind of mistake this taxonomy exists to prevent. WebQTrn-1294_a4b2006a was labelled verifier_fn on the theory that the verifier had checked a claim about a different candidate than the one the evaluator resolved. The run record refutes that: the draft itself reads "Marlon Brando also played in *Joy*", and the verifier rejected exactly what the drafter asserted. Brando was not in *Joy*; De Niro was, and the evaluator had resolved him. **The verifier was correct.** The

error is upstream, in the answerer, and the label was corrected to answer_selection. That case is also the clearest positive demonstration in the census of the verification layer doing its job, and Section 9.5 opens by pointing back to it.

9.3.3 Failing to Commit: Six Cases of Hedging on a Verified Fact

The other category was kept open for mechanisms the schema did not anticipate. All six of its cases turned out to be the same one, and it is not in the schema because nothing predicted it: *the evaluator resolves the exact gold value, the verifier returns* grounded, *and the drafted text hedges anyway*. No rejection is involved, so it is not a verifier error at all.

WebQTest-689 states "Spanish Language was spoken in Spain" and then that the first language spoken in Spain could not be determined, in one answer. WebQTrn-125_003c6a04 resolves Memphis, matching gold exactly, and the draft says the location could not be determined. WebQTrn-1392_d372995c names both of Eleanor Roosevelt's schools correctly and then declares the school undeterminable. WebQTest-989_6d0a1bf8 (Battle of Dunkirk), WebQTest-1923_a64ef0f5 (*The Last Song*), and WebQTrn-2721_2d207ed9 (Alois Hitler) are the same shape. In every case the entity list comes back empty and the question is scored as a hedge.

The contrast with the extraction bug is exact and worth holding: there the text is right and the entity list is wrong; here the text contradicts a fact it has just stated. Six of six instances share the mechanism, which makes it a defect rather than a distribution.

9.3.4 Context-Stripping: When Decomposition Discards the Question

Subtyping is thinner in this category than elsewhere, and the shortfall is worth stating before the cases are read. decomposition_error holds 38 failures; the four named subtypes — extraction_bug (9), paraphrase_drift (9), context_stripping (3), and over_decomposition (1) — cover 22 of them. The other 16 are labelled at category level only. That is a real limit on how far the mechanisms below generalise: they are the mechanisms of the 22, not of all 38, and the subtype vocabulary was left open rather than stretched to cover the residue.

paraphrase_drift (9 cases) and over_decomposition are the expected decomposition failures — a rewritten sub-objective scoring worse against the target relation than the raw question text, or a redundant step resolving an already-unambiguous entity. WebQTest-1829 is the clean over-decomposition case: the plan splits "what type of religion did massachusetts have" into ['find Massachusetts', 'find the religion type associated with #1'], burns 14 of 16 calls on three evaluator backtracks, and hedges, while the undecomposed run finds the same fact in three calls with no backtracks.

context_stripping (3 cases) is the mechanism this chapter would feature if it could feature only one, because it is a failure mode of decomposition that decomposition's own design cannot

see. Splitting a question into steps *removes* a qualifier that appears only in a later clause, so an early sub-objective resolves the wrong sense of an ambiguous term before the disambiguating information ever arrives.

WebQTest-1367 asks where Glastonbury, England is. The plan's first step anchors on 'find Glastonbury' alone, dropping the question's own disambiguator, and the explorer resolves toward the identically-named Connecticut town.

WebQTrn-2615_48a6ac56 asks what instruments the author of *Fela!* plays. The first sub-objective, "find the author of *Fela!*", is scored with no mention of instruments, so it resolves the wrong sense of an ambiguous authorship relation before the constraint that would have disambiguated it appears in step two. The undecomposed run answers all five instruments correctly, in fewer calls.

WebQTrn-2570_d63877a4 is the strongest specimen in the census, and its trace shows every link of the mechanism. The question — "This 33rd president was the nation's leader during WW2" — is a single entity under two descriptions. The planner emits ['find the 33rd president', "find the nation's leader during WW2"], with no #1 reference between them, modelling two descriptors of one entity as though they were a chain. Because sub-objective one is the active scoring context, the anchor World War II scores 0.128–0.133 — junk tier — while presidency-hub relations score 0.46–0.54. Decomposition actively *suppressed* the one anchor that could have discriminated. Objective one then never completes (objective_done: false at every round), so objective two never activates and the WW2 constraint is never used at any point in the search. The agent spends its depth budget wandering the presidency hub and drafts a plausible-sounding president: Woodrow Wilson, the 28th, and nowhere near the war. The undecomposed run answers Harry S. Truman correctly in three calls, because "WW2" stayed in the scoring context.

Two further observations belong to that trace. No ordinal relation appears anywhere in it, so the "33rd" constraint played no part either — consistent with the environment's lack of ordinal encoding (Section 9.6). And the verifier returned grounded on the Wilson claim, which Section 9.5 takes up as a boundary case rather than a defect.

Three instances of one mechanism, two of them surfaced by the ablation contrast and one by ordinary reading, is what makes context-stripping a named finding rather than an anecdote — and it is the concrete evidence behind the adaptive-gating proposal of Section 10.3.

A fourth case belongs beside these and is filed as plain decomposition_error, since no subtype fits it. WebQTest-869 asks "where is ancient phoenician". The planner rewrote the sole sub-objective as ['find ancient Phoenician'], so the explorer re-identified the topic instead of locating it and the run answered "Phoenicia" — the question's own subject, grounded, and useless. The undecomposed run answered "Ancient Phoenicia was in Lebanon" in three calls against six.

What separates it from the three cases above is *what* the rewrite discarded. They drop a constraint that arrives in a later clause; this one drops the interrogative, the word that says what is being asked for. The result lands in the graph as an echo (Section 9.7): with the ask removed, the nearest

well-grounded entity to a topic is the topic. Both losses are visible to a planner that compares the rewrite against the original question, which is why the gating proposal of Section 10.3 checks the rewrite for the interrogative as well as for redundancy.

9.4 Relation-Selection and Navigation Errors

relation_selection is the largest category at 65 and carries no subtypes, because the mechanism is uniform: a wrong relation was explored, or the right one was explored and abandoned. Its interest is not in the individual cases but in the fact that several of them are the *same* gap seen from different angles.

WebQTest-1730 asks where the Paraná river flows. The system answers with the rivers it flows *into*, having explored geography.river.mouth and origin; a location-containment relation was never tried, and gold is the continent.

Three separate questions across both datasets — WebQTrn-1758_477d7040, WebQTest-1226, WebQTest-314 — all reach for government.governmental_jurisdiction.government, the organisation entity, when the question asks for a form of government and the answer lives on government.form_of_government.countries. One relation confusion, reproduced three times on unrelated questions.

The sharpest instance is a triplet on CWQ. All three questions sharing the stem WebQTrn-1770 appear, under different perturbations — _540abec8 via a Beyoncé documentary, _6325ee89 via the actor who played Etta James, _6a7c160a via the composer of a Beyoncé track — and all three resolve Beyoncé correctly, all three never once try people.person.children, the single relation that answers each of them, and all three hedge. Three independent entry points into the same resolver hitting the identical dead end is the cleanest evidence in the corpus that a relation gap is *systemic* rather than an artefact of one question's phrasing. It is a property of the scorer's ranking over that entity's relation set, not of the question.

9.4.1 The Evaluator Abandons a Good Relation

premature_termination is small (8 cases) and splits into two mechanisms that must not be merged.

Five carry the evaluator subtype and are the *same* systemic pattern: the correct relation is explored with a solid score and the evaluator backtracks away from it and never returns. WebQTest-1226 (0.451), WebQTest-517_e7d8b401 (0.428), WebQTest-1635 (0.631), WebQTrn-710_e3d40457 (0.702), and WebQTest-314 (0.731) all show the identical shape across entirely unrelated domains — government form, constructed languages, sports drafts, mascots. A relation scoring 0.73 being explored and discarded is not a reasoning gap; it reads as a threshold or backtracking-policy defect, and it is one of the most directly fixable findings in the census.

The other three carry budget and are genuine exhaustion: WebQTest-1170 exhausts its backtrack budget wandering person-attribute relations after the direct edge returns nothing usable; WebQTest-1736_125140bf hits two real dead ends before the plan's second hop can run. The third, WebQTest-1630, is instructive because it belongs to two families at once: all five gold names surfaced across repeated evaluator rounds with a grounded verifier, and the run never transitioned from continue to answer before exhausting its step budget — a commitment failure of the kind Section 9.3 documents, expressed through the budget.

9.5 Verifier Rejection and Acceptance Errors

Before the errors, the correct rejections. Section 9.3.2 narrates one in full: the verifier rejected a claim the drafter had fabricated about the wrong actor, and the layer worked exactly as designed. The 52 questions on which the verifier fired unsupported are not a population of mistakes.

The two error polarities are, however, very unevenly evidenced — five cases against one — and that imbalance is not a fact about the verifier. It is a fact about the instrumentation, and it is the finding of this section.

9.5.1 Defining the Error Polarities

The verifier's positive class is "the claim is supported". A *false positive* is therefore a claim wrongly **accepted**, and a *false negative* a claim wrongly **rejected**. Because the verifier's own outputs are labelled supported and unsupported, "positive" has no stable referent for a reader, and the rest of this thesis uses the plain-English terms.

9.5.2 Wrongly Rejected: Semantically Correct, Structurally Unmatched

Five cases, all sharing one mechanism and all carrying the subtype structural_not_semantic: the claim is true and the answer correct, but no single triple states it in the phrasing the drafter used, so a transitively-true or differently-anchored fact is rejected and the retry loses the answer.

WebQTest-1133 resolves William Henry Smith across every backtrack round; the verifier flags the claim unsupported both times and the run hedges on what is very likely the right answer. WebQTest-38 resolves *all four* gold campaign entities in one attempt; the rejection forces a retry that narrows to two names and fails anyway. WebQTest-725 resolves both gold states, California and Nevada, then loses them across re-exploration and is rejected. WebQTest-1348_b7598df9 resolves Houston Oilers correctly and grounded; the verifier rejects the phrasing "Peyton Manning's dad played for Houston Oilers" because the underlying triple is stated in terms of Archie Manning, and the retry recovers only the person's name. WebQTrn-568_f4dd5e48 resolves

Mecklenburg County — the exact gold — through a sound two-hop chain from newspaper to city to county, is rejected, retries, resolves it again, and is rejected again.

The mechanism is one thing: the structural check demands a single edge asserting the drafted claim, and a transitively-true multi-hop fact does not present one. This is the same limitation Section 6.3.1 described by construction, now observed at census scale, and Section 10.3 names the semantic-level check that would address it.

9.5.3 Wrongly Accepted: One Specimen and a Boundary Case

Exactly one clean specimen exists in the whole census. WebQTrn-1597_8adc06ba asserts a Seth MacFarlane connection to *ThunderCats*, the verifier returns grounded, and a direct Cypher query confirms that no edge from MacFarlane to Lion-O or *ThunderCats* exists in the environment. The claim was accepted with nothing behind it.

The instructive companion is *not* a second defect. WebQTest-1620 asks when President Wilson was in office; the environment exposes inauguration events but no date literal, so the system answers with two inauguration mentions. The asserted entities exist, the edges are real, and the structural check passed *correctly*. The answer is nonetheless wrong. The same holds for the WebQTrn-2570_d63877a4 case of Section 9.3: Woodrow Wilson was retrieved, the claim's endpoints genuinely connect in the graph, and the verifier was right to say so — while the answer was wrong for the question.

That contrast is the most useful point in this section, and it is the same argument Section 2.3.3 makes and Section 1.3 builds on: structural grounding is a claim about edge existence, not about answer adequacy. A verifier that checks whether a claim is *in* the graph cannot, by construction, check whether it is *the answer*. Table 8.6's Tier-2 column is the measurement of that gap; these two cases are what it looks like in a single trajectory.

9.5.4 A Rate for One Polarity, and a Blank for the Other

The verifier fires unsupported on 16 of 400 WebQSP questions and 36 of 400 CWQ questions — 52 firings against 748 grounded outcomes. The five wrongly-rejected cases are drawn from those 52, so that polarity has a denominator: roughly one firing in ten cost a correct answer. Even that is a lower bound, since a wrongful rejection that did not change the final answer never entered the census, which reads only failures.

Wrongful acceptance has *no logged population at all*. Accepted claims are never persisted — Section 5.9 records that the verifier logs only the claims it rejects — so the wrongly-accepted cases sit undifferentiated inside the 748 grounded outcomes with nothing marking them. The single specimen surfaced only because the ablation-discordance reading happened to walk that trajectory by hand.

The two polarities are therefore reported asymmetrically and deliberately: a rate for rejection, an explicit blank for acceptance, and no average of the two. A combined "verifier error rate" computed from these numbers would be an artefact of what the logs happen to record. Section 9.9.3 treats this as the instrumentation gap it is, and Section 10.3 gives the one-line fix that would convert the blank into a rate.

9.6 Knowledge-Graph Gaps: Literals, Temporal Qualifiers, and Ordinals

kg_gap is the third-largest category at 44 and CWQ's single largest hedge category at 24. It invokes the unanswerable-in-environment class defined in Section 4.7.4, and its five subtypes cluster by mechanism. Those five account for 39 of the 44; the remaining 5 carry no subtype, because the environment gap in each was specific enough that grouping it with anything else would have been an invention. They are counted in the category total throughout and are not silently dropped.

ordinal (9 cases) — superlatives such as "first", "last", "smallest", "earliest". Every instance shares one shape: the relevant relation is explored, candidates come back, and the minimum-or-maximum comparison across them simply never happens, because no component performs it. WebQTest-245, on the year of the first Miss America pageant, is the clearest illustration — the system answers 1925, then 1980, then 1929 across successive rounds, genuinely different years each time, because nothing sorts and takes the extreme.

numeric_literal (16 cases, the largest subtype) — internal identifiers such as tvrage_id, thetvdb_id, ISO alpha-3 codes, and Freebase's own located identifiers, which CWQ's templates invoke constantly and which no relation in the explored schema exposes. Several of these hedges are unusually honest: WebQTrn-849_f0ffa147, asking which country bordering Germany has ISO code CHE, names Switzerland in the answer text and then states explicitly that the code itself could not be confirmed from the available facts — a correct inference, correctly caveated, rather than a false assertion.

date_literal (4) and temporal_qualifier (4) — the environment exposes inauguration and election *events* but not the date values themselves, so a question about when an office was held returns event mentions (WebQTest-1620), and a question about who was president in a given year substitutes government.election.winner for a term-range check it cannot perform. WebQTest-749 answers "George H. W. Bush" for 1988: the winner of that November's election, who took office the following January.

data_error (6) — genuine coverage holes rather than expressiveness limits. WebQTrn-261_7bc18657 returns four consecutive rounds of explored=[] (max_score 0.0) because the image reference the question names resolves to nothing in the graph at all.

The distinction between the first four subtypes and the last matters for what follows from them. data_error is a gap in this particular import; ordinal, numeric_literal, date_literal, and temporal_qualifier are gaps in what a triple-and-traversal architecture can express at all, and no larger graph fixes them. They need an operator, which is why Section 10.3 lists literal and ordinal support alongside the architectural items rather than among the repairs.

9.7 The Echo Attractor: Answering with the Topic or an Intermediate Entity

echo names a mechanism that recurs independently of the two mechanical categories above: the system resolves a real, well-grounded entity that sits *near* the correct answer in the graph's structure, and the verifier passes it because it genuinely is grounded. The fact is true. It is not what was asked.

Three shapes appear, and their boundary is not always crisp. topic (6 cases) is confusion between entities sharing a name or a tight cluster: WebQTest-1707 asks who plays Lex Luthor on *Smallville* and answers "John Glover plays Lionel Luthor" — a different character in the same fictional cluster, with Lionel, Alexander, and Lucas Luthor variants all surfacing in the trace before the run gives up. granularity (7 cases) is the right entity family at the wrong level: WebQTest-86 answers "Kingdom of Denmark" where gold is "Denmark" — a genuinely distinct node, the formal sovereign-state entity rather than the common-name one. WebQTrn-1405_ced20d84 asks which school founded in 1636 President Kennedy attended, has Harvard College in its very first candidate list, and narrows to Harvard University instead — whose founding-date claim the verifier then correctly rejects. intermediate — stopping at a hop along the way rather than continuing to the target — has no rows of its own in this census, because the reading pass filed those cases under granularity; the two are best treated as one family.

The throughline is what makes this a named mechanism rather than a residue category. None of these are cases where the system knew nothing. Every one resolves a real, verifiably-true fact through a real edge. The failure is entirely in *which* true fact gets surfaced — architecturally different from picking a wrong edge and from there being no edge at all. The right edge exists, is used, and lands one node away.

That is also why the echo attractor is invisible to any evaluation treating systems as independent. Section 9.8 shows all five systems of the main matrix falling into it together, on the same questions.

9.8 Benchmark Defects: Gold Noise and Ambiguous Questions

Two numbers must be kept apart. The consensus pass of Section 7.7 *flagged* 58 WebQSP and 47 CWQ questions where at least three of the five systems agreed on the same non-gold answer. Adjudication confirmed 22 and 19 as genuine defects. The gap between those figures is not measurement noise — it is the echo attractor of Section 9.7, operating across systems.

Read what the flagged consensus answers have in common. Five systems answer "rick scott" where gold lists his professions, "thor heyerdahl" where gold lists his occupations, "amelia earhart" where gold is Pilot and Writer — the topic entity returned as the answer. Five answer "belgium" where gold is the containing region, "dominican republic" where gold is Greater Antilles, "san francisco giants" where gold is the 2014 World Series — the composition's intermediate rather than its final hop. Five answer "maryland" for counties, "nevada" for Las Vegas, "united arab emirates" for Dubai — one level too coarse.

Cross-system consensus against gold is dominated not by label errors but by a shared attractor, and that is a finding about evaluation methodology as much as about these systems. A policy of rescoring whenever three systems agree — which is a natural thing to want — would have silently converted this system's most characteristic failure mode into apparent correctness. The manual adjudication step is what stands between the two, and it is the reason the pass is reported with both numbers rather than one.

9.8.1 The Confirmed Defects

What adjudication did confirm is nonetheless substantial: 58 questions across the two datasets where the benchmark, not the system, needed correcting. Two disjoint counts make that total — 41 excluded before the census read anything (22 + 19), and 17 more still visible in the census as gold_noise or ambiguous_question rows (3 WebQSP + 14 CWQ). One question moved between those two sets rather than adding to them: WebQTrn-64_d8e43a02 began as a census row, was adjudicated a type mismatch between question and gold, and was promoted to a formal exclusion mid-project. It is counted once, inside the 41, and it is why the census's own label file carries one dropped row.

Two specimens carry the point. WebQTest-958 — "what are some famous people from el salvador" — has 116 gold entities, including raw Freebase machine identifiers leaked into the answer strings: a "list some famous X" template ballooning gold past anything a finite answer could match under any metric. WebQTest-55_54e856d3 — "Who had a concert named Crazy Love Tour?" — asks for a person and receives gold listing the artist's professions, the annotation query having traversed one relation too far past Michael Bublé.

The Vicksburg pair is worth reading side by side, because two sibling questions share a gold

string and receive opposite diagnoses. WebQTest-1797_5a1c66f4 asks who was president during the battle of Vicksburg, with gold Ulysses S. Grant — who took office six years afterwards; the president in 1863 was Lincoln. The full system answers that the claim could not be verified and its verifier flags it unsupported; the no-planner ablation asserts Grant confidently, with zero backtracks, and is scored a *hit* purely for restating a gold value that answers a different question. WebQTest-1797_dece4dda asks which government-position holder fought at Vicksburg — coherent as asked, since Grant was a serving general — and the full system genuinely fails, answering with the Confederate commander John C. Pemberton, a real combatant who held no government position. The qualifier is present in the sub-objective text and never filters the candidate frontier.

Same battle, same gold string: one question unanswerable and one a real composite_claim failure. The pipeline that hedged on the first was more epistemically correct than the one that "succeeded".

One process observation belongs on the record. WebQTrn-64_d8e43a02 was adjudicated gold_ok by the consensus pass — several systems converging on "Tupac Shakur" read as a topic echo — and the opposite way by the census reading of the same evidence: the question literally asks for an actor's name, and gold gives a character name from a different role in the same film. The second reading held, and the row was promoted to a formal exclusion. Two independent adjudication passes can legitimately disagree, and keeping a gold_noise category open in the taxonomy is what let that be resolved rather than settled by whichever ran first.

9.8.2 Where Bad Gold Comes From: Flattened Multi-Valued Relations

The San Antonio case shows the provenance of one defect precisely enough to generalise. WebQTest-634_4f2c0d9e asks which county contains the place where the San Antonio City Council sits. Four systems, spanning parametric and graph-based knowledge, answer Bexar County; gold says Comal. Adjudication recorded the verdict gold_wrong: San Antonio and its city council are in Bexar. But the census reading of the same case recorded something more specific, and more useful — kg_gap, with the note that this is a containment ambiguity in the source data, arising from annexed territory or from Freebase carrying multiple imprecise containment edges.

That is neither a pure label error nor a pure graph error. It is a **multi-valued containment relation flattened to a single value**, with the annotation pipeline then selecting the marginal one. A city can lie in more than one county; a benchmark whose gold field holds one answer per relation must choose; and the choice is made by a SPARQL query with no notion of which value is the principal one. Framing it this way explains why the consensus diagnostic caught it at all — the systems and the world agree on the principal value while the label holds the marginal one — and it predicts where else such defects will cluster: any relation that is genuinely one-to-many

but stored as though it were one-to-one.

9.9 Discussion

9.9.1 What Agency Buys, and What It Costs

Agency buys the deep tail. On CWQ the accuracy of the agentic system *rises* across hop strata while every static system decays, and it buys that at roughly an order of magnitude in tokens over one-shot retrieval. Against the other agentic system it buys the same accuracy for half the language-model calls, and never exhausts a budget that clips the baseline on 44% of multi-hop questions — the difference between a guided search and a blind one, priced.

What it costs, beyond tokens, is a longer failure surface. Every mechanism in this chapter except kg_gap and the benchmark defects is a failure the static baselines cannot have, because they have no plan to mis-decompose, no frontier to abandon, and no draft to check. decomposition_error at 38 cases is the clearest instance: a component added to help contributed a failure family of its own, and Section 8.8 shows it is a net negative on the shallower benchmark. The honest summary is that agency converts a retrieval problem into a control problem, and control problems fail in more ways.

9.9.2 When Verification Helps and When It Over-Hedges

The verification layer's contribution is precision and auditability, and its cost is recall. Both directions are visible in this census at case level.

It helps when a draft asserts something the traversal never supported: the De Niro case of Section 9.3.2, where a fabricated actor was rejected before it could be emitted, and the Vicksburg-president case of Section 9.8, where declining to assert was the epistemically correct act and the metric scored it as a loss.

It over-hedges when a true claim has no single edge stating it in the drafted phrasing — the five wrongly-rejected cases of Section 9.5, three of which had already resolved the exact gold answer before the rejection forced a retry that lost it. That is the same trade the development set measured at Section 8.1, where the layer cost one correct answer out of 80 and removed none that were wrong, and the same one the ablation of Section 8.8 could not detect at $n \approx 200$.

The scope of the claim this evidence supports is therefore narrow and should be stated narrowly. Verification does not make this system more accurate. It makes the answers it does give attributable, at a small and measurable cost in recall, and it provides no protection at all against the failure mode this chapter finds most often — an answer that is grounded, connected, true, and not what was asked.

9.9.3 Threats to Validity

The environment is friendlier than the knowledge base it stands in for, and this bounds every absolute number. It is the union of per-question subgraphs that the RoG distribution pre-extracted *so as to contain their own question's answer* (Section 4.2). Taking the union restores distractor mass — an agent exploring one question's topic meets edges contributed by thousands of unrelated questions — but it cannot restore what was never extracted. That is why answer reachability is certified at 97.3% on WebQSP and 99.7% on CWQ (Section 4.7): a BFS extraction from full Freebase under the same hop bound would not reach those rates, and no system here is ever asked to search a graph that might simply not contain the answer. Accuracies measured on this environment are therefore *not* comparable to evaluations over full Freebase, and the ceilings should be read as part of the result rather than as background. The mitigation is the one Section 4.2 states and it is real but narrow: every system compared here navigates the identical environment, so the *relative* ordering is sound even where the absolute level is optimistic. Every claim in Section 8.9 is relative for this reason.

Entities are keyed by name, so homonyms are merged. The distribution's triples are keyed by human-readable name rather than by machine identifier, so distinct real-world entities sharing a surface form collapse into one node. Two consequences run in opposite directions and neither is measured here. A merged node can manufacture a path between things that are not actually connected, which inflates reachability and could let a wrong answer verify as grounded — the structural check asks whether an edge exists, and a spurious edge exists. It can equally destroy a distinction the question depends on. This is inherent to name-keying rather than a defect of the construction, but it means structural groundedness is a claim about *this* graph's node identity, not about the world.

The instrumentation gap is the most serious of the remaining items. The verifier logs only the claims it rejects, never the ones it accepts with their matching triples. Wrongful acceptance is therefore not self-diagnosing: the single specimen in this census surfaced by manual trace reconstruction, and there is no way to know how many others sit inside the 748 grounded outcomes. One polarity is reported at census scale and the other anecdotally, and that asymmetry is a property of the logging decision rather than of the verifier. Section 10.3 gives the fix.

The judge fell short of its pre-registered bar. Cohen's $\kappa = 0.6995$ against a threshold of 0.7 (Section 7.6). Agreement is substantial by conventional reading, and the miss is 0.0005, but the Tier-2 numbers are reported as corroborating rather than decisive, and no conclusion here rests on them alone. The judge also runs on the same backbone as the systems it judges.

Topic entities are given, not linked. Both graph-navigating systems seed from the dataset's annotated topic entities. This isolates navigation from entity linking, which is the intent, but it means the reported accuracies assume a linking step that a deployed system would have to perform and that would introduce errors of its own.

Ablations are underpowered. Three of four conditions detect nothing at $n \approx 200$, and "no detectable effect" is not "no effect" (Section 8.8).

One relabelling was made after seeing the result it affects. A development question originally classified as mediator-heavy was reclassified as unanswerable-in-environment after its outcome was known; Chapter 6 flags it in place, and the aggregates were not rescored.

The reported accuracy has an unmeasured floor. The extraction bug of Section 9.3 converts correct reasoning into a wrong entity list on a specific question shape, so the measured accuracy on that shape is a lower bound rather than an estimate.

Benchmark defects were adjudicated by the author. The exclusion set of Section 7.7 rests on single-annotator judgements, made with the consensus signal visible. The scope limits are recorded there; the accuracy tables are not rescored, and the defects run in both directions.

Chapter 10

Conclusion

10.1 Summary of Contributions

This thesis asked what an agent gains by navigating a knowledge graph rather than retrieving from it, and what a claim-level check against the traversed subgraph adds on top. The answers are not uniform, and the value of the work lies partly in where they diverge.

Chapter 1 located the problem: hallucination in knowledge-grounded question answering is not merely a matter of asserting false things but of asserting things with no basis in the knowledge the system was given, and the standard evaluation convention — Hits@1 over a multi-valued gold set — cannot see the difference. Chapter 2 fixed the vocabulary the technical chapters assume, including the operational definition of ungroundedness this thesis measures against, and withdrew the proposal's characterisation of the navigation strategy as MCTS-inspired: what the method is, is guided best-first search with a bounded beam and backtracking, and it is described that way throughout.

Chapter 3 positioned the work against the agentic KGQA literature and identified the attribution gap it addresses: published systems bundle decomposition, search, and verification into single reported numbers, so the field has no evidence about which component earns its cost.

Chapter 4 built the knowledge environment and, more importantly, *certified* it: 2.59 million entities and 8.31 million triples with a measured answer-reachability ceiling of 97.3% and 99.7% over the full test splits; a three-stage entity resolver; and an explicit inventory of what it cannot express — no date literals, no numeric literals, no ordinals — which Chapter 9 later shows accounts for the third-largest failure category in the census. Certifying the environment before measuring on it is what allows a failure to be attributed to the system rather than to the substrate.

Chapter 5 specified the framework: a state machine over a constrained, deterministic graph tool API in which the program orchestrates and the language model only decides, with a hybrid scoring function blending embedding similarity against a batched language-model judgement, a signal-maximum backtracking trigger that fires only when every available signal is weak, and an

explicit budget meter that makes cost a first-class, logged quantity.

Chapter 6 specified the Structural Verification Layer, and reported its behaviour honestly enough to bound its own claims: a draft is decomposed into atomic claims, each checked first against traversed adjacency and then, if necessary, by batched entailment, with unsupported claims triggering targeted re-exploration or deterministic removal from the answer. The chapter also records that the repair path never executed on the development set, that ten of eighty answers were certified with no claims at all, and that entities appearing in no claim are never checked — the layer's guarantees stated as they are rather than as designed.

Chapter 7 fixed the experimental design in advance and reports the decisions that went the other way: a backbone selected, reversed, and reselected under a pre-registered tiebreaker after the first determinism estimate proved to be noise; a test-set build that read the wrong split and was caught by a contamination guard before any system ran; and a judge-validation threshold missed by 0.0005 and reported as missed.

Chapter 8 reported the measurements. AGR leads every cell of the main matrix and every one of the eight paired comparisons rejects, including both against a reimplementation of the agentic baseline running on the same graph with the same model under the same budget. That last margin is located rather than merely claimed: on the questions where the shared cap lets the baseline finish, the baseline is slightly ahead, and the whole margin comes from the 29% and 44% of questions where the cap truncates it (Section 8.4). The finding is therefore about affordability rather than about resolution quality — beam search cannot complete under a budget a deployed system would impose, and guided search can. On ComplexWebQuestions AGR's accuracy *rises* with hop count while every other system decays — the strongest available form of the multi-hop claim. It runs at half the language-model calls of the agentic baseline and never exhausts a budget that clips that baseline on 44% of multi-hop questions. Structural hallucination is eliminated: over both datasets, zero ungrounded assertions in $1{,}709$ asserted entities, against 22.1% of the $1{,}001$ the parametric control asserts. And three of four ablations detect nothing, which is reported at the same length as the one that does.

Chapter 9 read all 259 remaining failures and named each mechanism. The findings that generalise beyond this system are three. *Context-stripping*: decomposing a question can discard a qualifier that appears only in a later clause, so an early sub-objective resolves the wrong sense of an ambiguous term before the disambiguating information arrives — and in the strongest specimen, decomposition actively suppressed the one anchor that could have discriminated, driving its score into the junk tier. *The echo attractor*: the most common failure of a graph-navigating system is not to invent an entity but to surface a real, well-grounded, verifiably-true entity that sits one node away from the answer, which the verifier passes because it genuinely is grounded. And *shared attractors break consensus-based evaluation*: five independent systems converge on the same wrong answer often enough that a policy of rescoring on majority agreement would have converted this failure mode into apparent correctness.

Four contributions follow from that body of work.

The framework and its verification layer. An agentic KGQA system that returns its answer paired with the triples supporting it, built so that the supporting evidence is a byproduct of the control flow rather than a post-hoc reconstruction.

A component-level ablation. Four single-deviation conditions with paired significance testing, closing the attribution gap of Chapter 3 for this architecture — and finding that only one of the four components has a detectable accuracy effect.

Stratum-dependent decomposition. The literature treats decomposition as straightforwardly beneficial. It is not: removing the planner *improves* WebQSP accuracy by 0.083 F1 at $p = 0.006$ while cutting tokens by 30%, and trends toward harming CWQ. The asymmetry, not the pooled average, is the result, and it says that decomposition should be gated on question structure rather than applied unconditionally.

Quantified benchmark defect rates. Fifty-eight questions across the two test samples where the benchmark, not the system, needed correcting — with the two evidence signatures that distinguish a world-fact error from an annotation-pipeline error, and the provenance argument that explains where one class of them comes from.

10.2 Limitations

The limitations that most constrain what may be concluded are, in order of severity:

Wrongful acceptance is unmeasured, and the evidence is not persisted. The verifier persists only the claims it rejects. One polarity of its error is therefore reported at census scale and the other from a single specimen that surfaced by manual trace reconstruction, and no rate exists for the polarity that matters most to a hallucination claim. The same decision costs the output contract its auditability: run records store the *count* of supporting triples rather than the triples themselves (Section 6.6), so an examiner can verify that the answers came from an evidence-tracking system but cannot inspect the evidence. This is an instrumentation decision constraining a scientific conclusion, and it is the first item in Section 10.3.

Three of four ablations are underpowered. At $n \approx 200$, a component that changes five questions in a hundred cannot be resolved. "No detectable effect at this sample size" is what the data supports, and it is not the same statement as "no effect".

Structural grounding is not answer adequacy. Both navigating systems reach zero structural hallucination, so that property belongs to the navigation paradigm rather than to this thesis's verification layer, and the semantic-support metric that would separate them is validated at $\kappa = 0.6995$ on samples of 60 — underpowered for the 5-point gap it measures. The strongest evidence for the layer is therefore the precision column of the main matrix and the case-level analyses, not a groundedness differential against the agentic baseline.

The evaluation is single-environment, single-backbone, and single-annotator. One knowledge graph built from one distribution, one frozen backbone at $\approx 75\%$ trajectory stability, two benchmarks sharing a lineage, and a failure census and benchmark-defect adjudication performed by the author. Each of those is a defensible scoping decision and each bounds generality.

The agentic baseline prunes from a narrower candidate set. It considers 40 relations and 20 neighbours per step against AGR's 300 and 200, cuts that bind on roughly a third of its expansions and on almost none of AGR's (Section 7.4.4). This cannot account for the budget clipping on which Section 8.4 rests, since a narrower set is cheaper rather than dearer, but it does mean the unclipped subset understates what that baseline could resolve at equal width. Equalising the widths is specified in Section 10.3.8.

The static graph baseline retrieves one logical hop, and samples it. Its decay across hop strata is therefore partly a property of its retrieval radius and not only of static retrieval as a paradigm, and Section 7.4.3 scopes what may be concluded from it accordingly. A second bound compounds the first: it takes at most 100 edges per topic entity with no ordering imposed, and on 72.5% of the questions at least one topic entity exceeds that degree (Section 7.4.3), so on those it answers from an arbitrary sample of the neighbourhood. Both bounds depress it, and neither is a property of static retrieval as such. The per-stratum argument does not rest on it: the comparator that cannot represent a chain *at any radius* is Vector-RAG, whose context is a single verbalised triple.

Entity linking is assumed. Topic entities are given by the datasets, so the reported accuracies presume a linking step a deployed system would have to perform.

One measured defect depresses the reported accuracy. The entity-extraction bug of Section 9.3 converts complete, verifier-certified reasoning into an empty or wrong entity list on a specific question shape. The headline numbers are a floor for that shape.

10.3 Future Work

The census makes it possible to separate proposals that are backed by a counted, repeated pattern from proposals that are backed by an argument. Both are listed; they are not the same kind of claim.

10.3.1 Repairs the Census Earned

Three defects each recur with a uniform mechanism across unrelated questions, and each is small.

Entity extraction (nine instances): the step converting drafted answer text into the scored entity list defaults to the sentence's grammatical subject rather than its predicate values. Fixing it

requires the claim decomposer to be told which argument of the drafted sentence is the answer — an instruction, not an architecture.

Evaluator abandonment (five instances): relations scoring as high as 0.73 are explored and then backtracked away from, never revisited, across four unrelated domains. This reads as a threshold or backtracking-policy defect and would be diagnosed by logging the evaluator's decision against the score of the relation it discards.

Drafting commitment (six instances, all six of the other category): when the evaluator has resolved the exact gold value and the verifier has returned grounded, the drafter should commit to it rather than stating the fact and declaring it undeterminable in the same sentence.

10.3.2 Logging Accepted Claims to Make Wrongful Acceptance Measurable

Persist each accepted claim together with the triples that matched it. This converts the wrongly-accepted class from an anecdote into a rate with a denominator, makes grounded-but-wrong answers self-diagnosing from the logs rather than by manual trace reconstruction, and would let the entity-level and claim-level guarantees of Chapter 6 be audited rather than argued. The same change closes a second gap: the run record currently stores the *number* of supporting triples rather than the triples (Section 6.6), so the thesis's own output contract — answer paired with its evidence — holds inside the run but cannot be checked afterwards from any committed file. It is the highest-value change in this list relative to its cost, and it is a logging change.

10.3.3 Adaptive Decomposition Gating

The planner ablation and the context-stripping cases together specify the fix rather than merely motivating it. A decomposition step should be gated on two checks: does the rewritten sub-objective preserve the question's actual interrogative — the Phoenician case of Section 9.3 lost the word "where" entirely — and does the step resolve an entity that was already unambiguous, which is what over-decomposition does. A third check falls out of the strongest specimen: sub-objectives that carry no reference to each other are descriptors of one entity rather than a chain, and should not be sequenced as one.

10.3.4 Semantic-Level Verification

All five wrongly-rejected cases share one mechanism: the structural check requires a single edge asserting the drafted claim, and a transitively-true multi-hop fact does not present one. A check that accepts a bounded relation chain expressing the claim — rather than requiring one literal triple — would recover those answers. The relation-path formulation of RoG [9] is the natural

source for what such a chain may look like, and using it as an acceptance criterion rather than as a retrieval plan is the change being proposed. The Tier-2 judge of Section 7.5.3 is a prototype of exactly this judgement, which suggests folding a bounded form of it into the loop; the κ of 0.6995 is also a caution about how reliable such a check would be.

10.3.5 A Set-Intersection Operator for Compound Questions

composite_claim is 47 cases and CWQ's largest category. Its no_set_intersection subtype names the gap precisely: the architecture has no AND-join primitive, so a "find the overlap" sub-objective degrades into another single relation-scoring pass over an incomplete first-hop set. An explicit operator that resolves each clause of a compound question into a candidate set and intersects them is a well-specified addition to the tool API rather than a change to the agent.

10.3.6 Ordinal and Literal-Value Support

kg_gap is 44 cases and CWQ's single largest hedge category. Two of its subtypes need an operator rather than a bigger graph: a minimum-or-maximum-by-value operation for superlatives, and a literal-extraction path for numeric identifiers, ISO codes, and dates that the environment exposes as event or entity nodes rather than as plain values. Neither exists in the current tool API, and every instance in the census fails for that reason alone.

10.3.7 Path Fidelity Against Gold SPARQL Relation Chains

The third evaluation criterion of the approved proposal, deferred for the reason given in Section 7.1.2: it needs the benchmarks' gold SPARQL relation chains, which the distribution used here does not carry. Reconstructing them from the upstream releases and aligning them to this environment's relation vocabulary would measure whether the system reaches the right answer *by the right path* — a question this thesis's groundedness metrics can only approximate, and one that bears directly on the echo attractor, where a right-looking answer is reached by a path one hop short of the correct one.

10.3.8 Baselines at Equal Width and Equal Radius

Two of the three baseline bounds named in Section 10.2 are cheap to lift, and both are re-runs rather than redesigns.

Equalise the candidate widths. Re-run the agentic baseline with 300 relations and 200 neighbours per step — AGR's widths — leaving beam width, depth, budget, prompts, and scoring untouched. This is the experiment that decides whether the unclipped-subset comparison

of Section 8.4 was measuring search strategy or candidate supply, and it is the more informative of the two because that subset is the only place the thesis compares the two agentic methods at parity. Both outcomes are usable: if the baseline improves on the unclipped subset, the honest reading is that the affordability finding is the whole finding and the resolution comparison was never fair; if it does not, the resolution comparison stands at equal width, which is a stronger statement than the thesis can currently make. Note that it will also raise the clip rate, because wider candidate sets make each pruning call dearer, so the two effects have to be read apart — report the split as Section 8.4 does rather than the aggregate.

Widen the static retriever's radius, and order its fanout. The static graph baseline follows GraphRAG [4] in retrieving structured context in one shot, and expands one logical hop (Section 7.4.3), so the cleanest reading of its collapse on deep CWQ questions — that a fixed radius cannot hold a chain — is confounded with the particular radius chosen. The experiment that separates them is cheap and specified: re-run the same baseline with neighbours expanded a second time, everything else frozen, and report the same strata. Two outcomes are informative and neither is bad for this thesis. If CWQ `h2` recovers substantially, the honest conclusion is that the reported gap over static retrieval on that stratum was partly a radius artefact, and the paradigm claim narrows to what Section 7.4.3 already states. If it does not recover, or recovers while precision falls as the window fills with hub noise, the paradigm claim is strengthened by an experiment rather than by an implementation detail. The same pass should impose an ordering on the fanout cap — descending fanout, or an embedding pre-rank — so that the 100 edges kept per topic entity are a stated selection rather than an arbitrary one (Section 7.4.3); until then, the two limits are confounded and only their sum is measured. Running it costs one pass over 800 questions at baseline rates; not running it is the reason the claim is stated narrowly here.

10.3.9 Beyond This Environment

Four directions extend the work rather than repairing it. *LLM-constructed knowledge graphs* as a controlled comparison would separate what the framework contributes from what a curated environment contributes, by holding the agent fixed while varying the substrate's provenance and noise profile. *Rollout-based value estimation* would supply what Section 2.5.3 explicitly disclaims: the current scorer evaluates each candidate once and never revises it in light of what is found later, and genuine tree search over this same tool API is a well-defined next system rather than a rebranding of this one. *Temporal and multi-source environments* address the failure family that recurs throughout Chapter 9 — questions with a time qualifier the graph cannot express — and raise the question of what a claim check means when sources disagree. And *transfer to high-stakes domains* is where the auditability property, rather than the accuracy, is the point: an answer paired with the triples supporting it is worth more in a setting where a reader must be able to check it than in one where the reader only wants the answer.

10.4 Closing Remark

The result this work would most want carried forward is not the accuracy margin. It is the shape of the gap that remains. A system that traverses a real graph and copies its answers out of real triples can be made to assert nothing ungrounded — that problem is solved, and it is solved by the navigation paradigm rather than by any checking layer built on top of it. What remains, and what this thesis's census found more often than any other mechanism, is an answer that is grounded, connected, and true, and is not the answer to the question that was asked. Fact verification against a knowledge graph turns out to be the easier half of the problem; *relevance* verification is the half still open.

References

[1] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, pp. 1–55, 2025.

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[3] Y. Li, Z. Li, P. Wang, J. Li, X. Sun, H. Cheng, and J. X. Yu, "A survey of graph meets large language model: Progress and future directions," *arXiv preprint arXiv:2311.12399*, 2023.

[4] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.

[5] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems*, vol. 37, pp. 59532–59569, 2024.

[6] D. Li, Y. Niu, Y. Ai, X. Zou, B. Qi, and J. Liu, "T-GRAG: A dynamic GraphRAG framework for resolving temporal conflicts and redundancy in knowledge retrieval," in *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 11880–11889, 2025.

[7] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, L. M. Ni, H.-Y. Shum, and J. Guo, "Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph," in *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2307.07697.

[8] L. Chen, P. Tong, Z. Jin, Y. Sun, J. Ye, and H. Xiong, "Plan-on-graph: Self-correcting adaptive planning of large language model on knowledge graphs," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. arXiv:2410.23875.

[9] L. Luo, Y.-F. Li, G. Haffari, and S. Pan, "Reasoning on graphs: Faithful and interpretable large language model reasoning," in *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.01061.

[10] J. Jiang, K. Zhou, W. X. Zhao, Y. Song, C. Zhu, H. Zhu, and J.-R. Wen, "KG-Agent: An efficient autonomous agent framework for complex reasoning over knowledge graph," *arXiv preprint arXiv:2402.11163*, 2024.

[11] Y. Xu, S. He, J. Chen, Z. Wang, Y. Song, H. Tong, K. Liu, and J. Zhao, "Generate-on-graph: Treat LLM as both agent and KG for incomplete knowledge graph question answering," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. arXiv:2404.14741.

[12] J. Jiang, K. Zhou, Z. Dong, K. Ye, W. X. Zhao, and J.-R. Wen, "StructGPT: A general framework for large language models to reason over structured data," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. arXiv:2305.09645.

[13] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 7, pp. 3580–3599, 2024.

[14] C. Ma, Y. Chen, T. Wu, A. Khan, and H. Wang, "Unifying large language models and knowledge graphs for question answering: Recent advances and opportunities," in *Proceedings of the 28th International Conference on Extending Database Technology (EDBT)*, pp. 1174–1177, 2025.

[15] S. Dhuliawala, M. Komeili, J. Xu, R. Raileanu, X. Li, A. Celikyilmaz, and J. Weston, "Chain-of-verification reduces hallucination in large language models," in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 3563–3578, 2024.

[16] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[17] Y. Hu, W. Xuan, Q. Zhou, Z. Li, Y. Li, J. Hu, and F. Fang, "A self-correcting agentic graph RAG for clinical decision support in hepatology," *Frontiers in Medicine*, vol. 12, p. 1716327, 2025.

[18] K. Bollacker, C. Evans, P. Paritosh, T. Sturge, and J. Taylor, "Freebase: A collaboratively created graph database for structuring human knowledge," in *Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data*, pp. 1247–1250, 2008.

[19] W.-t. Yih, M. Richardson, C. Meek, M.-W. Chang, and J. Suh, "The value of semantic parse labeling for knowledge base question answering," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 201–206, 2016.

[20] A. Talmor and J. Berant, "The web as a knowledge-base for answering complex questions," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 641–651, 2018.

[21] I. Robinson, J. Webber, and E. Eifrem, *Graph Databases: New Opportunities for Connected Data*. O'Reilly Media, 2nd ed., 2015.

[22] N. Francis, A. Green, P. Guagliardo, L. Libkin, T. Lindaaker, V. Marsault, S. Plantikow, M. Rydberg, P. Selmer, and A. Taylor, "Cypher: An evolving query language for property graphs," in *Proceedings of the 2018 International Conference on Management of Data*, pp. 1433–1445, 2018.

[23] J. Berant, A. Chou, R. Frostig, and P. Liang, "Semantic parsing on Freebase from question-answer pairs," in *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pp. 1533–1544, 2013.

[24] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

[25] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems*, vol. 35, pp. 24824–24837, 2022.

[26] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[27] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR)*, 2023.

[28] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pp. 6769–6781, 2020.

[29] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pp. 3982–3992, 2019.

[30] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[31] J. Pearl, *Heuristics: Intelligent Search Strategies for Computer Problem Solving*. Addison-Wesley, 1984.

[32] L. Kocsis and C. Szepesvári, "Bandit based monte-carlo planning," in *Proceedings of the 17th European Conference on Machine Learning*, pp. 282–293, 2006.

[33] C. B. Browne, E. Powley, D. Whitehouse, S. M. Lucas, P. I. Cowling, P. Rohlfshagen, S. Tavener, D. Perez, S. Samothrakis, and S. Colton, "A survey of monte carlo tree search methods," *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 4, no. 1, pp. 1–43, 2012.

[34] H. Luo, H. E, Y. Guo, Q. Lin, X. Wu, X. Mu, W. Liu, M. Song, Y. Zhu, and L. A. Tuan, "KBQA-o1: Agentic knowledge base question answering with monte carlo tree search," in *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2501.18922.

[35] T. Shen, J. Wang, X. Zhang, and E. Cambria, "Reasoning with trees: Faithful question answering over knowledge graph," in *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 3138–3157, 2025.

[36] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[37] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[38] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[39] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.

[40] X. Tan, X. Wang, Q. Liu, X. Xu, X. Yuan, and W. Zhang, "Paths-over-graph: Knowledge graph empowered large language model reasoning," *arXiv preprint arXiv:2410.14211*, 2024.

[41] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *International Conference on Learning Representations (ICLR)*, 2024.

[42] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[43] S. Min, K. Krishna, X. Lyu, M. Lewis, W.-t. Yih, P. W. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi, "FActScore: Fine-grained atomic evaluation of factual precision in long form text generation," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 12076–12100, 2023.

[44] F. F. Bayat, L. Zhang, S. Munir, and L. Wang, "FactBench: A dynamic benchmark for in-the-wild language model factuality evaluation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 33090–33110, 2025.

[45] Y. Gu, S. Kase, M. Vanni, B. Sadler, P. Liang, X. Yan, and Y. Su, "Beyond I.I.D.: Three levels of generalization for question answering on knowledge bases," in *Proceedings of the Web Conference 2021*, pp. 3477–3488, 2021.

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Appendix A

Prompt Templates

Seven prompts constitute the whole of AGR's interface to the language model. Six more serve the baselines: two shared by the three static baselines, and four belonging to the reimplemented agentic baseline. All thirteen are reproduced here verbatim from the source that sends them, in the order a question encounters them. The agentic baseline's four matter more to a sceptical reader than any of AGR's own, because Section 7.4.4 rests the central comparison on a reimplementation rather than on the authors' code, and these prompts are what that reimplementation actually is. Braces written `{like this}` are Python format fields; doubled braces `{{...}}` are literal braces in the emitted text.

This appendix carries a checker rather than a promise. scripts/check_appendix_prompts.py extracts each constant from the source and asserts that it still appears here. Nothing in the LaTeX build could detect the divergence on its own, and the reproducibility argument of Section 7.10 depends on the two not diverging: the response cache keys on prompt text, so a prompt edited in one place and not the other invalidates every cached run as well as this appendix.

Two properties of this set matter more than any individual wording. First, no prompt contains a configuration value. Neither α nor τ nor any budget number appears in any template, which is what allows one cached response to be shared across every condition of the α–τ sweep (Section 8.1). Second, no prompt asks the model to respect a budget. Budgets are counters in code (Section 5.8); the model is never asked to stop, only prevented from continuing.

A.1 Planner

Sent once per question, unless the planner is ablated. Section 5.4 describes the four-archetype few-shot design and the topic-mention restriction; this is the template that implements both. The instruction to use as few sub-objectives as possible is the provision Section 8.9 finds to have been insufficient on the shallower benchmark.

```
You decompose questions into a minimal ordered chain of sub-objectives for
↪
navigating a knowledge graph. Each sub-objective must be answerable by
following one relation (possibly through an event node). Use #N to
↪ reference
the result of sub-objective N. Use as FEW sub-objectives as possible.

For topic_mentions, extract ONLY specific named entities that appear
literally in the question (people, places, organizations, works, products)
↪ .
Do NOT extract generic type words such as "celebrity", "university",
"country", "senator", "airport", or "currency" -- these describe what
↪ kind
of answer is wanted, not where to start searching.

Example 1 (single hop -- do NOT decompose):
Question: "who is the president of France?"
{"sub_objectives": ["find the president of France"],
 "topic_mentions": ["France"]}

Example 2 (composition):
Question: "who directed the film that won Best Picture in 1998?"
{"sub_objectives": ["find the film that won Best Picture in 1998",
                    "find the director of #1"],
 "topic_mentions": ["Best Picture"]}

Example 3 (conjunction):
Question: "which countries border France and have Spanish as official
↪ language?"
{"sub_objectives": ["find countries that border France",
                    "find which of #1 have Spanish as an official
↪ language"],
 "topic_mentions": ["France", "Spanish"]}

Example 4 (superlative):
Question: "what is the largest city in the country where the Nile ends?"
{"sub_objectives": ["find the country where the Nile ends",
                    "find the largest city in #1"],
 "topic_mentions": ["Nile"]}

Question: "{question}"
```

The plan this returns is validated rather than trusted. Four checks run before it is accepted, and any failure installs the single-objective fallback described in Section 5.4: the sub-objective list must be non-empty, it must hold no more than four entries, every `#N` reference must point strictly backwards, and at least one topic mention must be present.

A.2 Relation Scorer

Sent once per expansion step, scoring all candidate edges across all anchors in a single call. This is the prompt that carries the property discussed in Section 5.5.2: it mentions neither α nor the anchor ordering, so its response is reusable across every sweep condition.

```
Sub-objective: {objective}

Candidate knowledge-graph edges (anchor entity --relation-->):
{candidates}

For EACH candidate, rate from 0.0 to 1.0 how likely following that edge
↪ from
that anchor leads toward completing the sub-objective. 1.0 = directly
answers it, 0.0 = irrelevant. Judge the RELATION MEANING in context of
↪ the
anchor, not surface word overlap.
```

The final instruction exists because of the failure it names. Judging surface word overlap is what an embedding does, and the blend of Section 5.5.2 already supplies that signal; the model is asked for the part the embedding cannot give.

A.3 Evaluator

Sent once per navigation cycle. The fact window is the thirty triples ranked most relevant to the current sub-objective, not the thirty most recent — the change forced by the failure recorded in Table 5.4.

```
Current sub-objective: {objective}
Recently retrieved facts:
{facts}
Remaining budget: {d_left} more expansions, {b_left} more backtracks.

Decide:
- "answer" if the facts complete the sub-objective (list satisfying
↪ entities
in "resolved", exactly as named in the facts),
- "continue" if this direction is promising but incomplete,
- "backtrack" if this direction is a dead end.
```

The remaining budget is stated for context, not for enforcement. A model that ignores it changes nothing: the counters in Section 5.8 terminate the run regardless of what the evaluator decides.

A.4 Drafter and Claim Decomposer

One call produces the draft answer, its asserted entity list, and the atomic claims the verification layer will check. Section 6.2 explains why these are one call rather than three: a draft and a claim set produced separately can disagree about what was asserted, and the layer would then verify something the reader never sees.

Four provisions here are load-bearing, and Section 6.2 gives the case that forced each. Copying entity names exactly is what keeps the graph's spelling — University Yale rather than Yale University — in the answer. The distinction between entities the draft *asserts* and entities it merely mentions is what stops a hedge from being scored as an assertion. The anti-vacuity instruction is what stops the model answering "a university" to a *which university* question. And requiring a hedge to carry zero claims is what makes an abstention verifiable rather than merely unverified.

```
Question: {question}
Facts retrieved from the knowledge graph:
{facts}
Entities identified as promising during exploration:
{candidates}

1. Draft an answer using ONLY the facts and candidates above. Use entity
   names exactly as written. If insufficient, say what could not be
   determined.
2. List the answer entities themselves in "answer_entities", copied
↪ EXACTLY
   as they are named in the facts above -- these are the entities your
↪ draft
   asserts as the answer, not every entity it mentions. If your draft is
↪ a
   hedge ("could not be determined"), "answer_entities" MUST be [].
   IMPORTANT: "answer_entities" must be specific entities of the kind the
   question asks for (a person for "who", a place for "where", etc.).
   Never restate the question or name entity types/categories as the
   answer. If you cannot name specific answer entities from the facts,
   your draft must be a hedge and "answer_entities" must be [].
3. Decompose your draft into atomic factual claims, each as
   {{"h": entity name, "r": short relation phrase, "t": entity name}}.
   Only include claims your draft actually asserts. A hedged draft has
↪ zero
   claims.
```

A.5 Entailment Check

Sent only for claims that the two structural routes of Section 6.3.1 could not settle. It is the semantic fallback, and the definition of *unsupported* is given explicitly because the default reading of the word is too permissive: a model asked merely whether a claim is supported will accept anything it believes to be true.

```
Retrieved knowledge-graph triples:
{facts}

For each claim below, answer whether it is directly supported by the
↪ triples
above (true) or not (false). Unsupported includes: plausible but absent,
contradicted, or only inferable via outside knowledge.
Claims:
{claims}
```

The three enumerated cases are the three ways a fluent model fails this task. *Plausible but absent* is the hallucination the layer exists to catch. *Contradicted* is rarer and easier. *Only inferable via outside knowledge* is the one that matters most for the thesis's claim, because a model drawing on parametric memory here would certify exactly the content the graph was supposed to constrain.

A.6 Rewriter

Sent when verification rejects at least one claim but at least one entity survives. It is the mechanism by which an unsupported claim is removed rather than merely flagged.

```
Question: {question}
Draft answer: {draft}
The following claims in the draft are NOT supported by the knowledge
↪ graph:
{unsupported}
The following entities remain verified and MUST be kept in the answer,
named exactly as written: {kept}

Rewrite the draft: keep the verified entities and their supported claims,
and remove or explicitly hedge the unsupported claims. If the kept list
↪ is
empty, state that the answer could not be verified against the knowledge
graph. Respond with prose only.
```

The kept list is computed in code by the deterministic filter of Section 6.5, not proposed by the model. The rewrite call may reword the answer but cannot change which entities it asserts, and Section 6.7 reads the case in which that separation decided whether a question was scored

correct.

A.7 Answerer

Sent on the path where verification is skipped, which is the draft-only ablation of Section 8.8, and on the give-up path where the budget drained before a draft existed.

```
Question: {question}
Facts retrieved from the knowledge graph:
{facts}

Entities identified as promising during exploration:
{candidates}

Answer using ONLY these facts and candidates. Give the answer entity name
↪ (s)
exactly as they appear above. If the information is insufficient, state
↪ what
could not be determined.
Also list in "answer_entities" the entities your answer
asserts (exactly as named above); [] if it asserts nothing.
```

Comparing this against the drafter of Section A.4 shows what the draft-only ablation removes. The anti-vacuity provision is absent here, and Section 8.8 reports the hedge-rate consequence.

A.8 Baseline Prompts

The baselines share the backbone, the budget, and the answer-entity contract of the full system, so that the comparison isolates retrieval rather than output format. All three emit `{"answer": "...", "answer_entities": ["..."]}`.

No-retrieval

The parametric control. It receives the question and nothing else, which is what makes it a measurement of memorisation rather than of retrieval (Section 7.4).

```
Answer this question with the name(s) of the answer entity or
entities. Be concise. If you do not know, say so and return [].

Question: {question}
```

Vector-RAG and GraphRAG

Both use the template below; GraphRAG imports it from the Vector-RAG module precisely so that the two differ only in how `{facts}` was retrieved. Section 7.4 records the consequence noted in Section 7.4.5: this prompt is more abstinent than the full system's drafter, and the resulting hedge rates are a property of the prompt as much as of the retrieval.

```
Facts (retrieved from a knowledge graph):
{facts}

Question: {question}
Answer using ONLY the facts above. Name entities exactly as written in
↪ the
facts. If the facts are insufficient, say so and return [].
```

A.9 Agentic Baseline (Think-on-Graph)

Four prompts implement the reimplementation of Section 7.4.4, at beam width 3 and depth 3. They are given in full because the thesis's central comparison is against this code rather than against published numbers (Section 8.4), and a reader who wants to challenge that comparison should be able to read what was actually run. The structure follows the original: prune relations, expand, prune entities, ask whether what has been gathered suffices, and answer if it does.

The contrast with AGR's scorer is worth noting while reading the first of them. This prompt *selects* a set of relations; AGR's scores every candidate on $[0, 1]$ and blends that with an embedding similarity (Section 5.5.2). Selection cannot express "all of these are poor", which is what the low-signal backtrack trigger needs to fire on.

Relation pruning

```
Question: {question}
Current entity: {entity}
Available relations (relation, direction, fanout):
{relations}
Select up to {w} relations most likely to lead to the answer.
Return {{"relations": [{{"rel": "...", "dir": "in|out"}}]}}.
```

Entity pruning

```
Question: {question}
Candidate entities reached (via {entity} --{rel}-->):
{entities}
```

```
Select up to {w} entities most likely on the path to the answer.
Return {{"entities": ["..."]}}.
```

Sufficiency check

This is the prompt the call budget most often interrupts. It runs once per depth, and a run clipped before it returns true answers from whatever the beam has gathered — the mechanism behind the clipped-subset accuracies of Table 8.4.

```
Question: {question}
Knowledge triples gathered so far:
{triples}
Can the question be answered from these triples alone?
Return {{"sufficient": true|false}}.
```

Answer

```
Question: {question}
Knowledge triples:
{triples}
Answer using ONLY these triples; name entities exactly as written.
If insufficient, say so and return [].
```

Appendix B

Tool API Specification and Cypher Templates

Section 5.2 argues that the model must never author a query. This appendix gives the five operations that follow from that decision, with the Cypher each one executes. Every query below is a fixed template. The model supplies values for the $-parameters and nothing else, so a malformed model output produces a failed lookup rather than a syntax error, and the error taxonomy of Chapter 9 never has to separate reasoning failures from query-authoring failures.

Two limits appear throughout. max_relations, set to 300, caps the relation list returned for an entity; max_fanout, set to 200, caps the neighbours returned for one edge. Table 5.4 records why the first was raised from its original value of 80.

Every invocation is appended to a JSONL log with its arguments, its result, and its wall-clock duration in milliseconds. That log is the per-call record the latency figures of Section 8.7 are computed from, and for the determinism probes it is the only surviving per-question artifact.

B.1 search_entity

Resolves a surface form to graph entities. This is the only entry point from text to graph, and the only operation that consults an embedding. It delegates to the three-tier resolver of Section 4.6 and returns the tier that produced the hit, so that a downstream failure can be attributed to resolution rather than to navigation.

```
search_entity(surface_form, k=5)
  -> [{id, name, score, tier}, ...]     up to k candidates

Tier 1  exact:     MATCH (e:Entity {name:$m}) RETURN e.id, e.name
Tier 2  lexical:   CALL db.index.fulltext.queryNodes('entity_name', $q)
                     YIELD node, score
                   RETURN node.id, node.name, score LIMIT $k
```

```
Tier 3  vector:    CALL db.index.vector.queryNodes('entity_embedding', $k,
↪  $v)
                     YIELD node, score
                   RETURN node.id, node.name, score
```

Tier 2 accepts its top hit only when the full-text score exceeds a threshold of 1.0. Without that gate the index returns a ranked list for almost any input, including inputs with no correct answer in the graph, and tier 3 would never be reached.

B.2 get_relations

Returns the relation types incident to an entity, in *both* directions, each with its fanout count. Direction is carried explicitly because Freebase edges are asserted in one direction only, and the answer to a question is as often reached by traversing an edge backwards as forwards.

```
get_relations(entity_id)
  -> [{rel, dir, n}, ...]     sorted by n descending, capped at
↪ max_relations

MATCH (e:Entity {id:$id})-[r]->()
RETURN r.fb_name AS rel, 'out' AS dir, count(*) AS n
UNION ALL
MATCH (e:Entity {id:$id})<-[r]-()
RETURN r.fb_name AS rel, 'in' AS dir, count(*) AS n
```

Administrative predicates are filtered before the cap is applied, by prefix: common.topic., common., freebase., type., kg., user., dataworld., rdf-schema# and owl#. These are schema bookkeeping rather than facts about the world. They also have very high fanout, so filtering before sorting is what keeps them from consuming the whole list.

B.3 get_neighbors

Expands one edge from one entity. This is the operation that makes Freebase's mediator nodes invisible to the agent: where a neighbour is a mediator, the tool hops through it in the same call and returns the entity on the far side, carrying the via chain that reached it.

```
get_neighbors(entity_id, relation, direction='out')
  -> {neighbors: [{id, name, via[, cvt]}, ...], truncated: bool}

MATCH (e:Entity {id:$id})
MATCH (e)-[r {fb_name:$rel}]->(n)        -- or <-[r]- when direction='in'
RETURN n.id, n.name, n.is_cvt LIMIT $cap
```

```
-- for each neighbour with is_cvt = true, one further hop:
MATCH (c:Entity {id:$cid})-[r2]-(m:Entity)
WHERE m.id <> $orig AND m.is_cvt = false
RETURN m.id, m.name, r2.fb_name AS rel2 LIMIT $cap
```

The via chain is what makes the traversal auditable. A two-element via means the result was reached through a mediator, and the cvt field names the mediator that was crossed. Section 4.3 states the two conventions this creates: a mediator counts as two edges wherever the thesis measures distance in the graph (the reachability gate, the hop strata), and as one step against the agent's depth budget, because the agent should not be charged twice for an artifact of the encoding.

The truncated flag was intended to report whether the fanout cap was hit, so that a truncated expansion — a plausible cause of a later failure — could be distinguished from a relation that was simply never explored. It does not deliver that, for the two reasons Section 5.8 records: it is returned to the caller but never written to the tool log, and it is computed on the row count before mediator expansion while the list is truncated after it. Nothing in this thesis reads it.

B.4 verify_triple

Asks whether a specific relation holds between two specific entities. It was built as the relation-aware structural route of the verification layer, and it is *not called by any node in the final design*: free-text claim relations do not map onto Freebase predicates, so requiring the relation to match rejected true claims, and verify_connection replaced it. Section 5.2.2 gives that history and Section 6.3.1 gives the two routes the verifier ended up with. It is reproduced here because it remains part of the tool API, is exercised by the integration tests, and is the natural primitive for the relation-aware check Section 10.3 proposes.

```
verify_triple(head_id, relation, tail_id)
  -> {direct, via_cvt, supported}

MATCH (h:Entity {id:$h}), (t:Entity {id:$t})
RETURN
  EXISTS { MATCH (h)-[{fb_name:$rel}]-(t) } AS direct,
  EXISTS { MATCH (h)-[{fb_name:$rel}]-(c:Entity {is_cvt:true})-[]-(t) }
    AS via_cvt
```

Both patterns are undirected. A claim is supported if either holds. The via_cvt arm exists for the same reason as the mediator hop in get_neighbors: a fact stated through a mediator is still a fact, and a verifier that missed it would reject true claims about exactly the compositional questions the thesis is about.

B.5 verify_connection

Asks only whether two entities are adjacent, directly or through one mediator. The relation is not named.

```
verify_connection(a_id, b_id)
  -> {direct, via_cvt, connected}

MATCH (a:Entity {id:$a}), (b:Entity {id:$b})
RETURN EXISTS { MATCH (a)-[]-(b) } AS direct,
       EXISTS { MATCH (a)-[]-(c:Entity {is_cvt:true})-[]-(b) } AS via_cvt
```

This is the fifth operation, and the one not in the original design. Section 5.2 records that it was added when the verification layer of Chapter 6 needed a relation-agnostic adjacency test. Its permissiveness is deliberate and is also its weakness: Section 6.8 analyses wrongful acceptance as a direct consequence of asking whether two entities are connected at all rather than whether they are connected in the way the claim asserts.

Appendix C

Sampled Question IDs and Run Manifests

Every number this thesis reports is computed over the question sets listed here. Section 7.2 describes how the two test samples were drawn and certified; this appendix records their contents, so that a reader can reproduce a reported figure over exactly the questions it was computed from. The identifiers are the dataset's own. A CWQ identifier carries the WebQSP identifier it was derived from as its prefix, which is why some of them are long.

Table C.1: The three question sets. The test samples are stratified by hop count, the development set by question shape.

Set	n	Strata
test_webqsp	400	h1 256, h2 128, h3plus 4, unreachable 12
test_cwq	400	h1 137, h2 211, h3plus 49, unreachable 3
dev80	80	conjunction 13, cvt_heavy 10, one_hop 18, two_hop 32, unanswerable_env 5, unanswerable_fake 2

C.1 WebQSP test sample ($n = 400$)

Source: results/phase4/test_webqsp.json.

```
WebQTest-0 WebQTest-100 WebQTest-1003 WebQTest-1007 WebQTest-1010 WebQTest-1011 WebQTest-1014
WebQTest-1020 WebQTest-1022 WebQTest-103 WebQTest-1035 WebQTest-1047 WebQTest-105
WebQTest-1057 WebQTest-1059 WebQTest-1061 WebQTest-1063 WebQTest-1067 WebQTest-1069
WebQTest-1077 WebQTest-1079 WebQTest-108 WebQTest-1080 WebQTest-1090 WebQTest-1095
WebQTest-1097 WebQTest-1099 WebQTest-1108 WebQTest-1110 WebQTest-1111 WebQTest-1114
WebQTest-1115 WebQTest-1120 WebQTest-1121 WebQTest-1130 WebQTest-1133 WebQTest-114
WebQTest-1147 WebQTest-1151 WebQTest-1169 WebQTest-1170 WebQTest-1177 WebQTest-1178
WebQTest-118 WebQTest-1187 WebQTest-1189 WebQTest-1199 WebQTest-121 WebQTest-1215 WebQTest-122
WebQTest-1226 WebQTest-1233 WebQTest-1239 WebQTest-124 WebQTest-1242 WebQTest-1259
WebQTest-1267 WebQTest-1269 WebQTest-127 WebQTest-1271 WebQTest-128 WebQTest-1281
WebQTest-1289 WebQTest-129 WebQTest-1291 WebQTest-1292 WebQTest-1296 WebQTest-1304
WebQTest-1309 WebQTest-1313 WebQTest-132 WebQTest-1321 WebQTest-1322 WebQTest-1324
WebQTest-1327 WebQTest-133 WebQTest-1330 WebQTest-1334 WebQTest-1339 WebQTest-1350
WebQTest-1353 WebQTest-1354 WebQTest-1362 WebQTest-1367 WebQTest-1369 WebQTest-1370
```

WebQTest-1373 WebQTest-1376 WebQTest-1381 WebQTest-1387 WebQTest-1399 WebQTest-1416
WebQTest-1422 WebQTest-1426 WebQTest-1445 WebQTest-1452 WebQTest-1456 WebQTest-1457
WebQTest-1458 WebQTest-1461 WebQTest-1464 WebQTest-1465 WebQTest-147 WebQTest-1470
WebQTest-1471 WebQTest-1477 WebQTest-1484 WebQTest-1485 WebQTest-1490 WebQTest-1493
WebQTest-150 WebQTest-1501 WebQTest-1520 WebQTest-1524 WebQTest-153 WebQTest-1534
WebQTest-1548 WebQTest-1550 WebQTest-1555 WebQTest-1556 WebQTest-1557 WebQTest-1560
WebQTest-1561 WebQTest-1564 WebQTest-1566 WebQTest-1567 WebQTest-1572 WebQTest-1575
WebQTest-1584 WebQTest-1585 WebQTest-1595 WebQTest-16 WebQTest-161 WebQTest-1611 WebQTest-1618
WebQTest-1620 WebQTest-1628 WebQTest-1629 WebQTest-1630 WebQTest-1633 WebQTest-1635
WebQTest-164 WebQTest-1646 WebQTest-165 WebQTest-1650 WebQTest-1655 WebQTest-1658
WebQTest-1667 WebQTest-1668 WebQTest-1669 WebQTest-1673 WebQTest-1674 WebQTest-1679
WebQTest-1680 WebQTest-1682 WebQTest-1687 WebQTest-1688 WebQTest-1698 WebQTest-1707
WebQTest-1711 WebQTest-172 WebQTest-1723 WebQTest-1724 WebQTest-1725 WebQTest-1730
WebQTest-1733 WebQTest-1736 WebQTest-1737 WebQTest-1744 WebQTest-1752 WebQTest-1758
WebQTest-1759 WebQTest-176 WebQTest-1760 WebQTest-1763 WebQTest-1764 WebQTest-1767
WebQTest-1768 WebQTest-177 WebQTest-1770 WebQTest-1782 WebQTest-1791 WebQTest-1793
WebQTest-1794 WebQTest-1795 WebQTest-1796 WebQTest-1813 WebQTest-1822 WebQTest-1824
WebQTest-1829 WebQTest-1835 WebQTest-1836 WebQTest-1840 WebQTest-1847 WebQTest-1857
WebQTest-186 WebQTest-1864 WebQTest-1868 WebQTest-1876 WebQTest-1877 WebQTest-1897
WebQTest-1898 WebQTest-1902 WebQTest-1907 WebQTest-1909 WebQTest-191 WebQTest-1914
WebQTest-1915 WebQTest-1919 WebQTest-1920 WebQTest-1923 WebQTest-1928 WebQTest-1929
WebQTest-193 WebQTest-1937 WebQTest-194 WebQTest-1940 WebQTest-1942 WebQTest-195 WebQTest-1954
WebQTest-1962 WebQTest-1967 WebQTest-1968 WebQTest-1969 WebQTest-1971 WebQTest-1972
WebQTest-1976 WebQTest-1978 WebQTest-1979 WebQTest-1981 WebQTest-1985 WebQTest-1993
WebQTest-200 WebQTest-2002 WebQTest-2003 WebQTest-2006 WebQTest-2017 WebQTest-2020
WebQTest-205 WebQTest-206 WebQTest-209 WebQTest-211 WebQTest-215 WebQTest-225 WebQTest-23
WebQTest-231 WebQTest-233 WebQTest-234 WebQTest-239 WebQTest-245 WebQTest-254 WebQTest-256
WebQTest-262 WebQTest-264 WebQTest-273 WebQTest-284 WebQTest-286 WebQTest-291 WebQTest-296
WebQTest-298 WebQTest-3 WebQTest-301 WebQTest-304 WebQTest-308 WebQTest-310 WebQTest-311
WebQTest-314 WebQTest-316 WebQTest-329 WebQTest-334 WebQTest-336 WebQTest-341 WebQTest-342
WebQTest-359 WebQTest-362 WebQTest-371 WebQTest-377 WebQTest-379 WebQTest-38 WebQTest-383
WebQTest-388 WebQTest-397 WebQTest-403 WebQTest-409 WebQTest-410 WebQTest-417 WebQTest-419
WebQTest-425 WebQTest-43 WebQTest-431 WebQTest-432 WebQTest-433 WebQTest-440 WebQTest-441
WebQTest-447 WebQTest-462 WebQTest-464 WebQTest-469 WebQTest-477 WebQTest-479 WebQTest-480
WebQTest-487 WebQTest-490 WebQTest-513 WebQTest-514 WebQTest-516 WebQTest-518 WebQTest-519
WebQTest-523 WebQTest-537 WebQTest-542 WebQTest-543 WebQTest-545 WebQTest-555 WebQTest-567
WebQTest-579 WebQTest-585 WebQTest-59 WebQTest-599 WebQTest-605 WebQTest-608 WebQTest-617
WebQTest-618 WebQTest-624 WebQTest-636 WebQTest-637 WebQTest-642 WebQTest-658 WebQTest-66
WebQTest-661 WebQTest-663 WebQTest-665 WebQTest-670 WebQTest-689 WebQTest-698 WebQTest-701
WebQTest-703 WebQTest-704 WebQTest-710 WebQTest-721 WebQTest-725 WebQTest-734 WebQTest-735
WebQTest-737 WebQTest-738 WebQTest-74 WebQTest-742 WebQTest-748 WebQTest-749 WebQTest-769
WebQTest-776 WebQTest-783 WebQTest-784 WebQTest-788 WebQTest-8 WebQTest-812 WebQTest-83
WebQTest-830 WebQTest-832 WebQTest-838 WebQTest-848 WebQTest-86 WebQTest-861 WebQTest-865
WebQTest-866 WebQTest-867 WebQTest-869 WebQTest-873 WebQTest-875 WebQTest-88 WebQTest-881
WebQTest-884 WebQTest-893 WebQTest-896 WebQTest-901 WebQTest-910 WebQTest-911 WebQTest-915
WebQTest-919 WebQTest-923 WebQTest-925 WebQTest-93 WebQTest-933 WebQTest-936 WebQTest-942
WebQTest-958 WebQTest-959 WebQTest-96 WebQTest-960 WebQTest-966 WebQTest-969 WebQTest-973
WebQTest-975 WebQTest-978 WebQTest-980 WebQTest-983 WebQTest-985 WebQTest-991 WebQTest-994
WebQTest-996

C.2 CWQ test sample ($n = 400$)

Source: results/phase4/test_cwq.json.

WebQTest-1000_0adacc5f8d85e317c77fbf2ffa83dae9 WebQTest-1000_1d3a86f839023bb35d439f56a673dbe6
WebQTest-1000_6099ea03f4fd2476605c4a513318d29c WebQTest-1000_7457bb008a1da743b19ff9ce3a5cac63

WebQTest-1002_806620bbf6ca7cf3ced5d4ff130c714d WebQTest-1002_9a330187030f30935208376dd179f40d
WebQTest-1012_131eaad9f6b360a7166467b5784470d4 WebQTest-1012_73741811f34519b29f7d19ccfd4d9553
WebQTest-1012_872253e47dd6ddaa213ff31eeda8783b WebQTest-106_8a7ae3422867fd4b2d40b06930915468
WebQTest-106_c8290c381954f40b4b920252a7df33da WebQTest-1072_92f27159a96f716c1691371e64e5a604
WebQTest-1091_61efa0b17cc53c2187422d88bf670f22 WebQTest-1120_5fd34c6d0f1cf393b2bd58d6e9690993
WebQTest-1120_c78405f6f2157645f1098f8a871f1bc2 WebQTest-1168_1591d6a0e6cef93c6863edb2eb0e198f
WebQTest-1171_a8ed8aa8e8eaf11e4b00b81b2ac2ac23 WebQTest-1178_09e61020934818b85641a22821e7a455
WebQTest-1251_cbf2f20f6caf754bc49d672ca7b150b7 WebQTest-1251_cf6cc4cc9ed790243a390f155ae72256
WebQTest-1260_018dc0c070e53555fa37f63d4b83ca87 WebQTest-1260_c4e06c3a9e4b4f10bd1ae97f1742c198
WebQTest-12_5373d9d04da65179a41e5d4669db121e WebQTest-12_7b54b31f3e5a6273f4fd2a20e565ec6d
WebQTest-12_974c964a1a2810facfeb4b1a996e878a WebQTest-1301_5714a7a5f1feec1348045800b0c0c28c
WebQTest-1320_2492cfb926882a4bf9c580ba4f7c700a WebQTest-1348_3f44e0c4fd23a8334a972ebb812e0e35
WebQTest-1348_9cd662e0ee5c2ffbbbb0dc86053b1680 WebQTest-1348_b7598df908bf8cbe941f82e1cefaec28
WebQTest-1379_8b1ff0551fb22f1d4e54d33d3656b8e8 WebQTest-1384_9440ac3bd6d8053ca4d553da408a7cb0
WebQTest-1387_e519712761951b558bb7cbf4ea39198a WebQTest-1470_888604e15ae965565efd2aa1eb615bab
WebQTest-1473_3d4ecc1165ae1c21499afb8f7b9e25a3 WebQTest-1513_5422fa602469c38418171ad342583759
WebQTest-1528_04810196a28df9032811889ca112f28b WebQTest-1528_3c737d4a4dbf3b5acb082a4a1e43d792
WebQTest-1528_505f2eda6be64caf7b48a166de833564 WebQTest-1528_6ecf5c92092ae9080307178a28b8d589
WebQTest-1528_ac9d054b50d6d55401844da4807076b2 WebQTest-1547_9a301711fdf36a2e2d6fdd7b21d65d3c
WebQTest-1560_210d8463ee35ff65448a711c4461f122 WebQTest-1560_9b121a33fc0ee97ee3a3c43ee4e02a78
WebQTest-1569_0b5333d98ef87008aa02d1fbc1554b05 WebQTest-1603_6f7bafc08148c28d022ab04b0e6a4aaf
WebQTest-1603_7ca9a8a41caa3634b454484965413fd3 WebQTest-1686_554811ebe1463287ee640a214683ea57
WebQTest-1686_80cf1e206617287917f0af7538f69ab8 WebQTest-1686_c374fe555611a2e3aba1ba9ffd739d13
WebQTest-1705_35c94dc6944d964963b44949d8ed512c WebQTest-1736_125140bfa1a60527bde6e40dce7fe54a
WebQTest-1785_23fbfc07989820c1a44f16de0e5291be WebQTest-1785_2fd3a482f02db22067954809fe7b222b
WebQTest-1785_6e1e55b0316051b2aec513f70a70b4e2 WebQTest-1785_759908f37fd43c40a532a73779dd793d
WebQTest-1785_98806527b045c387792db297c407c023 WebQTest-1785_9ecb1e0e419ee56c63da7b3da5e40ac9
WebQTest-1785_bd2684abe96767b0564cea78024983f2 WebQTest-1785_e0f5bd8c1ccde7ddfa3be8bdf3122ebc
WebQTest-1785_ee9df68984cde21c3be056caefd6613f WebQTest-1785_fdb7a181aeebb9a12bea9e01a41dd606
WebQTest-1797_194b220adab0abb76bdf49354cfdf3ff WebQTest-1797_202ded8e5df1c80ea9770154e24e5ffb
WebQTest-1797_2fb9e2823ccf35d2103fa8846d6f2ca8 WebQTest-1797_345a0887c7012fd7647ad0c6f9783e67
WebQTest-1797_5a1c66f408118c6342c4a0eb350b58de WebQTest-1797_dece4ddacc57788e31bf70edc99f1f37
WebQTest-1817_1bbd9ffd7b604510a3d6751b9f8ef796 WebQTest-1823_a8cd75f8533d96e3be293401d63fad4b
WebQTest-1840_3d474acada1d3230d94e79aad9206043 WebQTest-1923_037ebc0e903ab6eb5ef8ce44ecd3bd64
WebQTest-1923_29c81279ed9a982e12f82e764083db76 WebQTest-1923_2c5591f1dbe7402405004d7a0836dd3b
WebQTest-1923_3dac7aa955af6ee2ede38f6d42711a00 WebQTest-1923_44adab808bc1863fab0f07e6556afd67
WebQTest-1923_7084416ab9f72f1f1f8fc3ce7871ee4a WebQTest-1923_93e2fc83e55c4e0deaa4671a531ca523
WebQTest-1923_963f10fbb72072fc718bd8cc3e0ec12d WebQTest-1923_a64ef0f5ce397a5e1ef6fcd550ebfcfb
WebQTest-1923_a90c08a839ac4d0ac22ebbf436bb578b WebQTest-1923_b10bde702bb6c68ce3cdced85d5aff02
WebQTest-1923_eaeaff16b51e52669edc90501a373c61 WebQTest-212_7905d5d52ac1a17f68996d4a2245e682
WebQTest-213_024fd6ca0b4cb30927c22e93a552ae6c WebQTest-213_05bdda30187b7cfb104ec6e7d5ecec6a
WebQTest-213_3ec496c94ff4f83167e7d83479fa7082 WebQTest-213_4d64427814a804819d7bd73a4187dbe2
WebQTest-213_71621b15ae9777fca8c1becaa188a3bb WebQTest-213_7debc6a8247700de6db4ac7b0f4cc0e1
WebQTest-213_d9993d3e6f665d0c937cf7c0209ff9f8 WebQTest-302_88f3f649eed28a53b8a5479dc2c98cb1
WebQTest-361_5f8c05737061d5501b6a23a978b5589b WebQTest-389_27f6987805010183edbeba6b91a3cafd
WebQTest-389_544dcbf5d728f23ac97aa05f025a646a WebQTest-43_a29af73bf225f568206dcf810358236f
WebQTest-450_b68e8f1a3f19f80d7dfc10bf3796b71e WebQTest-484_db24dc739b7b00cdab7cf3d825975aed
WebQTest-493_3ce332b703593e71708a5604865c5758 WebQTest-517_e7d8b401e84db19b4965bb23aa43871d
WebQTest-537_4439b7c167b6ccf30c53b610934c4521 WebQTest-537_560ee86f4cb9196c8d757880602e8433
WebQTest-537_b7771109a84a9b56518460dc9e8878ff WebQTest-537_de17dd1afec3743e98d327e6b1912e99
WebQTest-537_e5da8cda32fb1aa37028f9f7f7b1d3a8 WebQTest-538_4767846f70218c1178e6fb0c4937e547
WebQTest-538_485b6c9f1bd7bd7972f3dd4fcc11b1e9 WebQTest-538_5f5da3e1d4ca7df9f19ce3fdfc5790e2
WebQTest-538_65e7de970f6844e8520d7785799ec19e WebQTest-538_d37989c422fef8a98e7289ca0144159c
WebQTest-538_fc5392eefef107aa88084b05ddd2e246 WebQTest-55_54e856d3c44e0ac4a33f0f0ebb7a67d6
WebQTest-576_495085ad9dd6274cce483df63474ca21 WebQTest-576_99c43635b648023af901dc341b3bab6e
WebQTest-61_5b52c9e057d0f0e6704f0b264e864dfc WebQTest-61_7bd6b37d01a372aa5af2a8d5ef53d827
WebQTest-61_9be8c18e016b713cbb7f7feb6badebc7 WebQTest-634_4f2c0d9eb022ed67ec2cc356d4d5113a
WebQTest-654_f4f38507bbec5d30bec54c8c8dd9a72b WebQTest-712_4e9134eaaafee4fd4023d41d4770f5eb

WebQTest-719_2883cc73d6e7cc887da948a32c5fd2bb WebQTest-719_402552b61d68fcf116111585da32583b
WebQTest-719_f1a83a32bf04983a5a5f9da914f867c8 WebQTest-743_2b39afadc21f0a2a46021216l6db1f8c
WebQTest-743_4d2f11c63fa0611b0e29bc00f41bf7a9 WebQTest-743_e8f21e0f0c4375d30317d26f9b95cd91
WebQTest-759_33c0a2138ef3f4a4f61dbee8dc9714e5 WebQTest-783_b181354faec01d121c5f050e0d0da2fd
WebQTest-832_87afbf2481df60df8476b920f00c4247 WebQTest-832_c1d5c9c777c291a80c29c70ec43c45ad
WebQTest-832_c334509bb5e02cacae1ba2e80c176499 WebQTest-837_8c90d5499a627a3478313407b1404ecf
WebQTest-837_c4e06c3a9e4b4f10bd1ae97f1742c198 WebQTest-918_a6b0e48e01f77546d20952acdc67b396
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C.3 Development set ($n = 80$)

Source: results/phase3/dev80.json.

WebQTest-1267_9eeb2f38ee223b499967ac117f641e3b WebQTest-1356_903876a283f3ce8764412a6cfd6a2c44
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WebQTrn-1066 WebQTrn-1067_ed395c43c14a93f3e41c1758ec7e1b8a
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WebQTrn-1549 WebQTrn-1566_63779e80b19cde0010c566ec0b4a0ed6 WebQTrn-1589
WebQTrn-1603_169d8c870fa88566ce1f62da123323d6 WebQTrn-1719_124a834e764c099d727ab87c1f80ccee
WebQTrn-1719_96914535d2c1ae6120a090361e60c965 WebQTrn-1751 WebQTrn-2027 WebQTrn-2028
WebQTrn-2134 WebQTrn-2400 WebQTrn-2436 WebQTrn-254_01e2da60a2779c4ae4b5d1547499a4f8
WebQTrn-2584_0c8bcc717d50c92c31ff802641504b43 WebQTrn-2617
WebQTrn-262_c1b166cd48ca0675b0bb9815289851f3 WebQTrn-2721
WebQTrn-274_5b8fa743161c465927586720934bef48 WebQTrn-279_fd6e419f422eb640f1195f506522023e
WebQTrn-2825_5bf6906c26a34660ad81a9e0506d36fc WebQTrn-2840_abf44b12ba7b11ea7e71a11cb540d258
WebQTrn-3009_a453d9b2a3a3816e8ba70b1a240a4bdd WebQTrn-304_4ec9be4c828025d788bc5fb5ebca9e6e
WebQTrn-3083 WebQTrn-3085 WebQTrn-3109 WebQTrn-334 WebQTrn-3375
WebQTrn-3381_44d67b357829d662abe1c6dc637e219d WebQTrn-3457_b08f65da81574b8397d5f76ea4138c3d
WebQTrn-3460_f6e8b7149f2e422640a49beb753be4ef WebQTrn-3468 WebQTrn-3502 WebQTrn-3525
WebQTrn-3604_fa4b1fc5cc9db1e3164c9b382fcdc4e7 WebQTrn-3617_258527684b01adb1fef0f02a9081ebe7
WebQTrn-3631 WebQTrn-3710 WebQTrn-430_aa25cef57eebcc069b1783e80e760069 WebQTrn-449

```
WebQTrn-528_9591187bdfee1d3adb757d74382745ce WebQTrn-546
WebQTrn-547_1f0f082cf9a15e8c58c9655386e1363f WebQTrn-559_9e3fc7126b544740b568757fd7ae1e88
WebQTrn-568 WebQTrn-569 WebQTrn-770 WebQTrn-90 dev-fake-01 dev-fake-02 smoke-20
```

Appendix D

Annotation Instructions and Label Schema

Two annotation tasks in this thesis produced hand labels: the gold-quality adjudication of Section 7.7, and the failure census of Chapter 9. This appendix gives the instructions and the label schema both were carried out against, so that the categories the census reports can be read as definitions rather than as summary words.

D.1 Judge Instructions

The semantic-support judge of Section 7.6 is shown three things: the question, *one* entity the system asserted as an answer to it, and up to three knowledge-graph paths connecting the question's topic entity to that entity. It decides one thing: *do these paths express the relationship the question asks about, such that this entity is supported as an answer?* Mere connectedness is not support, and the prompt says so in those words.

The unit of judgement is therefore an *entity together with its paths*, not a claim and not a set of triples. That distinction is what the following rules are about, and it is why several of them have no analogue in the claim-level structural check of Section 6.3.1: that check asks whether an edge exists, whereas this one asks whether the edges that exist mean what the question asked. The sampling frame is built from asserted entities, so an abstention never reaches this task at all.

Eight rules constrain the judgement. Each was forced by a case met during calibration, and the guideline was fixed *before* the blind annotation began. That order is worth stating explicitly, because the alternative reading would invalidate the statistic it produced: a guideline revised in response to the disagreements it is later measured on records how well two passes were reconciled, not whether they agreed. The sequence was guideline first, then the blind pass of Section 7.6 — one hundred items labelled without opening the judge's output or the answer key — and $\kappa = 0.6995$ against a pre-registered bar of 0.70 is what that pass returned. The outcome is reported there rather than here.

1. **Judge the asked relationship, under the graph's vocabulary.** The test is whether the

path relations express what the question asked, as the graph is able to express it — not whether a better-designed schema would have said it more directly. A path that is merely *about* the right entities does not qualify.

2. **A mediator-bridged pair of edges is one relation.** Freebase routes many binary facts through an unnamed compound-value node, so the path renders as two hops. It is judged as the single relationship it encodes, on the same contract the tool layer uses (Section 5.2.3).

3. **Any one supporting path is enough.** Up to three are shown. If one of them expresses the asked relationship, the entity is supported; the others need not, and a weak path alongside a good one is not evidence against.

4. **Judge relations by role, not by type.** A relation that is generic in most contexts counts when it is itself the thing asked about — a nationality edge is generic filler for a question about films and is the entire answer for a question about citizenship.

5. **For conjunctions, every asked relationship must be covered by some specific path, jointly rather than in one chain.** A question imposing two constraints is supported when the paths together establish both, even if no single path establishes both at once.

6. **Judge each entity independently of answer-set completeness.** Whether the system found the *other* correct answers is recall, and it is measured by F1 elsewhere. This task asks only whether *this* assertion is grounded.

7. **Ignore superlative and temporal filters the graph cannot express.** Where a question asks for the first, largest, or most recent of something and the environment encodes no ordinals or date literals (Section 4.7.4), judge only the base relationship. Penalising the path for a filter the environment cannot represent would measure the environment, not the assertion.

8. **A path expressing only the question's identifying descriptor, rather than its actual ask, is unsupported.** "The 33rd president who led the nation during the war" identifies its subject twice and asks for it once; a path that only re-establishes the description has answered nothing.

Rule 8 names the *composition-echo* case; Section 7.6 draws out why it matters that the failure census arrives at the same mechanism independently.

D.2 Failure Census Label Schema

Every packet in the census receives one category, an optional subtype, and a one-sentence note. The note pinpoints where the trajectory *first* left the correct path — plan, then explored

relations, then backtracks, then verifier — rather than the last symptom. That ordering is the whole discipline of the exercise: a wrong answer usually has several things wrong with it by the end, and only the first one is a cause.

Convention for the verifier categories

The verifier's positive class is *claim is supported*. A false negative is therefore a claim wrongly *rejected*, and a false positive is a claim wrongly *accepted*. This is the standard convention and the one the census labels use throughout. It is stated explicitly because the gloss in the working reference originally paired the two names in reverse, and two Stage A labels were filed against the reversed reading before the error was found. Both were corrected; Section D.3 records them individually.

Table D.1: Failure categories and their subtypes. One category per packet.

Category	Subtypes	Meaning
decomposition_error	over_decomposition, paraphrase_drift, context_stripping, extraction_bug	the planner or drafting pipeline caused it
relation_selection	—	the scorer picked the wrong edge
composite_claim	conjunction_uncovered, no_set_intersection	a multi-constraint question handled only partially
premature_termination	budget, evaluator	stopped before reaching the answer
verifier_fn / verifier_fp	structural_not_semantic	hedged a right claim / passed a wrong one
kg_gap	date_literal, numeric_literal, temporal_qualifier, ordinal, data_error	the environment cannot express what the question needs
answer_selection	—	the correct candidate was retrieved and the drafter chose another
echo	topic, intermediate, granularity	the shared-attractor pattern of Section 9.7
gold_noise / ambiguous_question	see Table D.2	a defect in the gold or the question that survived Section 7.7
other	—	fits none of the above

Gold-defect subtypes

Seventeen gold defects survive into the census itself — 3 on WebQSP and 14 on CWQ, filed as gold_noise (7) or ambiguous_question (10) and reported as such in Section 9.8. They are not subtyped there, because the census met each one incidentally while reading a failure, whereas the

adjudication pass of Section 7.7 met roughly 148 flagged rows and made exactly this judgement at volume. Subtyping a defect class is only worth doing where the boundaries were drawn against many cases, so the adjudication pass supplies the reference list and the census rows are counted against its categories rather than given a vocabulary of their own. Table D.2 gives them with the counts they were derived from.

Table D.2: Gold-defect and question-defect subtypes, with the counts observed during adjudication across both datasets. The unit is the *flagged row*: 63 of the 148 rows the consensus pass surfaced were adjudicated defective, and these are their subtypes. That is not the unit of Table 7.2, which counts the 41 *questions* the adjudication excluded — a defective row does not always produce an exclusion, and Section 7.7 keeps the two apart deliberately.

Subtype	**Meaning**	**n**
gold_noise		
wrong_gold	the gold answer is factually incorrect	16
incomplete_gold	gold omits valid answers that systems correctly found	16
type_mismatch	gold answers at the wrong entity type or granularity, though the right type is in the graph	7
malformed_gold	gold is structurally corrupted, such as absurd entity counts or machine identifiers leaked into answer strings	3
ambiguous_question		
temporal	the question's time reference does not uniquely pick out gold's answer	9
container_vs_member	readable as asking about a container or about its members	3
place_vs_lineage	"where does X come from" answerable as a place or as an ancestry chain	3
malformed_question	two or more sub-questions conflated into one	2
birthplace_vs_upbringing	"where is X from" genuinely ambiguous between the two	1
variety	gold picks one defensible variety or register of an entity over another	1
place_name	a named place has competing referents and gold picks one	1
underdetermined	the question does not determine a single correct answer	1

D.3 Label Provenance and Regeneration

The census is hand labour recorded in files that scripts also write, so which file is authoritative for which field is not obvious from inspection. Table D.3 settles it.

After any relabelling, three scripts must run in order: pull_noplanner_categories.py, then synthesize_census.py, then build_thesis_numbers.py. Skipping the last leaves the thesis quoting the previous histogram.

Table D.3: Where each label lives, and what regenerating will do to it.

File	Status
labels_{ds}.csv	The hand-labelled census. dump_failure_packets.py rewrites the file but carries existing labels forward by question ID, so hand edits survive a re-run.
noplanner_categories_{ds}.csv	Generated, with no carry-forward. Labels belong in noplanner_discordant_{ds}.md, which pull_noplanner_categories.py transcribes into this CSV. An edit made directly to the CSV is overwritten on the next run.
labels_{ds}_dropped.csv	An input to the histogram rather than an archive: synthesize_census.py merges its rows into the totals. Nothing regenerates it, because it is written only at the moment a row leaves the population. If it is lost, recover it from version control.
sampling_manifest.json	Written but never read back. Its carried, to-read and dropped fields are progress counters. The thesis cites only the population and skip counts, which are stable across re-runs.

Corrections made after the first labelling pass

Three changes were made to labels after the initial pass, and are recorded here rather than silently absorbed.

- The false-positive and false-negative gloss in the working reference paired the two names in reverse of the standard convention. It was corrected, and the convention is now stated explicitly above. The census labels themselves had followed the standard convention throughout and were not changed.
- WebQTrn-1294 moved from verifier_fn to answer_selection. The verifier had been correct; the error was upstream, in the answerer.
- WebQTrn-1597 moved from verifier_fn to verifier_fp, the original label having used the reversed convention.

The histogram was regenerated afterwards. The totals are unchanged; three CWQ hedge rows moved between categories.

Appendix E

Implementation Notes

Section 7.10 names three practices that carry more reproducibility weight than the dependency list, and promises they are recorded here as techniques rather than as incidents. Those are the first three sections below. Three further techniques follow, which govern how the agent itself is built and which the earlier three depend on. In each case the defect that forced the practice is given as evidence rather than as narrative.

E.1 Records Self-Describe

Every record written by any run carries, alongside its result, the backbone description, a hash of the budget configuration, the full run configuration including the system name, and the source commit.

The cost is a few hundred bytes per question. The benefit is that provenance never depends on where a file sits or what it is called. A record found in the wrong directory, or copied out of one, still states what produced it. This is what made the cache-replay anomaly of Section 7.3 diagnosable: the question was which configuration had produced a set of records, and the records answered it themselves.

The generalisation is that a file path is not metadata. Directory layout is edited, archived and reorganised over a project's life, and every reorganisation is an opportunity to detach a result from the conditions that produced it. Anything a reader must know in order to interpret a record belongs inside the record.

E.2 Frozen Artifacts Are Committed; Regenerable Ones Are Not

The distinction is not about file size. It is about whether the artifact can be recreated.

The certified question sets, their half-splits, and every result file embody spent money and runs that cannot be repeated identically. They are committed. The graph import, the embedding matrices and the triple index are deterministic products of committed scripts run over public data, and are not committed even though rebuilding them takes hours.

Applying the rule requires honesty about the second category. An artifact is only regenerable if the script that generates it is committed *and* its inputs are still obtainable. Where either fails, the artifact is frozen regardless of how it was made, and Section 7.10 records the one case in this project where that judgement had to be made after the fact: a generated summary in the development-set archive that predates a fix applied to the runs it summarises.

E.3 Cache-Identity Verification

This is the technique of most methodological interest, and the one most readily reusable outside this thesis.

The problem it solves is stated easily and answered badly by ordinary testing. A refactor is made to code on the frozen experimental path. Tests pass. Have the reported numbers changed? Re-running the experiment answers the question, but a re-run that produces the same numbers is weak evidence, because it is exactly what a re-run would produce if the change had also altered which questions happened to succeed.

The stronger check exploits the response cache. Every model call is keyed by a hash over the model identifier, the sampling parameters, and the full prompt text. A code change that does not perturb the frozen path therefore issues *byte-identical prompts*, which means every call must be a cache hit. So:

> Re-run the affected questions and assert that the number of cache hits equals the number of model calls, and that both are non-zero.

A single cache miss proves that some prompt differed, and therefore that the change was not behaviour-preserving — before any output is compared. The assertion is stronger than output comparison because it tests the input to the non-deterministic component rather than the output of it, and it is cheap because a fully cached re-run costs nothing.

For this to work the cache hit must be indistinguishable from a live call inside the agent. The budget check still runs, the call counter still increments, and the original token counts are replayed into the meter from the stored record. A cached re-run therefore reproduces the original budget snapshots exactly rather than reporting zeros, which is what makes the identity assertion meaningful. A separate counter records hits, so live and replayed runs remain distinguishable in analysis.

Section 7.1.1 reports the one occasion this certificate was demanded and returned a clean result, for the use_planner flag that must be a no-op when disabled; Section 5.9 specifies the cache whose behaviour makes the certificate meaningful.

E.4 Config Injection and the Partial-Edit Hazard

The agent is a graph of nodes whose signatures the framework, not the author, controls. Passing a run configuration into a node therefore means either closing over it at construction time or threading it through the framework's own injection mechanism. The first is easier and is a trap: a node that closes over a configuration value captures whatever was in scope when the graph was built, which for a sweep means the first condition's value silently serving every condition.

The practice that follows is narrow and worth stating plainly. *A configuration value must reach the code that uses it by exactly one route.* Where a value can arrive by two routes, an edit that updates one of them leaves the other in place, and the resulting run is neither the old condition nor the new one. This is the partial-edit hazard, and it is not caught by tests, because each route works in isolation.

The defence used here is an assertion at the entry of every node that receives a configuration, checking the type of what actually arrived. It converts a silent wrong-condition run into a crash on the first question. A run that dies immediately costs minutes; a run that completes under the wrong condition contaminates a table.

E.5 Routers Read, Nodes Write

The state object is mutated by nodes and inspected by routers. Allowing routers to write as well produces control flow that cannot be reconstructed from the trace, because the state a node observes was modified by the routing decision that reached it.

The discipline is one sentence: *routers may read state and must not write it; nodes may write state and must not route.* It costs a small amount of duplication — a value a router needs must be written by the node before it, under a name the router agrees on — and buys the property that the trace is a complete account of the run. Every claim in Chapter 9 about where a trajectory went wrong depends on that being true, because the census reads traces, not code.

Two categories of scratch key exist in the state for this reason. Node outputs are the record. Router inputs, prefixed with an underscore, exist so that a router can decide without recomputing what a node already knew.

E.6 Instrumentation Decides What Can Be Concluded

The verifier persists the claims it rejects. It does not persist the claims it accepts.

That decision was made early, on the reasonable ground that rejections are the interesting events: they are what the layer does, and they are few. Its consequence surfaced only at analysis time. Wrongful rejection — a true claim hedged — is recoverable from the logs at census scale, because every rejection is on record. Wrongful acceptance — a false claim passed — is not, because an accepted claim leaves no trace distinguishable from a claim that was never tested.

The thesis therefore reports one polarity of verifier error as a rate and the other from a single specimen recovered by manual trace reconstruction. Since wrongful acceptance is the polarity that bears directly on a hallucination claim, this is the limitation named first in Section 10.2.

The technique to extract from this is not a fix; the fix is obvious and is the first item of Section 10.3. It is the prior question:

> Before freezing an instrument, ask which claim it will be asked to support, and whether the records it keeps can support the negation of that claim as well as its assertion.

Logging the rejections was sufficient to describe what the layer does. It was not sufficient to bound what the layer misses, and only the second supports a claim about hallucination. The cost of logging both was negligible; the cost of not doing so was a limitation that no amount of later analysis could remove.

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